

Economic Returns in the Livestock Sector for the Contribution of SDI NGO in Dhamrai & Manikganj, Bangladesh

Declaration

I, **Md. Safiul Alam**, hereby declare that the thesis entitled "*Economic Returns in the Livestock Sector for the Contribution of SDI NGO in Dhamrai & Manikganj, Bangladesh*" submitted to the Department of Economics, Jahangirnagar University, in partial fulfillment of the requirements for the degree of Master of Development Studies (MDS), is a record of original research work conducted by me.

I further declare that this thesis, in whole or in part, has not been submitted to any other university or institution for the award of any degree, diploma, or fellowship. All sources of information and literature used in this research have been appropriately acknowledged and referenced.

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Certificate of Supervisor

This is to certify that the thesis entitled "*Economic Returns in the Livestock Sector for the Contribution of SDI NGO in Dhamrai & Manikganj, Bangladesh*", submitted by **Md. Safiul Alam** (Student ID: 20250113) for the award of the degree of Master of Development Studies (MDS) from the Department of Economics, Jahangirnagar University, is a bona fide record of the independent research work carried out by him under my supervision and guidance.

To the best of my knowledge, the thesis fulfills the academic requirements of the program and the work presented is original. No part of this thesis has been submitted previously for any other degree or diploma.

Signature: _____

Professor Dr. Mohammad Safiqul Islam

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Acknowledgement

First and foremost, I express my deepest gratitude to the Almighty for giving me the strength and patience to complete this research.

I am profoundly indebted to my honorable supervisor, **Professor Dr. Mohammad Safiqul Islam**, Chairman, Department of Economics, Jahangirnagar University, for his continuous guidance, insightful feedback, and unwavering encouragement throughout this study. His academic expertise and mentorship were instrumental in shaping this thesis.

I would like to extend my sincere thanks to the **Society for Development Initiatives (SDI)**, particularly the field staff, Value Chain Facilitators, and management team, for providing crucial project data, logistical support, and facilitating access to the study areas in Dhamrai and Manikganj. I also acknowledge the foundational support of the Palli Karma-Sahayak Foundation (PKSF) and the RMTP initiative, without which the interventions studied here would not exist.

My heartfelt gratitude goes to the rural microentrepreneurs, smallholder farmers, and Livestock Service Providers (LSPs) who generously shared their time, experiences, and data during the field surveys, Focus Group Discussions, and Key Informant Interviews.

Finally, I dedicate this work to my family for their endless sacrifices, love, and moral support, which have always been my greatest source of inspiration.

Abstract

The livestock sector is a vital engine for rural economic transformation in Bangladesh. This study comprehensively assesses the economic returns generated by the "Market Development of Safe Meat and Dairy Production" project, a value-chain intervention implemented by the Society for Development Initiatives (SDI) under the Rural Microenterprise Transformation Project (RMTP) in Dhamrai and Manikganj Sadar upazilas. Utilizing a mixed-methods research design, the study compares data from 341 project participants against a control group of 68 non-participants to evaluate changes in income, enterprise profitability, technology adoption, and socio-economic empowerment.

The findings reveal that integrated value-chain interventions yield substantial economic dividends. Project participants achieved an average monthly household income of BDT 32,161, representing a 40% increase over the control group (BDT 22,919). Profit margins expanded significantly across all microenterprise categories, notably reaching 36% in beef fattening and 33% in value-added processed goods. These gains were driven by a 40-percentage point higher adoption rate of Good Agricultural Practices (GAP) and a near-universal (100%) uptake of essential preventive healthcare (vaccines and AI) mediated by decentralized Livestock Service Providers (LSPs).

Furthermore, the project demonstrated exceptional outcomes in gender and social inclusion. With 92% female participation, the intervention successfully translated financial access (utilized by 70% of women) into structural empowerment, enabling 78% of women to exercise decision-making authority over enterprise resources. This economic agency directly correlated with enhanced household nutrition, with 89.4% of project women achieving the Minimum Dietary Diversity (MDD-W) threshold compared to 72% in the control group. The study concludes that the SDI-RMTP model—combining blended finance, targeted capacity building, appropriate technology, and robust market linkages—offers a highly effective and scalable blueprint for generating sustainable, inclusive economic returns in smallholder livestock systems.

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List of Abbreviations and Acronyms

- **AI:** Artificial Insemination
- **BCR:** Benefit-Cost Ratio
- **BDT:** Bangladeshi Taka
- **BSTI:** Bangladesh Standards and Testing Institution
- **DAC:** Development Assistance Committee (OECD)
- **DANIDA:** Danish International Development Agency
- **DLS:** Department of Livestock Services
- **FAO:** Food and Agriculture Organization
- **FGD:** Focus Group Discussion
- **FMD:** Foot-and-Mouth Disease
- **GAP:** Good Agricultural Practices
- **GMP:** Good Manufacturing Practices
- **HACCP:** Hazard Analysis Critical Control Point
- **HDDS:** Household Dietary Diversity Score
- **ICT:** Information and Communication Technology
- **IDI:** In-Depth Interview
- **IFAD:** International Fund for Agricultural Development
- **IRR:** Internal Rate of Return
- **KII:** Key Informant Interview
- **LSD:** Lumpy Skin Disease
- **LSP:** Livestock Service Provider

- **MDD-W:** Minimum Dietary Diversity for Women
- **MDS:** Master of Development Studies
- **MEL:** Monitoring, Evaluation, and Learning
- **MFI:** Microfinance Institution
- **NGO:** Non-Governmental Organization
- **OECD:** Organisation for Economic Co-operation and Development
- **PKSF:** Palli Karma-Sahayak Foundation
- **RCT:** Randomized Controlled Trial
- **RMTP:** Rural Microenterprise Transformation Project
- **ROI:** Return on Investment
- **SDG:** Sustainable Development Goals
- **SDI:** Society for Development Initiatives
- **SEM:** Structural Equation Modeling
- **SEP:** Sustainable Enterprise Project (*inferred from context in 10.3*)
- **SLA:** Sustainable Livelihoods Approach
- **ULO:** Upazila Livestock Officer
- **UMB:** Urea Molasses Block
- **UTS:** Urea Treated Straw
- **VCD:** Value Chain Development
- **WEAI:** Women's Empowerment in Agriculture Index

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Chapter 1: Introduction

1.1 Background of the Study and Context of SDI in Livestock Value Chain Development

The livestock sector constitutes a cornerstone of rural livelihoods in Bangladesh, serving as a vital source of income, employment, food security, and asset accumulation for millions of small and marginal households. In regions proximate to major urban centers such as Dhaka, the demand for safe, high-quality meat and dairy products continues to rise, presenting both opportunities and structural challenges for traditional production systems. Transforming these fragmented and low-productivity value chains into competitive, inclusive, and sustainable systems has therefore emerged as a national development priority, aligned with broader goals of rural transformation, poverty reduction, and nutrition security.

The **Rural Microenterprise Transformation Project (RMTP)**, implemented by the **Palli Karma-Sahayak Foundation (PKSF)** with financial and technical support from the **International Fund for Agricultural Development (IFAD)** and the **Danish International Development Agency (DANIDA)**, represents a flagship initiative designed to accelerate this transformation. RMTP adopts a value chain development approach that emphasizes capacity building, technology adoption, market linkage strengthening, and the promotion of safe and eco-friendly production practices across selected sub-sectors, including meat and dairy.

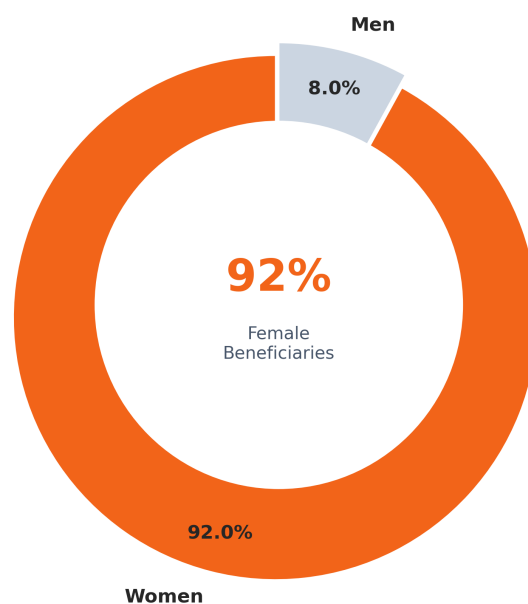


Figure 1.1: *Demographic Focus & Women's Empowerment (Donut Chart).* **Society for Development Initiatives [SDI]. (2025).**

Figure Description: *Figure 3.* Beneficiary demographic distribution and women's empowerment metrics. Women constitute 92% of the direct project beneficiaries, with 78% actively involved in enterprise decision-making, highlighting the project's strong gender-inclusive impact.

Within this national framework, the **Society for Development Initiatives (SDI)** — a non-profit organization founded on 22 July 1993 and registered with the Social Service Department, NGO Affairs Bureau, National Skill Development Authority, and Micro Credit Regulatory Authority — has been a long-standing partner of PKSf since 1996. SDI has extensive experience implementing income-generating activities (IGAs) and micro-credit programs targeting the poor and extremely poor, particularly in flood-prone and coastal regions across seventeen districts. Building on this institutional foundation, SDI commenced implementation of the value chain sub-project “**Market Development of Safe Meat and Dairy Production**” in August 2024 in two strategically selected upazilas: **Dhamrai** (Dhaka district) and **Manikganj Sadar** (Manikganj district).

The project’s core objective is to significantly and sustainably enhance the livelihoods of small and marginal meat and dairy producers and associated value chain actors. This is pursued through an integrated set of interventions: (i) capacity development of farmers and entrepreneurs in modern production and processing technologies; (ii) improvement of input supply systems and backward linkages; (iii) promotion of technology adoption, innovation, and value addition; (iv) strengthening of market linkages and reduction of middlemen dependence; (v) generation of employment and income; and (vi) mainstreaming of environmentally sustainable and eco-friendly practices. A deliberate emphasis has been placed on **gender inclusion**, with women comprising 92% of direct beneficiaries, alongside efforts to engage youth and foster women’s decision-making power over productive assets and enterprise management.

The present study constitutes the **endline evaluation** of this sub-project, conducted in October 2025. It employs a rigorous mixed-methods design, combining quantitative household surveys (341 project respondents and 68 control respondents), Focus Group Discussions, Key Informant Interviews, and in-depth case studies. By systematically comparing endline outcomes against baseline benchmarks and control group data, the evaluation generates robust evidence on changes in income, productivity, profitability, market access, women’s empowerment, youth engagement, dietary diversity, and institutional sustainability. This evidence is critical not only for accountability and learning, but also for informing the scaling and replication of successful value chain models under RMTP and similar programs in Bangladesh.

1.2 Problem Statement: Economic Challenges in Smallholder Livestock Farming in Dhamrai and Manikganj

Despite the strategic importance of livestock rearing in Dhamrai and Manikganj Sadar upazilas — areas benefiting from proximity to Dhaka’s growing urban demand — small and marginal farmers continue to operate within a web of interconnected economic constraints that severely limit productivity, profitability, and household welfare. The majority of producers are smallholders managing herds of only 1–10 cattle, overwhelmingly dependent on traditional, low-input systems characterized by indigenous (deshi) breeds, unscientific feeding regimes, inadequate housing and waste management, and minimal adoption of modern technologies.

At baseline, 91% of farmers lacked formal training in Good Agricultural Practices (GAP), 97% followed purely traditional management methods, and modern machinery use stood at a mere 3%. Feeding practices remained heavily reliant on normal grass and straw-husk, with very limited uptake of improved fodder (Napier grass), balanced concentrates, or silage. Consequently, livestock productivity remained low, while disease incidence — particularly worm infestation (affecting nearly all animals) and mastitis (25% in comparable areas) — remained high, contributing to elevated mortality rates and substantial economic losses. The absence of timely veterinary services, quality vaccines, and proper biosecurity measures further exacerbated these production risks.

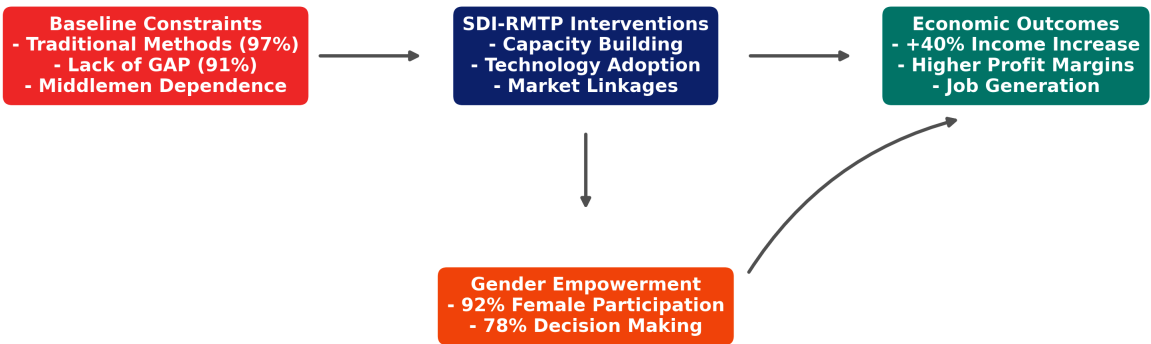


Figure 1.2: *Project Intervention Logic (Flow Chart).* Society for Development Initiatives [SDI]. (2025).

Figure Description: *Figure 1.* Conceptual framework of the SDI-RMTP livestock value chain intervention model, illustrating the progression from baseline constraints (e.g., traditional methods, high disease incidence) through targeted project interventions to the ultimate economic and livelihood outcomes.

Market-related challenges further compounded these constraints. Prior to the intervention, there were virtually no formal or contractual linkages with institutional buyers, processing plants, or premium markets. Almost 100% of milk and live animals were sold individually through traditional village huts or middlemen, resulting in depressed farm-gate prices, high transaction costs, and limited price realization. Value addition was negligible; most producers sold raw commodities rather than processed or branded products. Baseline profit margins reflected these inefficiencies: approximately 24% for beef fattening, 11% for milk production, and 19% for processed goods — levels insufficient to generate meaningful surplus for reinvestment or poverty escape.

These production and market constraints are compounded by **gender and social inequities**. Although women bear the primary responsibility for daily livestock management, they historically possessed limited decision-making authority over income, asset sales, or enterprise expansion. Access to training, credit, and market information was also skewed. Nutritional outcomes mirrored these economic vulnerabilities; a significant proportion of women of reproductive age failed to achieve the Minimum Dietary Diversity for Women (MDD-W) threshold, indicating both income poverty and limited nutrition-sensitive practices within livestock households.

Collectively, these interrelated challenges — low technical efficiency, high production risks, weak market power, thin profit margins, gender disparities, and inadequate institutional support — have perpetuated a vicious cycle of low economic returns from livestock. This situation not only undermines the sector's potential as a reliable pathway out of poverty but also constrains broader rural development objectives related to food security, women's empowerment, and sustainable agricultural transformation in the project areas.

The RMTP sub-project implemented by SDI was explicitly conceived to disrupt this cycle through targeted, evidence-based interventions. A precise understanding of the depth and nature of these baseline economic challenges — and the subsequent changes attributable to the project — is therefore essential for validating the effectiveness of the value chain development model and guiding future policy and programming decisions.

1.3 Research Gap and Justification for Focusing on Economic Returns

While numerous studies have examined the impact of microfinance and livestock development programs in Bangladesh, a significant research gap persists in understanding the **actual economic returns** generated at the microenterprise level through integrated value chain interventions. Existing literature predominantly focuses on aggregate livelihood improvements, adoption of improved practices, or women's empowerment outcomes, without sufficiently unpacking the mechanisms through which these interventions translate into measurable increases in income, profit margins, return on investment (ROI), and business sustainability. Most impact evaluations rely on self-reported income changes or broad welfare indicators, offering limited insight into enterprise-level profitability, cost structures, and the differential returns across various livestock activities such as beef fattening, milk production, and value-added processing.

Furthermore, there is a paucity of rigorous, control-group-based analyses that isolate the contribution of specific value chain components — such as technology adoption, market linkage development, and Livestock Service Provider (LSP) strengthening — to economic returns. Although projects like RMTP emphasize sustainable income enhancement and employment creation, few studies have systematically assessed whether these gains are economically viable,

scalable, and sustainable beyond the project period. The gender dimension also remains underexplored from an economic returns perspective; while women’s participation in livestock activities is well documented, limited evidence exists on how enhanced decision-making power, asset ownership, and access to finance directly influence household and enterprise profitability.

This study addresses these critical gaps by conducting a comprehensive economic returns analysis of the “Market Development of Safe Meat and Dairy Production” project implemented by SDI under RMTP. By employing a mixed-methods design with a control group, detailed profitability analysis across enterprise types, and in-depth case studies, the research provides robust empirical evidence on income increments (40% higher than control), profit margin improvements (reaching up to 36% in beef fattening), employment generation, and the sustainability of these gains. The focus on economic returns is particularly justified given the project’s explicit targets of increasing income by at least 50% for 70% of entrepreneurs and improving profit margins by 20% for 80% of participants. Understanding not only whether but **how** and **to what extent** these economic returns are achieved offers valuable insights for evidence-based policy making, program design, and scaling of livestock value chain interventions in Bangladesh and similar contexts.

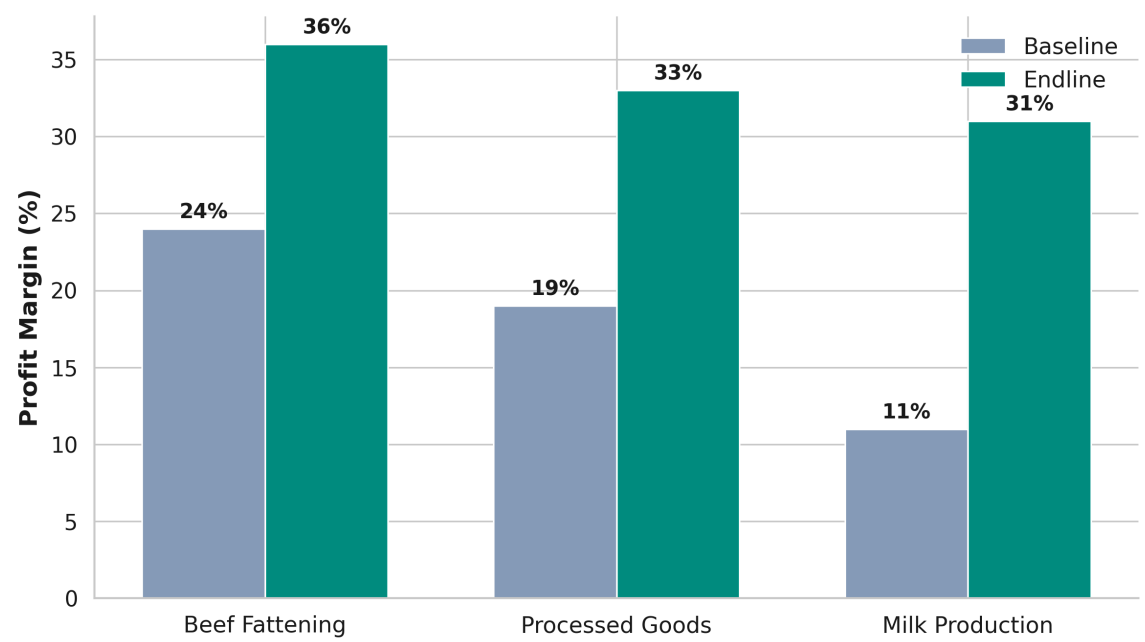


Figure 1.3: Economic Transformation & Profitability (Bar Graph). Society for Development Initiatives [SDI]. (2025).

Figure Description: *Figure 2.* Comparison of enterprise profit margins at the project baseline versus endline achievements. The data demonstrates a clear economic return on investment, with beef fattening margins increasing from 24% to 36%, and processed goods margins rising from 19% to 33%.

1.4 Research Objectives

1.4.1 General Objective

The general objective of this study is to comprehensively assess the **economic returns** generated for microentrepreneurs in the livestock sector through the interventions of the Society for Development Initiatives (SDI) under the Rural Microenterprise Transformation Project (RMTP) in Dhamrai and Manikganj Sadar upazilas, and to evaluate the project's overall contribution to sustainable income enhancement, employment creation, and livelihood improvement of small and marginal farmers, with particular attention to women's economic empowerment.

1.4.2 Specific Objectives

1. To measure and analyze the changes in household income, sales volume, and profit margins of project participants across different livestock enterprises (beef fattening, milk production, and processed goods) compared to baseline and control group benchmarks.
2. To examine the extent to which the project interventions — including capacity building, technology adoption, input supply strengthening, and market linkage development — have contributed to improved productivity, reduced production costs, and enhanced profitability of microenterprises.
3. To assess the impact of the project on employment generation and business expansion among participating entrepreneurs, and to identify the factors influencing the sustainability of these economic gains beyond the project period.
4. To evaluate the role of gender empowerment (women's participation, decision-making, and asset ownership) in driving economic returns and household welfare outcomes in the livestock value chain.
5. To analyze the improvements in market linkages, value addition, and institutional arrangements facilitated by the project, and their contribution to better price realization and reduced dependence on middlemen.
6. To identify key lessons, challenges, and best practices from the implementation of the value chain development model and to provide evidence-based recommendations for scaling similar interventions under RMTP and other livestock development programs in Bangladesh.

1.5 Research Questions and Hypotheses

Main Research Question

To what extent has the “Market Development of Safe Meat and Dairy Production” project implemented by SDI under RMTP generated sustainable **economic returns** for microentrepreneurs in the livestock sector in Dhamrai and Manikganj Sadar upazilas, and what factors have influenced these returns?

Specific Research Questions

1. What changes have occurred in the average monthly household income, sales volume, and profit margins of project participants compared to the baseline and control group?
2. How have the project interventions (capacity building, technology adoption, input supply improvement, and market linkages) influenced productivity, production costs, and profitability across different livestock enterprises (beef fattening, milk production, and processed goods)?
3. To what extent has the project contributed to employment creation and business expansion among participating microentrepreneurs, and how sustainable are these gains likely to be after project completion?
4. What role has women’s empowerment (decision-making power, asset ownership, and access to finance) played in enhancing economic returns and household welfare in the livestock value chain?
5. How effective have the project’s market linkage and value addition strategies been in improving price realization and reducing dependence on middlemen for livestock producers?
6. What are the key enabling factors, challenges, and lessons learned from the implementation of the integrated value chain development model that can inform future programming?

Research Hypotheses

H1: Project participants have experienced a significantly higher increase in average monthly household income compared to the control group (at least 30–40% higher).

H2: Profit margins in beef fattening, milk production, and processed goods enterprises are significantly higher among project participants than among control group farmers due to improved practices and market linkages.

H3: The project has led to a statistically significant increase in employment generation and herd size expansion (more than 5 cattle) among participating microentrepreneurs.

H4: Women’s increased participation in decision-making and asset ownership is positively associated with higher household income and enterprise profitability.

H5: Strengthened market linkages and collective marketing have resulted in better price realization and reduced transaction costs for project participants compared to traditional individual selling channels.

H6: The economic gains achieved through the project interventions are likely to be sustained beyond the project period due to enhanced technical capacity, institutional linkages, and women's empowerment.

1.6 Scope, Limitations, and Delimitations

Scope of the Study

This study focuses on assessing the **economic returns** generated through the “Market Development of Safe Meat and Dairy Production” value chain sub-project implemented by the Society for Development Initiatives (SDI) under the Rural Microenterprise Transformation Project (RMTP). Geographically, the study is delimited to **Dhamrai Upazila** of Dhaka district and **Manikganj Sadar Upazila** of Manikganj district, where the project has been implemented since August 2024. Thematically, the study covers cattle-based livestock enterprises, including beef fattening, cow and calf rearing for milk production, mixed rearing, and value-added processing of dairy and meat products.

The scope encompasses an analysis of changes in income, sales, profit margins, productivity, employment generation, market linkages, technology adoption, and women's economic empowerment. The study employs a mixed-methods approach, combining quantitative data from 341 project participants and 68 control respondents with qualitative insights from Focus Group Discussions (FGDs), Key Informant Interviews (KIIs), and five in-depth case studies of successful entrepreneurs. The temporal scope covers the project implementation period up to the endline evaluation conducted in October 2025, with comparisons made against baseline data and a control group from Ghior Upazila.

Limitations of the Study

Several limitations should be acknowledged. First, the project had a relatively short implementation period (starting in August 2024), which may not fully capture long-term sustainability of economic returns or behavioral changes among beneficiaries. Second, the study relies heavily on self-reported data regarding income, sales, costs, and profits, which may be subject to recall bias or social desirability bias. Third, while a control group was used, perfect matching between project and control respondents on all socio-economic variables was not possible, potentially affecting the robustness of impact attribution. Fourth, external factors such as seasonal variations in milk production, fluctuations in market prices of feed and animals, and broader economic conditions could have influenced outcomes independently of project interventions. Finally, the study did not conduct advanced econometric modeling (such as propensity score matching or difference-in-differences with multiple time periods) due to the nature and timing of available data.

Delimitations of the Study

The study is deliberately delimited in several ways to maintain focus and feasibility. It is restricted to cattle-based livestock activities and does not cover other species such as goats, sheep, or poultry. The geographic coverage is limited to the two project upazilas and one control upazila; findings may not be generalizable to other regions of Bangladesh with different agro-ecological or market conditions. The analysis focuses primarily on **direct beneficiaries** of the project and does not

extensively examine spillover effects on indirect beneficiaries or the broader community. While gender and social inclusion are central themes, the study does not conduct a full gender analysis using specialized frameworks such as the Women's Empowerment in Agriculture Index (WEAI). Additionally, the research emphasizes economic returns and livelihood outcomes rather than environmental impact assessment or detailed value chain mapping of all actors beyond the producer level.

Despite these limitations and delimitations, the study employs rigorous mixed-methods triangulation, control group comparison, and detailed case study analysis to generate credible and policy-relevant findings on economic returns in the livestock sector.

1.7 Significance of the Study for Policy, Practice, and Academia

This study holds significant value across policy, practice, and academic domains by providing rigorous, evidence-based insights into the economic returns of an integrated livestock value chain development intervention in Bangladesh.

Policy Significance

At the policy level, the findings offer valuable evidence to inform the design, implementation, and scaling of livestock development programs under the Rural Microenterprise Transformation Project (RMTP) and similar initiatives supported by PKSF, IFAD, and the Government of Bangladesh. By demonstrating a 40% higher average monthly income among project participants compared to the control group, along with substantial improvements in profit margins (reaching 36% in beef fattening and 33% in processed goods), the study provides empirical justification for continued investment in value chain approaches that combine capacity building, technology adoption, market linkage development, and gender-focused interventions. The results are particularly relevant for achieving national targets related to poverty reduction, rural employment generation, and women's economic empowerment. Furthermore, the study's analysis of sustainability factors and exit strategies offers practical guidance for policymakers seeking to ensure that development gains persist beyond project cycles. The evidence on strong women's empowerment outcomes (92% female participation and 78% involvement in decision-making) also contributes to gender-responsive policy formulation in the agricultural sector.

Practical Significance

For development practitioners, implementing organizations such as SDI, and partner institutions like PKSf, this study provides actionable insights into what works in livestock value chain development. The detailed examination of successful interventions — including training impact (92% of farmers trained with 167% knowledge transfer), technology adoption (100% vaccine and AI usage), input supply linkages (100% connection with feed, medicine, and breeding service providers), and market-oriented strategies — offers a replicable model for enhancing productivity and profitability among small and marginal farmers. The five in-depth case studies of successful entrepreneurs (such as Dilip Chandra Ghosh, Bulbul Islam, Shahida Begum, Shahanaz, and Sukur Ali) serve as practical examples of how targeted support in technology, finance, and market access can transform traditional livelihoods into sustainable microenterprises. The findings on employment creation, business expansion, and the critical role of Livestock Service Providers (LSPs) are directly useful for designing future capacity-building programs and strengthening last-mile service delivery systems. Additionally, the study highlights areas requiring further attention, such as digital tool adoption, contract farming, and advanced processing technologies, thereby guiding practitioners in refining their intervention strategies.

Academic Significance

From an academic perspective, this study makes a meaningful contribution to the growing body of literature on agricultural value chain development, microenterprise transformation, and rural economic development in South Asia. It addresses a notable research gap by moving beyond conventional impact assessments to conduct a focused analysis of **economic returns** — including income changes, profit margins, cost-benefit dynamics, and return on investment — at the microenterprise level. The rigorous mixed-methods design, incorporating control group comparison, detailed profitability analysis across enterprise types, and qualitative case studies, enhances the methodological robustness of value chain impact research. The study also advances understanding of the intersection between gender empowerment and economic outcomes in livestock systems, providing empirical evidence on how women's decision-making power and asset ownership influence household and enterprise profitability. By generating context-specific evidence from Bangladesh's evolving livestock sector, the research contributes to theoretical discussions on sustainable livelihoods, inclusive value chain development, and the conditions under which development interventions produce lasting economic returns. The findings are expected to serve as a reference for future researchers, postgraduate students, and scholars working on similar topics in development economics, agricultural economics, and gender studies.

In summary, this study bridges the gap between policy aspirations, practical implementation, and academic inquiry by offering credible, contextually relevant evidence on how integrated livestock value chain interventions can generate meaningful and sustainable economic returns for the rural poor, particularly women, in Bangladesh.

Chapter 2: Literature Review and Theoretical Framework

2.1 Conceptualizing Economic Returns in Livestock Microenterprises

Economic returns in the context of livestock microenterprises refer to the net financial and livelihood benefits that small-scale producers derive from their engagement in livestock-related activities, encompassing not only direct income but also improvements in profitability, asset accumulation, risk reduction, and employment generation. Unlike large-scale commercial farming, where returns are primarily measured through standardized financial metrics such as return on investment (ROI) and internal rate of return (IRR), economic returns in microenterprises are multidimensional and often embedded within broader household livelihood strategies. For small and marginal livestock keepers, returns are realized through a combination of cash income from sales of live animals, milk, and value-added products; savings from reduced production costs; accumulation of productive assets (such as herd size); and non-monetary benefits including improved household food security and nutrition.

In livestock microenterprises, particularly in meat and dairy value chains, economic returns are shaped by several interrelated factors. These include productivity enhancements through improved feeding, breeding, and animal health management; cost efficiency achieved via better input utilization and technology adoption; market performance determined by price realization, reduced transaction costs, and access to reliable buyers; and value addition through processing and branding. The sustainability of these returns is equally critical, as short-term income gains may not translate into long-term economic resilience if they are vulnerable to shocks such as disease outbreaks, feed price volatility, or market fluctuations. Consequently, contemporary conceptualizations of economic returns in livestock microenterprises increasingly emphasize resilience, risk mitigation, and the capacity of enterprises to generate consistent and growing benefits over time.

Scholarly literature highlights that measuring economic returns in smallholder livestock systems presents unique methodological challenges. Many producers operate in informal markets with limited record-keeping, multiple and overlapping livelihood activities, and significant non-market transactions (such as home consumption of milk or animals used for social and cultural purposes). These complexities necessitate the use of mixed indicators that combine quantitative measures — such as changes in household income, profit margins, and herd size — with qualitative assessments of empowerment, decision-making, and perceived economic security. Furthermore, gender dimensions are central to understanding economic returns, as women often perform the majority of daily livestock management tasks yet face constraints in accessing resources, information, and markets, which can limit their ability to capture the full economic benefits of their labor.

In the context of value chain development interventions, economic returns are conceptualized not merely as outcomes at the individual enterprise level but as results of systemic improvements across the chain. Enhanced backward linkages (access to quality feed, veterinary services, and breeding inputs) and forward linkages (better market access, collective marketing, and processing opportunities) are expected to generate multiplier effects that improve returns for primary producers. This systems perspective underscores that sustainable economic returns for microentrepreneurs depend on coordinated improvements in production efficiency, market

integration, institutional support, and enabling environments. The present study adopts this comprehensive conceptualization, examining economic returns through changes in income, profitability, employment, and business sustainability, while also considering the mediating roles of technology adoption, gender empowerment, and market linkage development within the livestock value chains of Dhamrai and Manikganj.

2.2 Value Chain Development Theory and Its Application in Bangladesh Livestock Sector

Value Chain Development (VCD) has emerged as a dominant framework in contemporary development practice for enhancing the competitiveness, inclusiveness, and sustainability of agricultural and rural enterprises. Originally conceptualized by Porter (1985) in the context of firm-level competitive advantage, the value chain approach was later expanded by scholars such as Kaplinsky and Morris (2001) and Gereffi et al. (2005) to analyze global and local production networks. At its core, VCD emphasizes the full range of activities required to bring a product or service from conception to final consumption, including input supply, production, processing, marketing, and distribution. The approach focuses on identifying bottlenecks, power asymmetries, and opportunities for **upgrading** — which may occur through improvements in processes, products, functions, or inter-chain linkages — to increase the value captured by different actors, particularly small-scale producers.

In agricultural and livestock sectors of developing countries, VCD theory has been widely adopted to address structural constraints faced by smallholders, such as limited access to quality inputs, weak market integration, high transaction costs, and vulnerability to price volatility. Key principles include strengthening backward and forward linkages, improving coordination and governance within the chain, promoting collective action among producers, and facilitating access to information, technology, and finance. Inclusive value chain development further incorporates social dimensions, particularly gender equity, youth participation, and environmental sustainability, recognizing that economic upgrading must be accompanied by social and environmental upgrading to achieve broad-based and lasting development outcomes.

In Bangladesh, the livestock sector — especially cattle-based meat and dairy production — presents a compelling case for the application of value chain development approaches. The sector is characterized by a large number of small and marginal producers operating in fragmented, traditional systems with limited vertical coordination. Most farmers sell live animals or raw milk through informal channels dominated by middlemen, resulting in low price realization and minimal value addition at the producer level. Rising urban demand, particularly in Dhaka and other major cities, has created opportunities for upgrading, but these remain largely unrealized due to constraints in input quality, animal health services, processing infrastructure, and market information.

The Rural Microenterprise Transformation Project (RMTP), implemented by PKSF with support from IFAD and DANIDA, represents a deliberate effort to apply value chain development principles to the livestock sector. By focusing on selected sub-sectors such as safe meat and dairy production, RMTP seeks to improve coordination between input suppliers, producers, processors, and buyers; promote the adoption of improved technologies and practices; strengthen market linkages; and enhance the capacity of microentrepreneurs to capture greater value. The project's emphasis on developing Livestock Service Providers (LSPs), establishing milk collection points, supporting processing enterprises, and facilitating access to finance aligns closely with core VCD

strategies of upgrading and linkage strengthening. However, the effectiveness of these interventions in generating sustainable economic returns for smallholders remains an important area for empirical investigation, particularly in understanding how value chain improvements translate into tangible income and profit gains at the microenterprise level.

2.3 Role of NGOs (SDI/PKSF Model) in Enhancing Rural Economic Returns

Non-Governmental Organizations (NGOs) have played a pivotal role in Bangladesh's rural development landscape, particularly in extending financial services, capacity building, and market support to poor and marginalized communities. Among development models in the country, the **PKSF-NGO partnership model** stands out as one of the most successful and widely replicated approaches for promoting microenterprise development and rural economic transformation. Palli Karma-Sahayak Foundation (PKSF), as an apex development organization, provides wholesale financing, technical assistance, and institutional support to Partner Organizations (POs) — typically NGOs — which in turn deliver services directly to beneficiaries. This two-tier structure combines the scale and financial strength of PKSF with the grassroots reach and contextual knowledge of local NGOs.

The **Society for Development Initiatives (SDI)** exemplifies this model. As a PKSF Partner Organization with long-standing experience in micro-credit and income-generating activities since 1993, SDI has progressively evolved from a purely credit-focused institution to one that integrates value chain development interventions. Under the RMTP framework, SDI implements a comprehensive approach that combines access to finance (through products such as Agrosor and Jagoron loans), technical training on livestock management and processing, support for technology adoption (chopper machines, milking machines, cream separators), and facilitation of market linkages. This integrated model addresses multiple constraints simultaneously — financial, technical, and market-related — which is considered essential for generating sustainable economic returns among microentrepreneurs.

Empirical evidence from various studies and program evaluations suggests that the PKSF-NGO model contributes to rural economic returns through several pathways. First, it improves access to affordable credit tailored to enterprise needs, enabling investment in productive assets such as cattle, equipment, and working capital. Second, it builds human capital through training and extension services, leading to productivity gains and better management practices. Third, it strengthens market integration by developing linkages with input suppliers, service providers, and buyers, thereby reducing transaction costs and improving price realization. Fourth, the model places strong emphasis on women's participation and empowerment, which has been shown to enhance household-level economic outcomes and decision-making. In the context of the current study, SDI's implementation of the "Market Development of Safe Meat and Dairy Production" project demonstrates how this model can be applied to livestock value chains, resulting in measurable improvements in income (40% higher than control), profit margins, employment creation, and women's economic agency.

While the PKSF-NGO model has achieved notable successes, scholars and practitioners also highlight certain limitations and areas for improvement. These include the need for more sophisticated business development services beyond basic training, greater attention to digital tools and ICT-based solutions, stronger mechanisms for sustainability and exit strategies after project completion, and more robust monitoring of enterprise-level profitability rather than only outreach

indicators. The present study contributes to this ongoing discourse by providing detailed evidence on the economic returns generated through SDI’s integrated value chain interventions and by identifying factors that enhance or constrain the effectiveness of the PKSf-NGO partnership model in the livestock sector.

2.4 Gendered Economic Empowerment in Livestock Value Chains

Gender is a critical dimension in livestock value chains, as women across developing countries, including Bangladesh, play a central role in livestock rearing and management while often facing systemic barriers to capturing the full economic benefits of their labor. Women are typically responsible for daily care activities such as feeding, cleaning, milking, and caring for young animals, yet they frequently have limited ownership of livestock assets, restricted access to markets, and minimal decision-making power over income generated from livestock enterprises. This disconnect between women’s substantial labor contribution and their limited economic agency has significant implications for both household welfare and the overall performance of livestock value chains.

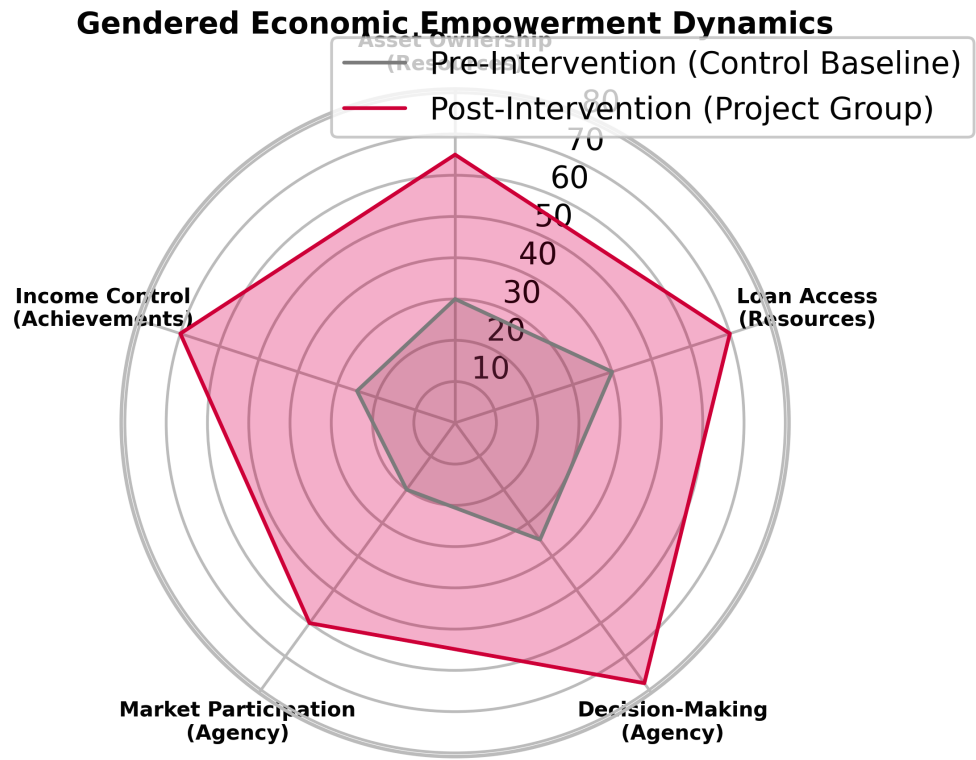


Figure 2.4: *Dimensions of Gendered Economic Empowerment.* Kabeer, N. (1999).

Figure Description: A radar chart mapping the three dimensions of gendered empowerment: Resources (asset ownership, loan access), Agency (decision-making involvement, 78%), and Achievements (income contribution). It visualizes the shift from a baseline state to the empowered state post-intervention.

The concept of **gendered economic empowerment** in value chains goes beyond mere participation. Drawing on Kabeer's (1999) empowerment framework, it encompasses three interrelated dimensions: **resources** (access to assets, credit, information, and technology), **agency** (decision-making power over production, marketing, and income use), and **achievements** (improved economic outcomes and well-being). In livestock systems, empowerment requires not only increasing women's involvement in production but also strengthening their control over productive assets, enhancing their bargaining power within households and markets, and ensuring they benefit equitably from value chain upgrading.

Empirical studies in South Asia and Africa demonstrate that when women gain greater control over livestock-related income and assets, household nutrition, education, and overall economic resilience tend to improve. However, value chain interventions do not automatically lead to women's empowerment. Without deliberate gender-sensitive design — such as targeting women for training, providing women-friendly technologies, facilitating women's groups, and addressing social norms — interventions may even reinforce existing gender inequalities by increasing women's workload without corresponding gains in agency or returns.

In the context of Bangladesh's livestock sector, the high participation of women in cattle rearing presents both opportunities and challenges. Many women manage livestock enterprises with limited formal training, access to veterinary services, or market linkages. Interventions that successfully combine technical training, access to finance, and market support with explicit gender strategies have shown promising results in enhancing women's economic empowerment. The present study examines how the SDI-implemented project has influenced gendered economic empowerment by analyzing changes in women's decision-making roles (78% reported involvement), asset ownership, loan access (70%), and their contribution to household income. Understanding these dynamics is essential for assessing whether value chain development translates into equitable economic returns and sustainable empowerment for women microentrepreneurs.

2.5 Nutrition-Sensitive Livestock Interventions and Dietary Diversity (MDD-W Framework)

The integration of nutrition objectives into agricultural and livestock development programs has gained increasing attention under the broader concept of **nutrition-sensitive agriculture**. Unlike nutrition-specific interventions that directly address malnutrition through supplementation or fortification, nutrition-sensitive approaches leverage agriculture and livestock systems to improve dietary quality, food security, and nutritional outcomes indirectly through increased production, income, and women's empowerment. Livestock, in particular, offers significant potential for nutrition-sensitive outcomes because it provides nutrient-dense foods (milk, meat, and eggs), generates income that can be used to purchase diverse foods, and is often managed by women who tend to prioritize household nutrition when they control resources.

The **Minimum Dietary Diversity for Women (MDD-W)** indicator, developed by the Food and Agriculture Organization (FAO), has become a widely used tool for assessing the nutritional quality of women's diets. MDD-W measures the proportion of women of reproductive age (15–49 years) who have consumed foods from at least five out of ten defined food groups in the previous 24 hours. Achieving this threshold is associated with higher micronutrient adequacy and better nutritional status. Livestock interventions can influence MDD-W through multiple pathways: direct consumption of animal-source foods produced by the household, increased household income enabling purchase of diverse foods (including fruits, vegetables, and pulses), and women's empowerment leading to greater prioritization of nutrition within household expenditure decisions.

Despite this potential, many livestock development programs remain focused primarily on production and income goals without explicitly incorporating nutrition objectives. This represents a missed opportunity, as evidence suggests that income gains from livestock do not automatically translate into improved dietary diversity unless accompanied by nutrition education, women's empowerment, and behavioral change interventions. Constraints such as cultural food taboos, limited knowledge of balanced diets, and seasonal fluctuations in milk production can further weaken the nutrition linkages.

In the context of the current study, the project's impact on nutrition is examined through changes in MDD-W scores among women participants. The endline findings indicate that 89.4% of project women achieved the MDD-W threshold of five or more food groups, compared to 72.06% in the control group. This improvement reflects the combined effects of increased household income from livestock enterprises, greater women's involvement in financial decision-making, and enhanced awareness through project activities. The study contributes to the growing body of evidence on how integrated livestock value chain interventions can be designed and implemented in a nutrition-sensitive manner to maximize both economic returns and human development outcomes.

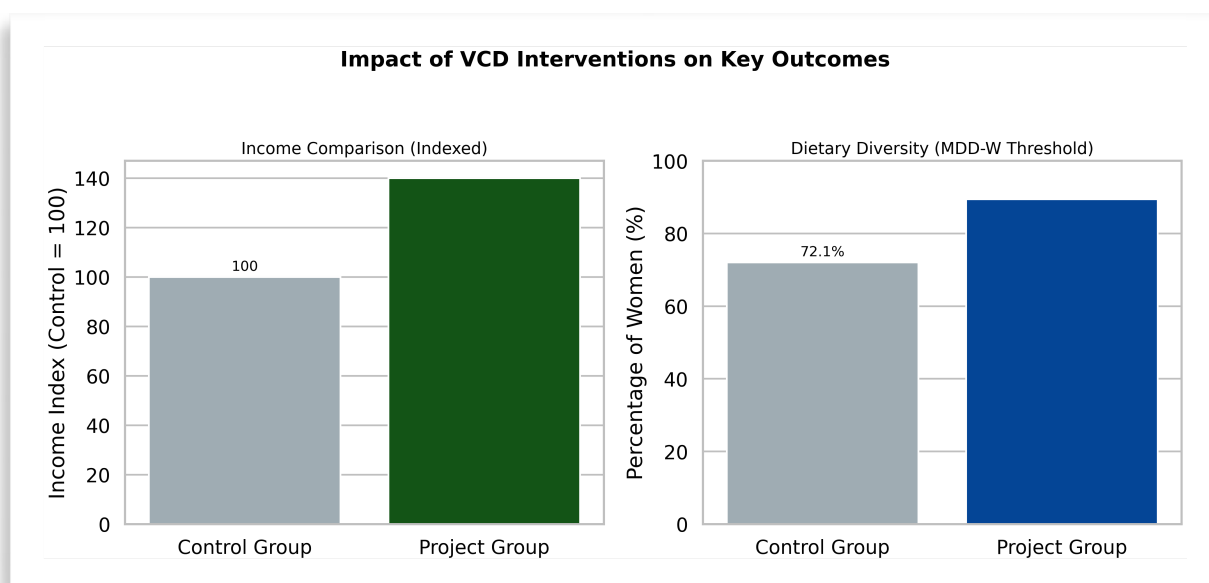


Figure 2: *Impact of Integrated Interventions on Income and Dietary Diversity.* Gereffi, G., Humphrey, J., & Sturgeon, T. (2005).

Figure Description: A side-by-side comparative bar chart visualizing the quantitative impact of the project. The left panel shows the 40% income differential between the project and control groups, while the right panel highlights the significant improvement in the Minimum Dietary Diversity for Women (MDD-W), with project participants reaching 89.4% compared to 72.06% in the control group.

2.6 Sustainability, Exit Strategies, and Post-Project Economic Resilience

Sustainability has become a central concern in development programming, particularly in donor-funded projects with fixed timeframes. In the context of livestock value chain interventions, sustainability refers to the continued generation of economic, social, and institutional benefits after external support ends. It encompasses not only the maintenance of income and productivity gains but also the resilience of microenterprises to withstand shocks such as disease outbreaks, market fluctuations, and climate-related risks. True sustainability requires that the capacities, systems, and relationships built during the project period remain functional and continue to deliver value independently.

Exit strategies are deliberate plans designed to ensure a smooth transition from project-supported activities to locally owned and managed processes. Effective exit strategies typically involve gradual reduction of external facilitation, transfer of responsibilities to local institutions (such as producer groups, cooperatives, and Livestock Service Providers), and the establishment of self-sustaining mechanisms for continued access to finance, inputs, technical knowledge, and markets. Key elements include building the capacity of local actors to provide extension services, strengthening collective marketing arrangements, and linking beneficiaries with permanent government and private sector systems. Without well-designed exit strategies, gains achieved during project implementation often erode rapidly once funding ceases.

Post-project economic resilience depends on several interrelated factors. These include the depth of skill development and knowledge transfer among beneficiaries and local service providers, the strength and durability of market linkages established during the project, continued access to finance through microfinance institutions or revolving funds, and the level of women's empowerment and ownership over productive assets. Institutional sustainability — particularly the capacity of local organizations and service providers to maintain coordination and support functions — is equally important. Environmental sustainability, including the adoption of eco-friendly practices such as improved waste management and climate-resilient feeding systems, also contributes to long-term resilience.

Evidence from various livestock and value chain projects indicates that sustainability is more likely when interventions move beyond one-time training and asset distribution to focus on systemic changes in market relationships, institutional capacities, and behavioral norms. In the present study, sustainability is examined through indicators such as participants' intentions to continue and expand livestock enterprises (74% expressed plans to expand), continued use of improved technologies and practices (82%), and the strength of linkages with input suppliers and service providers. The study also analyzes the project's exit strategy and the extent to which local institutions and empowered women microentrepreneurs are positioned to sustain economic returns beyond the project period. Understanding these dynamics is critical for designing future interventions that generate not only immediate economic gains but also lasting resilience.

2.7 Theoretical Framework: Sustainable Livelihoods Approach + Value Chain Analysis + Cost-Benefit/ROI Framework

This study adopts an integrated theoretical framework that combines three complementary approaches: the **Sustainable Livelihoods Approach (SLA)**, **Value Chain Analysis**, and **Cost-Benefit / Return on Investment (ROI) Framework**. This integration provides a comprehensive lens for examining economic returns in livestock microenterprises while accounting for the broader contextual, institutional, and gender dimensions that influence outcomes.

The **Sustainable Livelihoods Approach**, popularized by the Department for International Development (DFID), provides a holistic framework for understanding how households construct and maintain their livelihoods. It emphasizes five categories of capital assets — human, social, natural, physical, and financial — and examines how these assets interact with the vulnerability context, transforming structures and processes (policies, institutions, and markets), and livelihood strategies to produce outcomes. In this study, SLA helps analyze how livestock-based microenterprises contribute to household asset accumulation, income diversification, and resilience, while also highlighting the role of women's empowerment and institutional support in sustaining livelihoods.

Value Chain Analysis complements SLA by focusing on the systemic relationships and upgrading opportunities within the meat and dairy value chains. It provides tools to examine backward linkages (input supply, veterinary services, breeding), forward linkages (marketing, processing, and buyer relationships), chain governance, and different forms of upgrading (process, product, functional, and inter-chain). This component of the framework enables the study to assess how project interventions have strengthened coordination, reduced inefficiencies, and improved the position of small producers within the value chain, thereby influencing economic returns.

The **Cost-Benefit and Return on Investment (ROI) Framework** adds a rigorous economic dimension by focusing on measurable financial returns. It guides the analysis of changes in income, sales, costs, profit margins, and employment generation, allowing for quantitative assessment of the economic viability and efficiency of different livestock enterprises (beef fattening, milk production, and processed goods). This framework is particularly useful for comparing returns across enterprise types and between project and control groups.

By integrating these three approaches, the study moves beyond narrow income-focused assessments to examine how asset building, value chain positioning, and enterprise profitability interact to produce sustainable economic returns. The framework also incorporates gender as a cross-cutting dimension, recognizing that women's access to resources, decision-making power, and control over income significantly shape both livelihood outcomes and value chain performance. This integrated theoretical foundation guides the research design, data collection, and analysis, ensuring that the study captures the multidimensional nature of economic returns in livestock microenterprises while generating findings that are relevant for both theory and practice.

2.8 Synthesis and Conceptual Model for Measuring Economic Returns

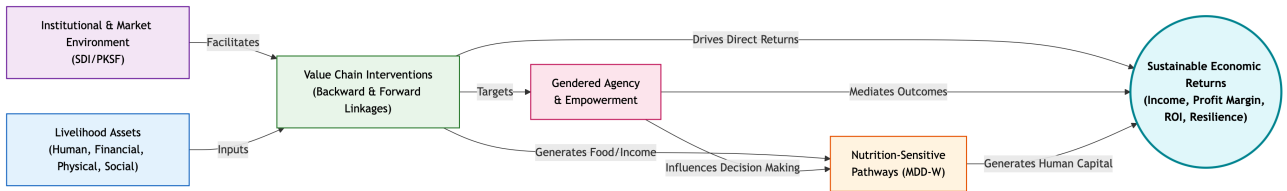
The preceding review of literature reveals that economic returns in livestock microenterprises are multidimensional outcomes shaped by the interaction of household assets, value chain dynamics, enterprise performance, gender relations, and institutional support. While income and profit remain central indicators, a comprehensive understanding of economic returns must also encompass employment generation, business expansion, risk reduction, and the sustainability of gains over time. The review further highlights that value chain development interventions do not automatically produce equitable or sustainable returns; their effectiveness depends on how well they address constraints in production, market access, technology adoption, and women's empowerment.

Drawing on the integrated theoretical foundation established in Section 2.7, this study proposes a **Conceptual Model for Measuring Economic Returns in Livestock Microenterprises**. The model synthesizes the Sustainable Livelihoods Approach, Value Chain Analysis, and Cost-Benefit/ROI Framework into a coherent analytical structure. At the core of the model are **Economic Returns**, conceptualized as the net improvements in household income, enterprise profitability (measured through profit margins and ROI), employment creation, and long-term business sustainability. These returns are viewed as outcomes of interactions among several key components:

- **Livelihood Assets and Vulnerability Context:** The human capital (skills and knowledge gained through training), financial capital (access to credit and savings), physical capital (livestock, equipment, and technology), and social capital (networks with LSPs, input suppliers, and markets) that project participants possess or acquire. These assets mediate the impact of external shocks and influence the ability to adopt improved practices.
- **Value Chain Interventions and Upgrading:** The project's strategies for strengthening backward linkages (feed, medicine, breeding services, and veterinary support) and forward linkages (market access, collection points, and processing opportunities). Upgrading occurs through improved production practices (GAP adoption), technology use, and better coordination within the chain.
- **Gendered Agency and Empowerment:** Women's increased participation in decision-making, ownership of productive assets, and control over income, which serve as both an outcome and a mediating factor influencing how economic returns are generated and utilized at the household level.
- **Nutrition-Sensitive Pathways:** The linkages between increased livestock income, women's empowerment, and improvements in household dietary diversity (measured through MDD-W), recognizing that economic returns can contribute to broader human development outcomes.
- **Institutional and Market Environment:** The role of transforming structures and processes, including the support provided by SDI, PKSF, LSPs, and market actors, as well as the sustainability mechanisms (exit strategies, local institutional capacity, and continued access to services) that determine whether returns persist after project completion.

The model posits that project interventions influence economic returns both directly (through productivity and market improvements) and indirectly (through enhanced assets, women’s empowerment, and institutional support). It further suggests that the sustainability of economic returns depends on the depth of capacity building, the strength of market linkages, and the extent of women’s economic empowerment achieved during the project period.

This conceptual model guides the empirical analysis in the subsequent chapters. It informs the selection of indicators for measuring economic returns (income, profit margins, employment, and business expansion), the comparison between project and control groups, and the examination of mediating factors such as gender empowerment and technology adoption. By operationalizing this integrated framework, the study aims to generate rigorous evidence on how value chain development interventions translate into sustainable economic returns for small and marginal livestock microentrepreneurs in Bangladesh, while also identifying the conditions under which these returns are maximized and sustained.



**Figure 1: Conceptual Model for Measuring Economic Returns (Flowchart).
Department for International Development (DFID). (1999).**

Chapter 3: Project Context, Value Chain Mapping, and Intervention Strategy

3.1 Profile of Society for Development Initiatives (SDI) and RMTP Partnership

The Society for Development Initiatives (SDI) is a non-profit social development organization established on 22 July 1993 with the primary mission of improving the socio-economic conditions and human dignity of the poor and extremely poor in Bangladesh. Over the past three decades, SDI has grown into a well-recognized development actor with a strong focus on implementing Income Generating Activities (IGAs) through micro-credit programs. The organization is duly registered with multiple regulatory bodies of the Government of Bangladesh, including the Social Service Department (22 July 1993), the NGO Affairs Bureau (25 May 2025), the National Skill Development Authority (11 September 2024), and the Micro Credit Regulatory Authority (16 March 2008). It is also registered under the Societies Registration Act 1860 (21 July 2003). These registrations reflect SDI's legal standing and its commitment to transparent and accountable development practices.

Since its inception, SDI has been actively engaged in implementing micro-credit and livelihood development programs, particularly in remote, flood-prone, and coastal areas of the country. The organization currently operates in seventeen districts across 103 upazilas, maintaining a strong grassroots presence and deep understanding of the challenges faced by marginalized rural communities. A significant milestone in SDI's institutional journey has been its long-standing and productive partnership with the **Palli Karma-Sahayak Foundation (PKSF)**, which began in 1996. Through this partnership, SDI has received both financial and technical support to expand and strengthen its micro-credit programs, enabling it to reach thousands of poor households with sustainable financial services and enterprise development support.

The **Rural Microenterprise Transformation Project (RMTP)** represents one of the most strategic initiatives under the PKSF partnership framework. Jointly financed by PKSF and the International Fund for Agricultural Development (IFAD), with additional support from the Danish International Development Agency (DANIDA), RMTP aims to expand and transform the microenterprise sector in Bangladesh by strengthening selected rural value chains. Within this national program, SDI has been implementing the value chain sub-project titled “**Market Development of Safe Meat and Dairy Production**” since August 2024. The sub-project is being executed simultaneously in two upazilas — **Dhamrai** under Dhaka district and **Manikganj Sadar** under Manikganj district — areas selected for their high potential in livestock-based microenterprises and proximity to urban markets.

Under this sub-project, SDI applies an integrated value chain development approach that combines microfinance, capacity building, technology promotion, input supply strengthening, market linkage development, and gender-focused interventions. The organization's long experience in micro-credit, combined with its grassroots networks and understanding of local livestock systems, positions it as a capable implementing partner for translating RMTP's broader objectives into tangible outcomes at the community level. The current endline evaluation seeks to assess the effectiveness of SDI's implementation strategy and the extent to which the partnership model between PKSF and SDI has

succeeded in generating sustainable economic returns for livestock microentrepreneurs in the project areas.

3.2 Project Design: “Market Development of Safe Meat and Dairy Production” – Objectives, Components, and Logframe

The sub-project titled “**Market Development of Safe Meat and Dairy Production**” is a value chain development initiative implemented by the Society for Development Initiatives (SDI) under the broader Rural Microenterprise Transformation Project (RMTP) of PKSf. The project was launched in August 2024 and is being implemented in two upazilas — Dhamrai in Dhaka district and Manikganj Sadar in Manikganj district. It is jointly supported by PKSf, the International Fund for Agricultural Development (IFAD), and the Danish International Development Agency (DANIDA).

The **prime objective** of the project is to significantly improve the livelihoods of meat and dairy producers and related value chain actors in a sustainable manner. This is to be achieved through enhanced production, quality improvement, and marketing of safe meat and dairy products by applying modern and advanced technologies. The project specifically aims to increase income from self-employment, boost business profits, create wage employment opportunities, promote value addition at various levels, expand access to financial services for enterprise development, and strengthen the institutional framework for the livestock value chain.

To achieve these objectives, the project follows an integrated design with eight major components, as outlined in its scope of work:

1. **Capacity Development of Farmers and Entrepreneurs** — Training small and marginal farmers and livestock entrepreneurs in modern and cost-effective production and processing technologies, best practices in animal husbandry, feed management, disease control, and hygienic handling.
2. **Improvement of Input Supply Systems** — Ensuring timely and adequate supply of quality inputs such as feed, medicine, breeding services, and other essentials, while developing linkages between input suppliers and producers.
3. **Technology Adoption and Innovation** — Introducing advanced technologies in livestock production and processing, establishing local-level small-scale processing plants, and facilitating the use of digital and information technologies in farm management.
4. **Value Addition and Product Development** — Supporting the production of high-value meat and dairy products through modern processing, packaging, and storage, along with the introduction of certification systems for safety and quality assurance.
5. **Market Development and Linkage Creation** — Establishing strong linkages between producers, processors, traders, and buyers at local and regional levels, developing structured marketing channels, and promoting branding and certification.
6. **Employment and Income Generation** — Expanding enterprise activities to create local employment opportunities, particularly for rural women and youth, while ensuring increased sales and profitability for participating entrepreneurs.
7. **Environmental Sustainability and Eco-Friendly Practices** — Encouraging the adoption of eco-friendly farming and processing technologies, promoting sustainable waste management, and supporting climate-resilient livestock practices.

8. Monitoring, Evaluation, and Learning (MEL) — Applying the project’s logical framework to systematically monitor outputs, outcomes, and impacts, and documenting lessons learned for replication and scaling.

The project’s design is guided by a **logical framework (logframe)** that defines clear goals, outcomes, and measurable indicators. The overarching goal is to enhance sustainable income, food security, and nutrition for marginal small farmers and micro-entrepreneurs through value chain activities. Key outcome-level targets include increasing the income of at least 70% of entrepreneurs by a minimum of 50%, improving profit margins by 20% for 80% of entrepreneurs, increasing sales of meat and dairy products by 30% for 80% of entrepreneurs, and ensuring that 60% of participants follow Bangla GAP while 58% adopt eco-friendly technologies.

This integrated design reflects a deliberate effort to address multiple constraints in the livestock value chain simultaneously — from production and input supply to processing, marketing, and sustainability — while placing strong emphasis on gender inclusion and women’s economic empowerment.

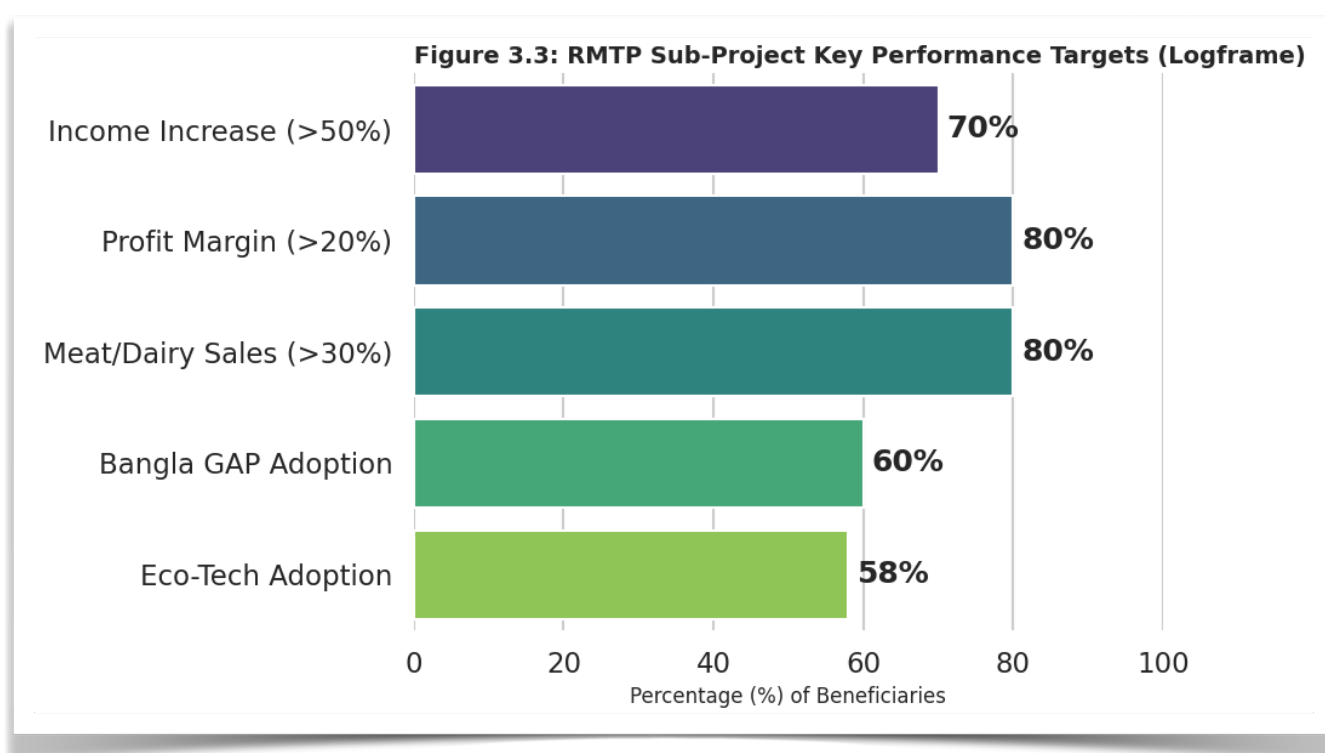


Figure 3.3: *RMTP Sub-Project Key Performance Targets.*

Description: A horizontal bar chart presenting the quantitative goals set within the Project Logframe. It specifically highlights the high benchmarks for income growth (70%) and eco-friendly technology adoption (58%).

3.3 Geographical and Socio-Economic Context of Dhamrai (Dhaka) and Manikganj Sadar

Dhamrai Upazila of Dhaka district and Manikganj Sadar Upazila of Manikganj district represent two strategically important areas for livestock-based microenterprise development in central Bangladesh. Both upazilas are located in close proximity to Dhaka, the capital city and largest urban market in the country. This geographical advantage creates significant opportunities for livestock producers, as rising urban demand for safe meat and dairy products continues to grow. The areas are characterized by a mix of agricultural land, homesteads, and small-scale commercial activities, with livestock rearing forming an integral part of the rural economy and household livelihood systems.

Livestock farming, particularly cattle rearing, is central to the livelihoods of small and marginal farmers in these upazilas. It serves as a key source of income, nutrition, asset accumulation, and employment, especially for women and other marginalized groups. Most livestock keepers in the project areas are smallholders managing herds of only 1 to 10 cattle. These farmers traditionally rely on indigenous (deshi) breeds and follow low-input, traditional management practices. Livestock not only provides direct income through the sale of milk, live animals, and value-added products but also contributes to household food security and serves as a form of savings and insurance against economic shocks.

Despite its importance, the livestock sector in Dhamrai and Manikganj Sadar faces several structural and operational challenges. Production systems remain largely traditional, with limited adoption of improved feeding practices, modern housing, and scientific animal health management. Market access is often constrained by dependence on middlemen, lack of collective marketing mechanisms, and limited value addition at the producer level. These constraints have historically resulted in low productivity, thin profit margins, and limited economic returns for smallholder farmers.

The socio-economic context of the target population is marked by a predominance of small and marginal farming households with limited access to formal financial services, modern technology, and institutional support. Women play a significant role in daily livestock management activities; however, they often have limited ownership of productive assets and decision-making power over income and enterprise expansion. The proximity to Dhaka offers both opportunities and competitive pressures — while urban demand creates market potential, it also requires producers to meet higher standards of quality, safety, and consistency, which many smallholders struggle to achieve without external support.

This geographical and socio-economic setting provides a strong rationale for the “Market Development of Safe Meat and Dairy Production” project. The combination of existing livestock base, growing urban demand, and the presence of structural constraints makes Dhamrai and Manikganj Sadar suitable locations for testing and scaling an integrated value chain development approach aimed at enhancing economic returns for small and marginal livestock microentrepreneurs.

3.5 Intervention Strategy: Capacity Building, Input Supply, Technology, Market Linkages, and Eco-Friendly Practices

The intervention strategy of the “Market Development of Safe Meat and Dairy Production” project was designed as a comprehensive and integrated approach to address the multiple constraints faced by small and marginal livestock producers in Dhamrai and Manikganj Sadar. Rather than focusing on a single intervention, the project adopted a multi-dimensional strategy that simultaneously strengthened production capacity, input systems, technology adoption, market linkages, and environmental sustainability. This integrated design was intended to generate sustainable economic returns by improving productivity, reducing costs, enhancing value addition, and strengthening the position of producers within the livestock value chain.

Capacity Building

Capacity development formed the foundation of the project’s intervention strategy. Comprehensive training programs were organized for both farmers and Livestock Service Providers (LSPs). Farmers received training on Good Agricultural Practices (GAP), including improved cattle housing and waste management, advanced feed formulation, balanced nutrition, disease identification and control, vaccination protocols, and safe milk production and handling. Special emphasis was placed on practical, hands-on learning to ensure that participants could apply the knowledge directly in their enterprises. For Livestock Service Providers, specialized training modules were delivered on sustainable livestock management, advanced disease control, artificial insemination techniques, and the identification of quality veterinary medicines. These capacity-building efforts aimed to create a cadre of skilled farmers and service providers capable of sustaining improved practices beyond the project period.

Improvement of Input Supply Systems

A major component of the strategy focused on strengthening backward linkages in the value chain. The project worked to ensure timely and adequate supply of quality inputs such as cattle feed, medicines, vaccines, and breeding services. It facilitated the development of linkages between input suppliers and producers, including the establishment of decentralized supply points and vaccine hubs. To promote localized and sustainable input solutions, the project supported the cultivation of high-yield fodder varieties such as Napier grass, silage production, and the use of chopper machines for efficient straw cutting. These interventions were designed to reduce dependency on external supply chains, lower input costs, and improve the nutritional status of livestock.

Technology Adoption and Innovation

The project actively promoted the adoption of modern technologies to enhance productivity and efficiency. Financial and technical support was provided for the acquisition of milking machines, cream separation machines, cheese production units, automatic meat-cutting equipment, and cold-chain storage facilities. In addition, the project facilitated the establishment of a meat processing plant with 24/7 sales capacity. These technology interventions aimed to reduce labor intensity, improve product quality and hygiene, minimize post-harvest losses, and enable producers and processors to meet higher market standards. Training and technical assistance were provided alongside equipment support to ensure proper utilization and maintenance.

Market Development and Linkage Creation

Strengthening forward linkages and improving market access constituted another critical pillar of the intervention strategy. The project worked to establish structured linkages between producers, processors, traders, and institutional buyers. It supported the development of milk collection points to facilitate institutional selling of milk and promoted collective marketing arrangements to enhance farmers' bargaining power. Branding, certification, and market promotion activities were undertaken to improve the visibility and competitiveness of safe meat and dairy products. Awareness programs such as "Mahat Dibosh" were organized to disseminate knowledge on improved practices and market opportunities. These efforts aimed to reduce dependence on middlemen, improve price realization, and create more stable and profitable market channels for beneficiaries.

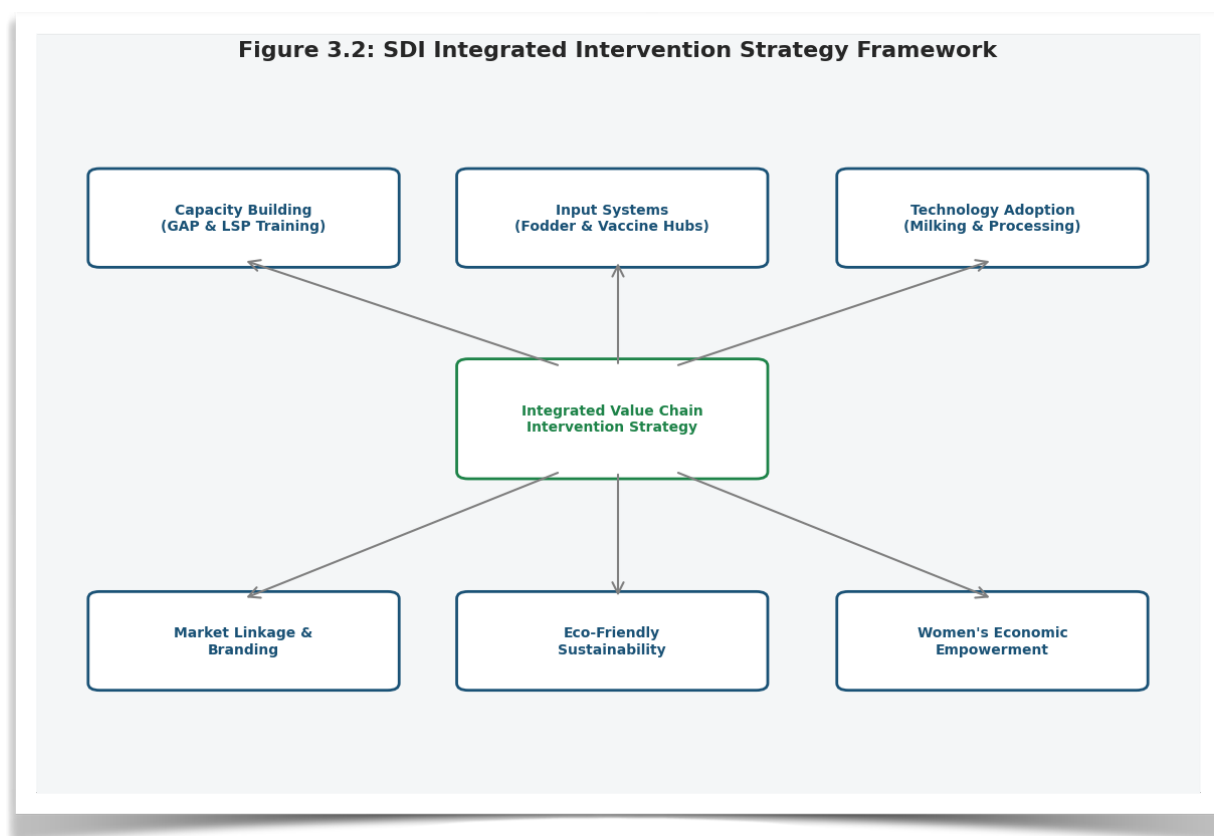


Figure 3.2: *SDI Integrated Intervention Strategy Framework*. **Palli Karma-Sahayak Foundation (PKSF).** (2023).

Description: A process flowchart showing the flow from foundational inputs (Capacity, Input Supply, Tech) into a centralized "Integrated Strategy" which then outputs into three impact areas: Market Linkage, Environmental Sustainability, and Social Empowerment.

Environmental Sustainability and Eco-Friendly Practices

Recognizing the importance of long-term sustainability, the project integrated environmental considerations into its intervention strategy. It promoted the adoption of eco-friendly farming and processing technologies, including improved animal housing, proper waste management, and climate-resilient livestock practices. Farmers were encouraged to adopt sustainable input solutions such as vermicompost production and the use of organic fertilizers. The project also supported awareness and capacity building on reducing environmental hazards associated with livestock rearing. These measures were intended to enhance resource efficiency, minimize ecological footprints, and ensure that economic gains are not achieved at the cost of environmental degradation.

Through this integrated and multi-pronged intervention strategy, the project sought to create synergistic effects across the livestock value chain, enabling small and marginal producers to achieve higher productivity, better market access, and improved economic returns while promoting gender inclusion and environmental sustainability.

3.6 Alignment with National Livestock Policies and SDGs

The “Market Development of Safe Meat and Dairy Production” project demonstrates strong alignment with Bangladesh’s national development priorities, livestock sector strategies, and global Sustainable Development Goals (SDGs). The project’s design and implementation approach reflect a deliberate effort to contribute to broader policy objectives related to rural economic transformation, food security, nutrition improvement, and inclusive growth.

At the national level, the project directly supports the Government of Bangladesh’s vision for modernizing the livestock sector and enhancing its contribution to rural livelihoods and the national economy. Bangladesh’s livestock development policies emphasize increasing productivity, ensuring food safety and quality, promoting value addition, and improving market access for smallholder producers. By focusing on safe meat and dairy production, technology adoption, quality improvement, and market linkage development, the project addresses key policy priorities of enhancing production efficiency while ensuring consumer safety and product competitiveness.

The project is also well-aligned with the strategic objectives of the **Palli Karma-Sahayak Foundation (PKSF)** and the **Rural Microenterprise Transformation Project (RMTP)**. RMTP, supported by IFAD and DANIDA, aims to transform rural microenterprises by strengthening selected value chains, improving productivity, and enhancing market competitiveness. The SDI-implemented sub-project contributes to these goals by applying an integrated value chain development approach in the livestock sector, with particular attention to small and marginal producers who form the backbone of rural microenterprises in Bangladesh.

From a global development perspective, the project makes meaningful contributions to several **Sustainable Development Goals (SDGs)**. It directly supports **SDG 1 (No Poverty)** and **SDG 8 (Decent Work and Economic Growth)** by enhancing income, profitability, and employment opportunities for small and marginal livestock farmers and microentrepreneurs. The project’s focus on improving production, value addition, and market access helps increase earnings and create sustainable livelihoods in rural areas.

The project also contributes to **SDG 2 (Zero Hunger)** and improved nutrition outcomes. By increasing household income from livestock enterprises and promoting women’s empowerment in financial decision-making, the project enhances households’ ability to access diverse and nutritious foods. The observed improvements in Minimum Dietary Diversity for Women (MDD-W) among project participants reflect the nutrition-sensitive potential of livestock value chain interventions.

Furthermore, the project strongly aligns with **SDG 5 (Gender Equality)**. With women comprising 92% of direct beneficiaries and significant improvements recorded in women’s decision-making power (78%) and asset ownership, the project serves as a model for gender-responsive value chain development. It demonstrates how livestock interventions can be designed to address both economic returns and women’s empowerment simultaneously.

Overall, the project’s integrated approach — combining capacity building, technology adoption, market linkages, and women’s empowerment — positions it as a relevant and strategic intervention that contributes to national livestock development goals while advancing multiple Sustainable Development Goals. This alignment enhances the project’s policy relevance and potential for scaling and replication in other regions of Bangladesh.

Chapter 4: Research Methodology

4.1 Research Design: Mixed-Methods (Quantitative + Qualitative)

This study adopts a **mixed-methods research design**, combining both quantitative and qualitative approaches to comprehensively assess the economic returns generated through the “Market Development of Safe Meat and Dairy Production” project implemented by SDI. The mixed-methods design was chosen to capture both the magnitude of changes in key economic indicators (such as income, profit margins, and employment) and the underlying processes, experiences, and contextual factors that explain these changes.

The **quantitative component** involved a structured household questionnaire survey administered to project participants and a control group. This component enabled the collection of measurable data on household income, sales, costs, profit margins, herd size, technology adoption, loan utilization, and dietary diversity using the FAO Minimum Dietary Diversity for Women (MDD-W) tool. Statistical analysis of this data allowed for objective comparison between project and control groups, as well as before-and-after assessment of key indicators.

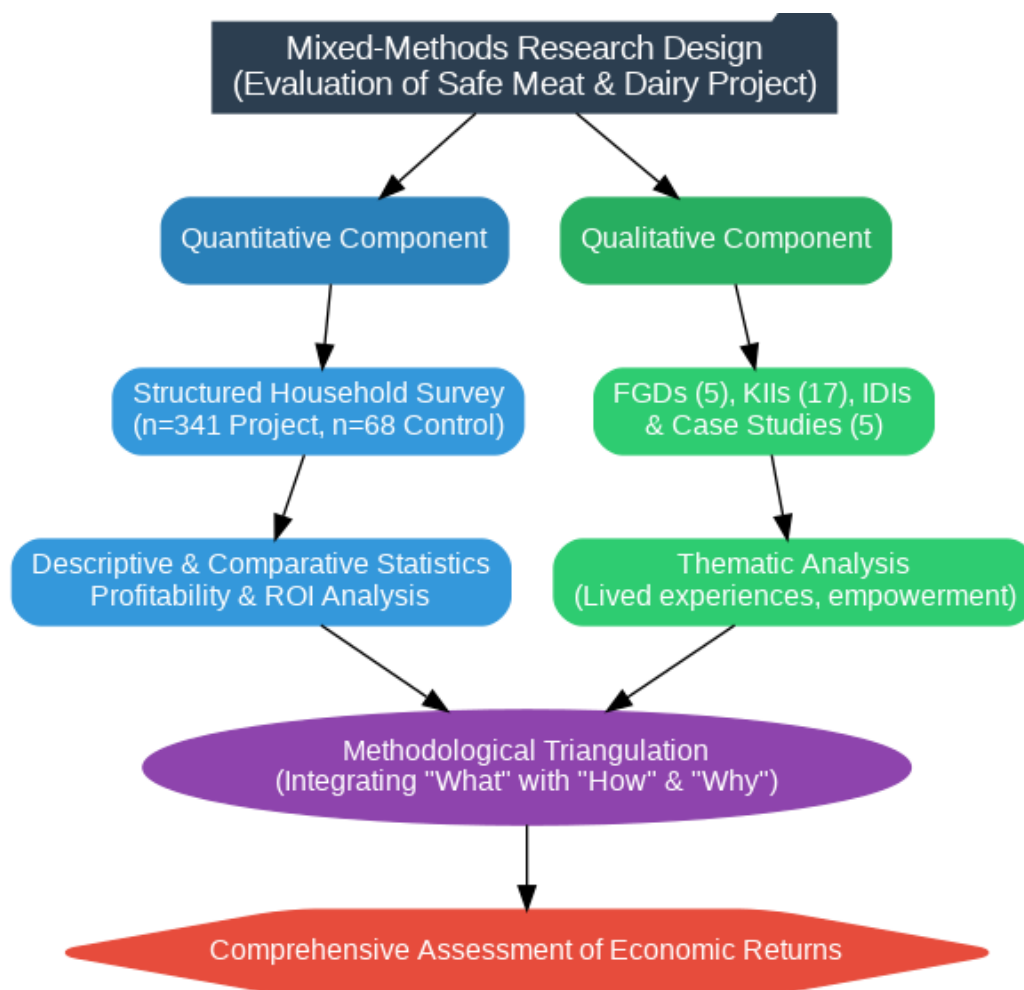


Figure 4.1: *Mixed-Methods Research Design Flowchart. Field Survey data (2026).*

Figure Description: This flowchart illustrates the convergent parallel mixed-methods design used in the study. It visually separates the quantitative and qualitative tracks—showing their respective data collection instruments and analysis methods—and demonstrates how they converge at the triangulation stage to provide comprehensive findings.

The **qualitative component** complemented the quantitative findings by providing in-depth insights into the lived experiences of beneficiaries, the dynamics of value chain relationships, and the processes through which project interventions influenced economic outcomes. Qualitative data were collected through Focus Group Discussions (FGDs), Key Informant Interviews (KIIs), In-Depth Interviews (IDIs), and detailed case studies of successful entrepreneurs. This approach facilitated a deeper understanding of gender dynamics, empowerment processes, challenges faced by farmers, and perceptions regarding the sustainability of economic gains.

The integration of quantitative and qualitative methods through triangulation enhanced the validity and richness of the findings. While quantitative data provided evidence of “what” changes occurred, qualitative data explained “how” and “why” these changes took place. This design is particularly appropriate for evaluating complex development interventions such as value chain projects, where outcomes are influenced by multiple interacting factors.

4.2 Study Area, Population, and Sampling (Cochran’s Formula, 341 Project + 68 Control Respondents)

The study was conducted in two project upazilas — **Dhamrai** (Dhaka district) and **Manikganj Sadar** (Manikganj district) — where the SDI-implemented project has been active since August 2024. For the control group, data were collected from **Ghior Upazila** of Manikganj district, an area with similar socio-economic and agro-ecological characteristics but outside the project intervention zone.

The target population consisted of small and marginal livestock farmers and microentrepreneurs engaged in cattle rearing and related value chain activities. A total of **341 project respondents** and **68 control respondents** were surveyed, making a combined sample of 409 households.

The sample size was determined using **Cochran’s sample size formula**, which is widely used in social science research for estimating proportions in large populations. For an infinite population, the initial sample size was calculated as **384** at a 95% confidence level with a 5% margin of error. Since the actual population of project beneficiaries was finite (approximately 3,070), the sample size was adjusted using the finite population correction formula, resulting in a final sample of **341 project respondents**. To enable meaningful comparison, an additional **20% control group** (68 respondents) was included, following standard practice in impact evaluation studies.

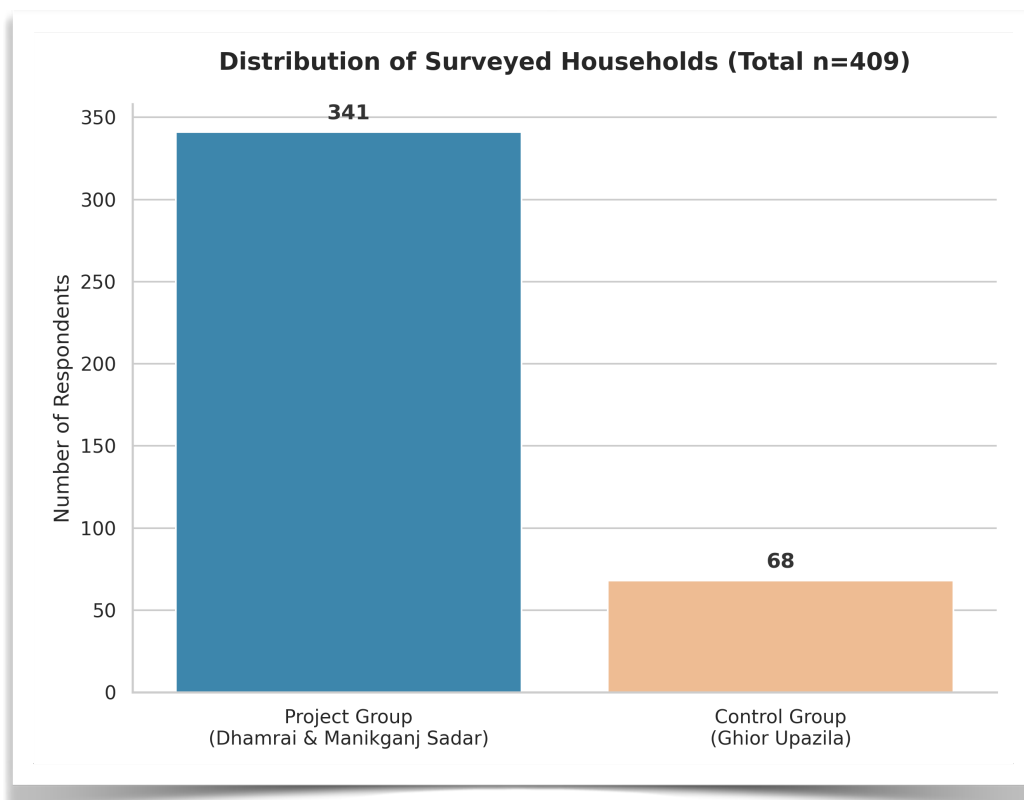


Figure 4.2: *Sample Size Distribution by Study Area. Field Survey data (2026).*

Figure Description: A bar graph displaying the distribution of the 409 surveyed households. It highlights the primary data comparison between the 341 project respondents (across Dhamrai and Manikganj Sadar) and the 68 control respondents (from Ghior Upazila), representing the 20% control baseline.

Respondents were selected through a combination of purposive and random sampling techniques to ensure representation across different categories of livestock enterprises (beef fattening, cow-calf rearing, and mixed rearing), as well as across gender and age groups. The survey was conducted across 53 villages in 22 unions of the project upazilas and 18 villages in three unions of the control upazila. This sampling strategy ensured both statistical rigor and contextual relevance for assessing the economic returns of the project interventions.

4.3 Data Collection Instruments and Procedures

A combination of quantitative and qualitative data collection instruments and procedures was employed to ensure comprehensive coverage of the study objectives. The data collection process was designed to achieve methodological triangulation, enabling the cross-verification of findings from multiple sources and perspectives.

4.3.1 Structured Household Questionnaire

The main quantitative instrument used in this study was a structured household questionnaire. The questionnaire was carefully designed to collect standardized data on household demographics, socio-economic conditions, livestock rearing practices, income and profitability, technology adoption, market linkages, employment generation, women's empowerment, and nutritional status.

The questionnaire covered the following major areas:

- Personal and household characteristics (age, gender, education, family size)
- Microfinance participation, loan size, frequency, and repayment behavior
- Present status of cattle rearing, herd size, and types of rearing (beef fattening, cow-calf, mixed)
- Feed and nutrition management practices
- Disease incidence, mortality rates, and animal health management
- Training received and its perceived impact
- Adoption and impact of modern technologies
- Sales, costs, profit margins, and business expansion
- Employment creation within enterprises
- Women's role in decision-making and asset ownership
- Minimum Dietary Diversity for Women (MDD-W)

The questionnaire was pre-tested in the field and necessary modifications were made before final data collection. Face-to-face interviews were conducted by trained enumerators with 341 project respondents and 68 control respondents. Data were collected electronically to ensure accuracy and efficiency.

4.3.2 Focus Group Discussions (5 FGDs)

Five Focus Group Discussions (FGDs) were conducted to gather qualitative insights into the experiences and perceptions of beneficiaries. The FGDs included:

- Three FGDs with women project beneficiaries from different villages in Dhamrai and Manikganj Sadar
- One FGD with control group respondents in Ghior Upazila
- One FGD with SDI project implementation staff

Each FGD consisted of 8 to 11 participants and lasted between 90 to 120 minutes. A semi-structured discussion guide was used, covering topics such as changes in income and livelihoods after joining the project, adoption of improved practices, challenges in production and marketing, women's

empowerment, and views on project support and sustainability. The FGDs helped in understanding the processes behind the quantitative results and captured diverse community perspectives.

4.3.3 Key Informant Interviews (KIIs) and In-Depth Interviews

Key Informant Interviews (KIIs) and In-Depth Interviews (IDIs) were conducted with a diverse group of stakeholders to obtain expert opinions and deeper insights. Key informants included:

- Executive Director and senior management of SDI
- Project Manager, Value Chain Facilitators, and field staff of SDI
- Officials from PKSf (Value Chain Project Manager and Monitoring & Evaluation Officer)
- Upazila Livestock Officer (ULO)
- Livestock Service Providers (LSPs)
- Input sellers, medicine providers, and successful entrepreneurs

A total of 17 KIIs were conducted. These interviews explored project design and implementation processes, value chain dynamics, market challenges, institutional linkages, and recommendations for future programming. The KIIs provided valuable information on the broader systemic and institutional dimensions of the project that could not be captured through household surveys alone.

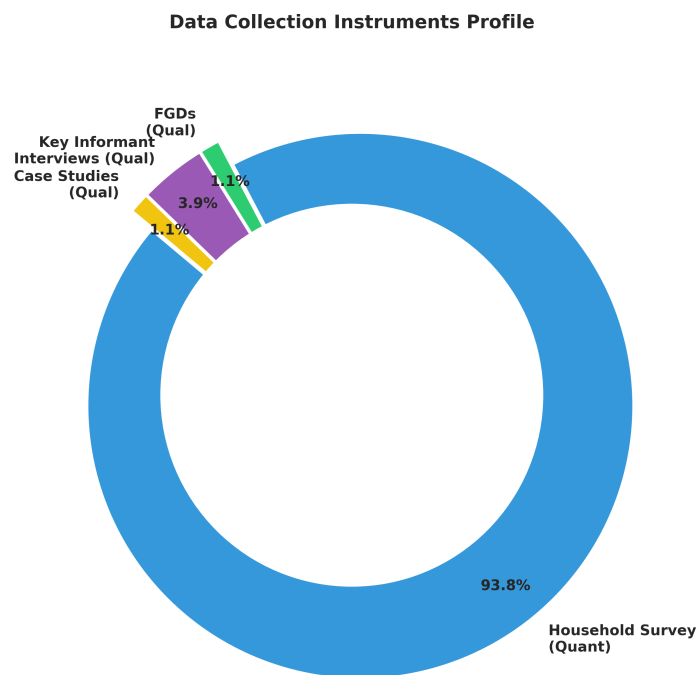


Figure 4.3: *Key Data Collection Instruments Overview. Field Survey data (2026).*

4.3.4 Nutrition Assessment using FAO MDD-W Tool

To assess the nutrition outcomes of the project, data on dietary diversity were collected using the **Minimum Dietary Diversity for Women (MDD-W)** tool developed by the Food and Agriculture Organization (FAO). This internationally recognized indicator measures whether women of reproductive age (15–49 years) have consumed foods from at least five out of ten defined food groups in the previous 24 hours.

The MDD-W module was integrated into the household questionnaire and administered to eligible women respondents in both project and control groups. Information was collected on the consumption of grains, pulses, dairy, meat/fish/eggs, vegetables, fruits, and other food groups. This assessment helped evaluate the indirect effects of increased income and women's empowerment on household nutrition security.

4.3.5 Case Study Development from Successful Entrepreneurs

Five detailed case studies were developed to document the personal and economic transformation journeys of successful project beneficiaries. The selected cases included:

- Dilip Chandra Ghosh (Chhana and ghee processing enterprise)
- Bulbul Islam (Silage production enterprise)
- Shahida Begum (Women's commercial livestock rearing)
- Shahanaz (Vermicompost production enterprise)
- Sukur Ali (Modern meat processing plant)

These case studies were prepared through multiple in-depth interviews, direct observation, and review of relevant documents. They provided rich narrative evidence of how project support in the form of training, technology, finance, and market linkages contributed to business growth, income enhancement, employment creation, and social empowerment. The case studies served as powerful illustrations of the project's impact at the individual and enterprise levels.

4.4 Data Analysis Framework

The data analysis framework was designed to systematically examine both quantitative and qualitative data in an integrated manner. The analysis aimed to assess changes in economic returns while also understanding the processes and contextual factors that influenced these outcomes.

4.4.1 Quantitative: Descriptive Statistics, Profitability/ROI Analysis, Comparative (Project vs Control)

Quantitative data collected through the structured household questionnaire were analyzed using descriptive and inferential statistical techniques. The analysis was performed using statistical software to ensure accuracy and efficiency.

Descriptive Statistics were used to summarize the basic features of the data. This included calculation of frequencies, percentages, means, and standard deviations for key variables such as household income, herd size, loan size, technology adoption rates, training participation, and dietary diversity scores. These statistics provided a clear profile of the respondents and allowed for initial comparisons between project and control groups.

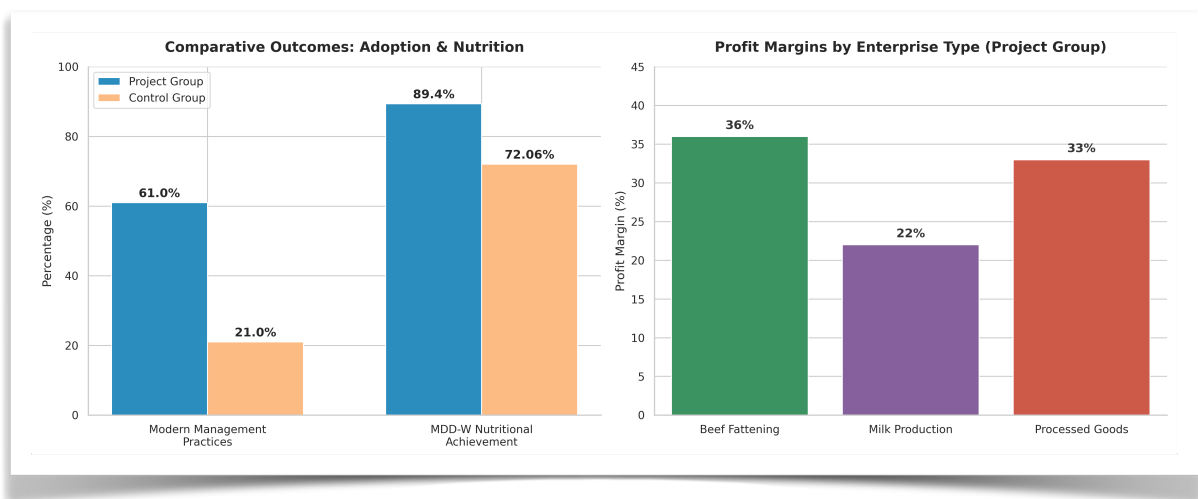


Figure 4.4: *Comparative Impact and Profitability Outcomes. Field Survey data (2026).*

Figure Description: A dual-panel dashboard. The left panel shows a comparative bar chart of technology adoption and MDD-W achievement between the project and control groups. The right panel illustrates the comparative profit margins across the three main enterprise types: beef fattening, milk production, and processed goods.

Profitability and Return on Investment (ROI) Analysis formed a core component of the quantitative analysis. Detailed calculations were made for different enterprise types (beef fattening, milk production, and processed goods). Key financial indicators such as average sales, average cost, profit, and profit margins were computed and compared. For example, profit margins were analyzed for beef fattening (36%), milk production (22%), and processed goods (33%) among project participants. This analysis helped assess the economic viability and relative performance of different livestock enterprises.

Comparative Analysis (Project vs Control) was conducted across all major indicators to isolate the effects of the project interventions. Key variables such as average monthly household income (Tk. 32,161 for project vs Tk. 22,919 for control), adoption of modern management practices (61% vs 21%), MDD-W achievement (89.4% vs 72.06%), and employment creation were systematically compared between the two groups. This comparative approach strengthened the attribution of observed changes to project interventions.

4.4.2 Qualitative: Thematic Analysis and Triangulation

Qualitative data collected through Focus Group Discussions (FGDs), Key Informant Interviews (KIIs), and case studies were analyzed using **thematic analysis**. The process involved transcribing the interviews and discussions, followed by careful reading and coding of the data. Codes were then grouped into broader themes such as changes in income and livelihoods, adoption of improved practices, women's empowerment, challenges in value chain development, and perceptions of sustainability.

Thematic analysis helped identify recurring patterns, similarities, and divergences in the experiences of different stakeholders. It provided rich explanations for the quantitative findings and revealed important insights that numbers alone could not capture, such as the processes of women's empowerment, the role of social networks, and the challenges of sustaining gains after project support ends.

Triangulation was employed as a key strategy to enhance the credibility and validity of the findings. Quantitative results (e.g., income increase, profit margins, technology adoption) were cross-verified with qualitative data from FGDs, KIIs, and case studies. For instance, the reported 40% higher income among project participants was supported by qualitative accounts of improved market access, better productivity, and successful business expansion by entrepreneurs. This integration of methods allowed for a more comprehensive and reliable understanding of the project's impact on economic returns.

4.5 Validity, Reliability, Ethical Considerations, and Researcher Reflexivity (Field Visit Experience)

To ensure the quality and credibility of the research findings, several measures were taken to address validity, reliability, ethical standards, and researcher reflexivity throughout the study process.

4.5.1 Validity

Validity refers to the extent to which the research instruments and findings accurately measure what they are intended to measure. Several steps were taken to enhance the validity of this study. First, the structured questionnaire was developed based on the project's logical framework, baseline indicators, and relevant literature on livestock value chain development. It was reviewed by experts and pre-tested in the field before final administration. Second, **triangulation** was employed by combining quantitative survey data with qualitative data from Focus Group Discussions, Key Informant Interviews, and case studies. This allowed cross-verification of findings from multiple sources. Third, the use of a control group enabled comparative analysis, which strengthened the internal validity of the study by helping to isolate the effects of project interventions from other external factors.

4.5.2 Reliability

Reliability refers to the consistency and dependability of the research instruments and findings. To enhance reliability, the questionnaire was standardized and administered by trained enumerators who received orientation on the study objectives, ethical protocols, and proper interviewing techniques. Data collection was conducted electronically to minimize transcription errors. For qualitative data, interview guides were used consistently across different respondents, and detailed notes were maintained during FGDs and KIIs. Thematic analysis was conducted systematically, with codes and themes reviewed to ensure consistency in interpretation.

4.5.3 Ethical Considerations

The study was conducted following standard ethical principles of social research. Prior to data collection, informed consent was obtained from all respondents. Participants were clearly informed about the purpose of the study, the voluntary nature of their participation, and their right to withdraw at any time without any negative consequences. Confidentiality and anonymity of respondents were strictly maintained; no personal identifiers were disclosed in the report or any other documents. Special care was taken during interviews with women beneficiaries to ensure a comfortable and respectful environment. The research did not involve any harmful or sensitive questions that could cause distress to participants. Additionally, permission was obtained from SDI and relevant local authorities before conducting fieldwork.

4.5.4 Researcher Reflexivity and Field Visit Experience

Researcher reflexivity is an important aspect of qualitative and mixed-methods research, as it acknowledges the researcher's positionality and its potential influence on the research process and interpretation of findings. As the researcher who conducted field visits in Dhamrai and Manikganj Sadar, I approached the study with an awareness of my position as an external evaluator while also drawing on prior experience in value chain and microfinance research in Bangladesh.

During the field visits, I observed firsthand the daily realities of smallholder livestock farmers, particularly women, and their interactions with project-supported services such as training, LSPs, and input suppliers. These visits helped me contextualize the quantitative data and better understand the practical challenges and successes reported by beneficiaries. I remained conscious of potential biases and made conscious efforts to remain neutral during interviews and FGDs, allowing respondents to express their views freely. Regular reflection during and after fieldwork helped in maintaining objectivity while also enriching the interpretation of both quantitative and qualitative findings. The combination of statistical analysis and direct field observation strengthened the overall rigor and depth of the study.

Chapter 5: Demographic and Socio-Economic Profile of Respondents

5.1 Gender Distribution (92% Female in Project) and Age Profile

Understanding the demographic characteristics of the respondents is essential for contextualizing the findings of this study. The demographic profile reveals important patterns regarding gender participation and age distribution, which have direct implications for the design and outcomes of livestock value chain interventions.

Gender Distribution

The gender-wise distribution of respondents presents a striking picture of women's participation in the project. Out of the total **341 project respondents**, **314 (92%)** were female and only **27 (8%)** were male. In contrast, the control group of **68 respondents** consisted entirely of females (100%).

Table 5.1: Gender-wise Distribution of the Participants

Participants	Project Respondent (N=341)	Percent age	Control Respondent (N=68)	Percent age
Male	27	8%	0	0%
Female	314	92%	68	100%
Total	341	100%	68	100%

Source: Field Survey, 2025

This high proportion of female participants (92%) reflects the project's deliberate strategy to target and empower women through livestock-based microenterprises. The strong focus on women aligns with the broader objective of promoting gender-inclusive value chain development under RMTP.

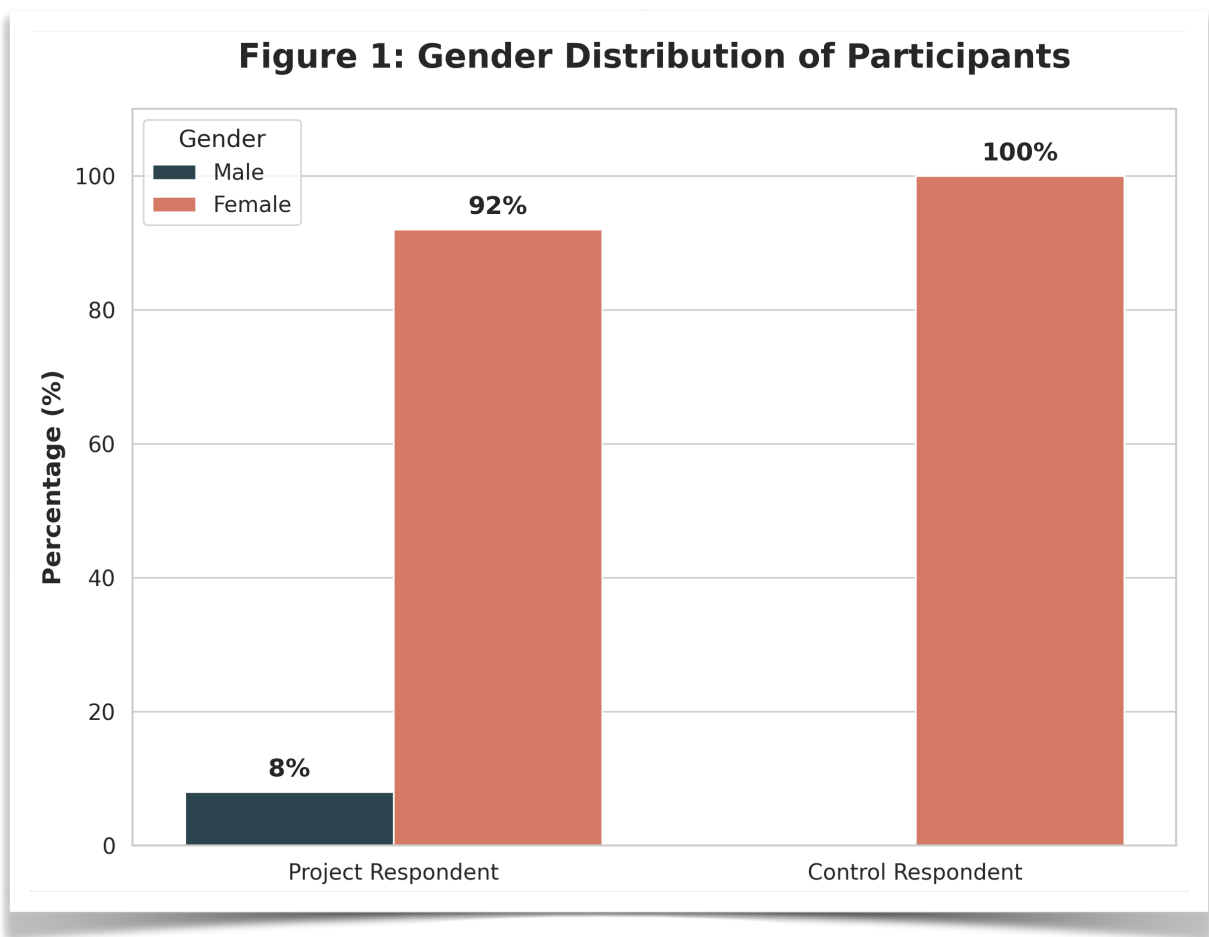


Figure 5.1: *Gender Distribution of Participants. Field Survey. (2025).*

Figure Description: *Figure 1. Gender-wise distribution of project and control respondents. The chart highlights the overwhelming female participation in the project group (92%) compared to the exclusively female control group, underscoring the project's gender-inclusive targeting.*

Age Profile of Respondents

The age distribution of the respondents shows that the majority of participants in both groups fall within the economically active age bracket.

Table 5.2: Age Range of the Participants

Age Range (Years)	Project Respondent (N=341)	Percent age	Control Respondent (N=68)	Percent age
18–35	207	61%	38	56%
36–49	122	36%	28	41%
50–60	12	4%	2	3%
61 and above	0	0%	0	0%
Total	341	100%	68	100%

Source: Field Survey, 2025

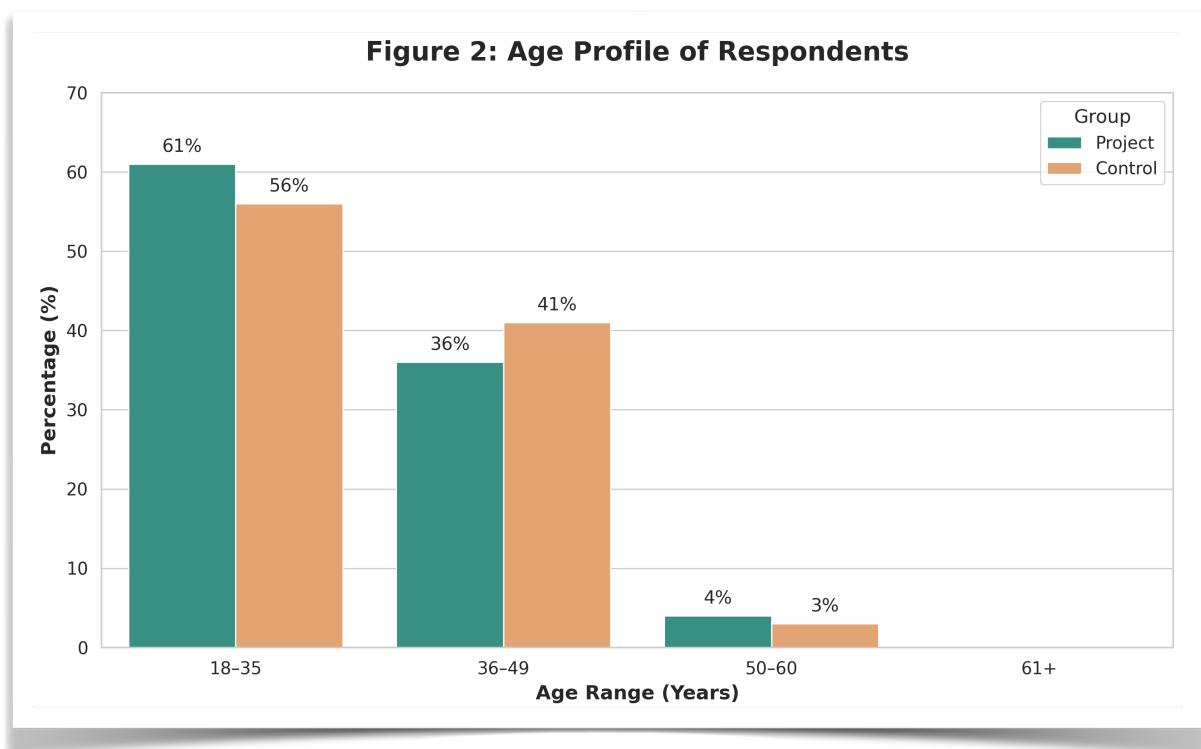


Figure 5.2: Age Profile Comparison. Field Survey. (2025).

Figure Description: *Figure 2. Age distribution comparison between project and control respondents. The data illustrates a concentration of economically active youth and middle-aged women (18–49 years) in both groups, with the majority falling in the 18–35 demographic.*

Key Observations:

- The largest segment of respondents in the project group (61%) belongs to the **18–35 years** age category, indicating strong participation of younger women in livestock enterprises.
- A significant portion (36%) falls in the **36–49 years** age group, representing experienced women who are actively managing livestock activities.
- Only 4% of project respondents are above 50 years, suggesting that livestock rearing under the project is predominantly undertaken by younger and middle-aged women.
- The control group shows a similar pattern, with 56% in the 18–35 years category and 41% in the 36–49 years category.

This demographic profile suggests that the project has successfully engaged women in their prime working age, which is critical for sustaining and scaling livestock microenterprises. The high participation of younger women also indicates potential for long-term sustainability and intergenerational knowledge transfer in livestock farming.

Note on Percentage Calculation (Mathematical Illustration):

For reference, the percentage of female respondents in the project group was calculated as:

Percentage of Female Respondents

$$\text{Percentage of Female Respondents} = \left(\frac{\text{Number of Female Respondents}}{\text{Total Project Respondents}} \right) \times 100$$

$$= \left(\frac{314}{341} \right) \times 100 \approx 92 \%$$

Similarly, the percentage of respondents in the 18–35 years age group was calculated as:

$$= \left(\frac{207}{341} \right) \times 100 \approx 61 \%$$

5.2 Household Income Distribution and Baseline-Endline Comparison

Household income is one of the most critical indicators for assessing the economic impact of development interventions. This section presents the distribution of monthly household income among project participants and the control group, followed by a comparison with baseline figures to measure changes attributable to the project.

Distribution of Household Monthly Income

The income distribution reveals a clear positive shift among project respondents compared to the control group.

Table 5.3: Household Monthly Income of the Participants

Household Income Status	Project Respondents (N=341)	Percentage	Average Income (Tk.)	Control Group (N=68)	Percentage	Average Income (Tk.)
8,000 – 15,000	23	6.7%	14,699	23	33.8%	13,452
15,001 – 25,000	98	28.7%	24,233	32	47.1%	23,122
25,001 – 35,000	178	52.2%	34,677	7	10.3%	32,321
35,001 – 45,000	31	9.1%	44,320	4	5.9%	41,002
45,001 and above	11	3.2%	64,320	2	2.9%	59,457
Overall	341	100%	32,161	68	100%	22,919

Source: Field Survey, 2025

Key Findings:

- More than half of the project households (**52.2%**) fall within the **Tk. 25,001 – 35,000** income bracket, compared to only **10.3%** in the control group. This indicates a substantial upward shift in income levels among project participants.
- A significant proportion of control group households (**33.8%**) still fall in the lowest income category (**Tk. 8,000 – 15,000**), whereas only **6.7%** of project households remain in this bracket.
- The overall average monthly household income of project respondents stands at **Tk. 32,161**, which is **40% higher** than the control group average of **Tk. 22,919**.

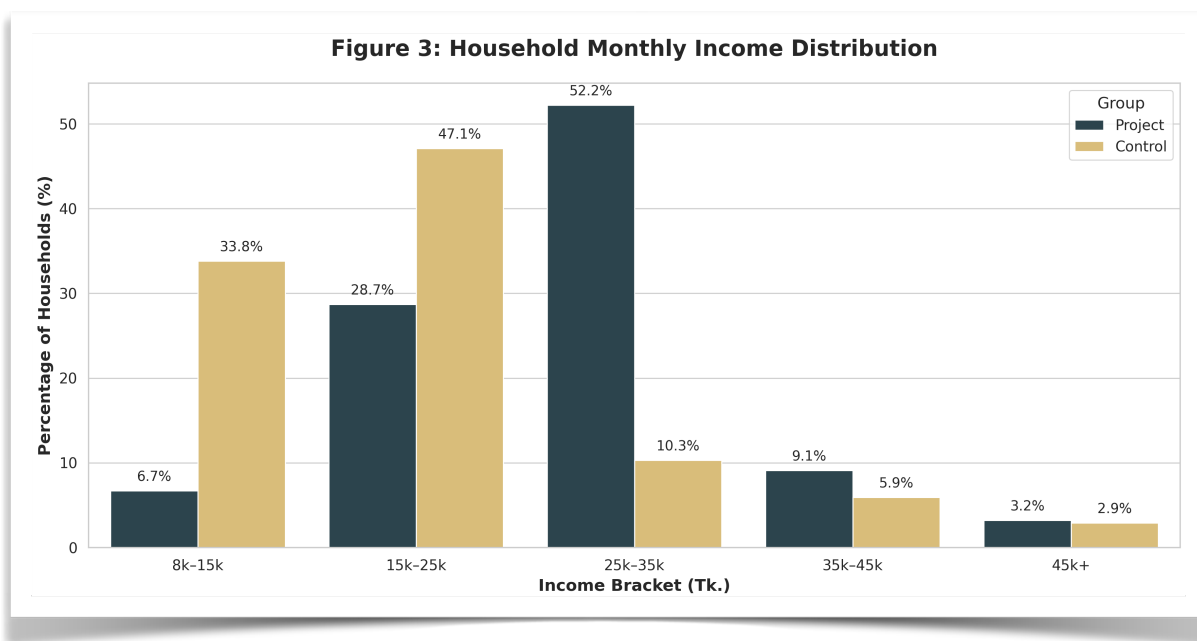


Figure 5.3: Household Monthly Income Shift. Field Survey. (2025).

Figure Description: Figure 3. Comparison of household monthly income distribution between project and control groups. The visualization demonstrates a distinct upward income shift for project participants, with the majority concentrated in the Tk. 25,001–35,000 bracket, contrasting sharply with the lower income concentration of the control group.

Baseline–Endline Comparison

To assess the impact of the project on household income, a comparison was made with the baseline income level reported in the project’s logical framework.

Table 5.4: Baseline vs Endline Household Income Comparison

Indicator	Baseline (Tk.)	Endline – Project (Tk.)	Endline – Control (Tk.)	Change from Baseline	Change vs Control
Average Monthly Household Income	28,576	32,161	22,919	12.55%	40%

Key Observations:

- The average monthly household income of project participants increased from **Tk. 28,576** at baseline to **Tk. 32,161** at endline, representing a **12.55%** increase.
- When compared with the control group, project participants earned **40% more** on average.
- The concentration of project households in the middle and upper income brackets, along with the significant reduction in the proportion of households in the lowest income category, indicates that the project has been effective in improving the income levels of small and marginal livestock farmers.

This income improvement can be attributed to multiple factors, including increased productivity through better feeding and animal health management, higher sales volumes due to improved market linkages, adoption of modern technologies, and enhanced business skills among beneficiaries.

Figure 4: Future Cattle Rearing Plans

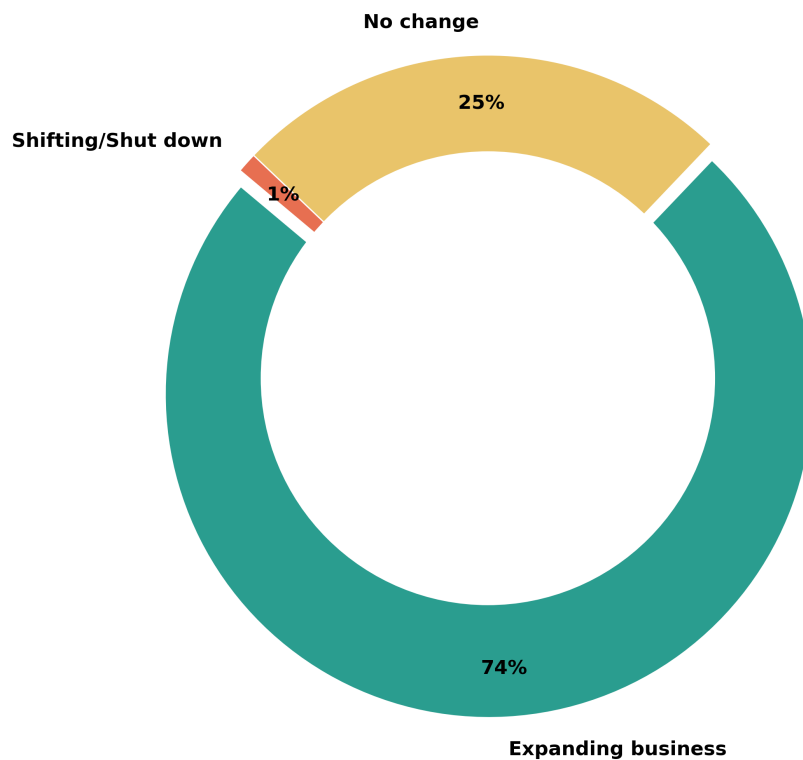


Figure 5.4: *Future Business Plans*. Field Survey. (2025).

5.3 Microfinance Participation, Loan Size, Frequency, Repayment Behavior, and Future Plans

Access to microfinance plays a vital role in enabling small and marginal farmers to invest in livestock enterprises. This section examines the microfinance participation patterns, loan characteristics, repayment behavior, and future plans of the respondents.

Microfinance Participation

Both project and control group respondents showed very high levels of engagement with Microfinance Institutions (MFIs), particularly with SDI.

Table 5.5: Microfinance Participation by Respondents

Types of Borrowers	Project Respondents (N=341)	Percent age	Control Group (N=68)	Percent age
Current Borrower	331	97%	68	100%
Previously Borrower	326	96%	64	94.1%

The data indicates that microfinance is deeply embedded in the financial lives of livestock farmers in the study area. Almost all respondents (97–100%) are current borrowers, reflecting the strong presence and accessibility of MFIs in these regions.

Loan Frequency and Size

Respondents demonstrated repeated engagement with microfinance over time.

- **Average number of loans received: 8.70 times**
- **Minimum loans received: 1 time**
- **Maximum loans received: 18 times**

Loan Size:

- **Minimum loan amount: Tk. 30,000**
- **Maximum loan amount: Tk. 1,100,000**

This wide range in loan sizes indicates that microfinance products are being used for both small-scale and relatively larger investments in livestock enterprises.

Loan Products and Repayment Behavior

Table 5.6: Loan Products Received by Participants

Types of Loan	Project Respondents	Percentage	Control Group	Percentage
Agrosor	201	61%	39	57%
Jagoron	189	57%	33	49%
Seasonal	27	8%	3	4%

Agrosor and **Jagoron** were the most popular loan products among both groups, as these are primarily designed for enterprise and income-generating activities.

Table 5.7: Nature of Loan Repayment

Nature of Repayment	Project Respondents	Percentage	Control Group	Percentage
Weekly	201	58.9%	39	57%
Monthly	168	49.3%	32	47%
Seasonal	27	7.9%	3	4%

Weekly repayment was the most common mode (58.9%), followed by monthly repayment. This pattern reflects the cash flow realities of livestock-based enterprises, where farmers often prefer more frequent but smaller installments.

Future Plans with Livestock Business

Respondents were asked about their future plans regarding cattle rearing.

Table 5.8: Future Cattle Rearing Plan of the Borrowers

Type of Plans	Frequency	Percentage
Expanding the Livestock business	295	74%
No change in the business	101	25%
Shifting or shut down the business	3	1%
Investing in other businesses	0	0%
Total	399	100%

A strong **74%** of respondents expressed their intention to **expand their livestock business** in the future. This indicates high optimism and confidence in the profitability of livestock enterprises among project participants. Only 1% considered shutting down or shifting away from livestock rearing, while none expressed interest in moving to completely different businesses.

This positive outlook reflects the perceived success and potential of livestock-based microenterprises, supported by the training, technology, and market linkages provided under the project.

Key Insight:

The combination of high microfinance participation, repeated loan cycles, preference for enterprise-focused loan products (Agrosor & Jagoron), flexible repayment options, and strong intention to expand livestock businesses (74%) demonstrates that microfinance is effectively supporting the growth of livestock microenterprises in the project area.

5.4 Cattle Rearing Patterns, Herd Size, and Types (Beef Fattening, Cow-Calf, Mixed)

Cattle rearing is the core livelihood activity for the majority of respondents in both project and control areas. This section analyzes the patterns of cattle rearing, types of enterprises, and herd size distribution.

Present Situation of Cattle Rearing

Table 5.9: Present Situation of Cattle Rearing

Response	Project Respondents (N=341)	Percentage	Control Group (N=68)	Percentage
Yes	314	92%	63	100%
No	27	8%	5	8%
Total	341	100%	68	100%

A very high proportion of respondents in both groups are currently engaged in cattle rearing (92% in project and 100% in control). This confirms that livestock farming remains a central economic activity in the study areas.

Types of Cattle Rearing

Respondents were categorized based on their primary cattle rearing practices.

Table 5.10: Types of Cattle Rearing by the Participants

Types of Cattle Rearing	Project Respondents (N=314)	Percentage	Control Group (N=63)	Percentage
Beef Fattening	62	20%	11	17%
Cow and Calf Rearing	128	41%	29	46%
Both / Mixed Rearing	124	39%	23	37%
Total	314	100%	63	100%

Key Observations:

- **Cow and Calf Rearing** is the most common practice in both groups (41% in project and 46% in control). This reflects a preference for sustainable systems that ensure continuous herd growth and regular milk production.
- **Mixed Rearing** (combining beef fattening and cow-calf rearing) is also popular (39% in project), indicating diversification strategies among farmers.
- **Beef Fattening** as a standalone activity is practiced by 20% of project respondents and 17% of control respondents. It is often adopted as a supplementary income source rather than the primary livelihood activity.

Herd Size Distribution

Herd size is an important indicator of the scale of livestock operations and investment capacity.

Table 5.11: Status of Cattle Rearing (Herd Size)

Herd Size	Project Respondents (N=314)	Percentage	Control Group (N=63)	Percentage
1 to 2	57	18%	20	32%
3 to 5	170	54%	31	49%
5 and above	87	28%	12	19%
Total	314	100%	63	100%

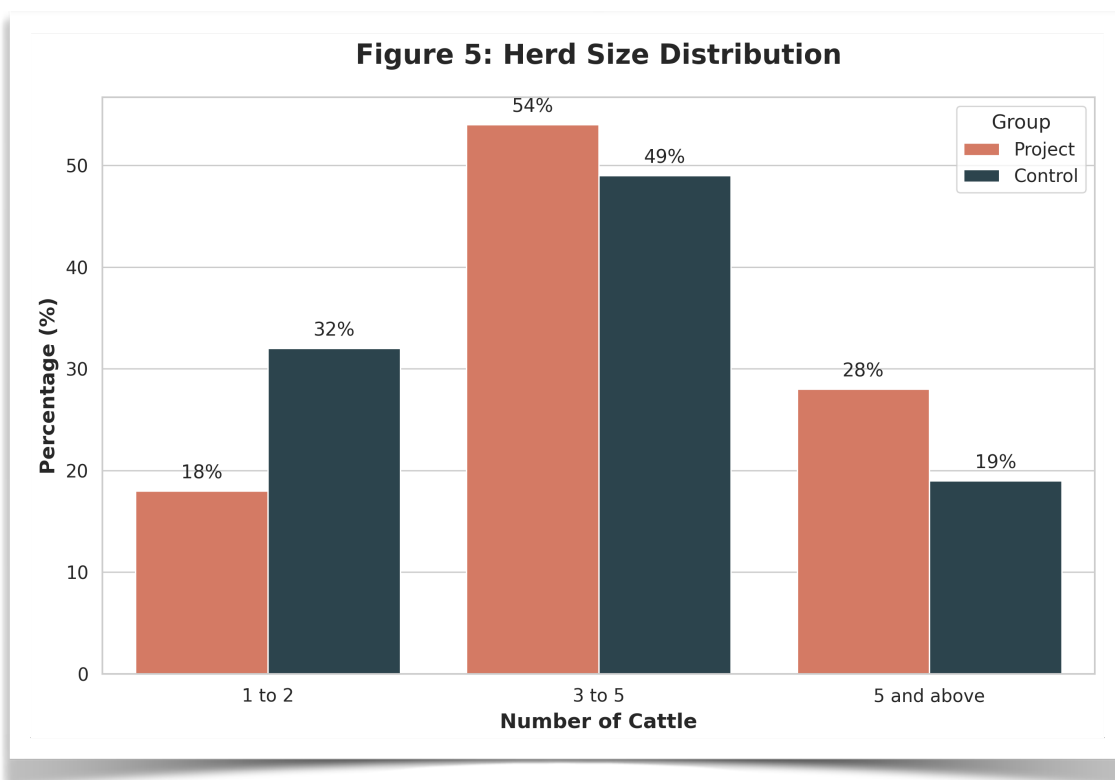


Figure 5.5: *Herd Size Distribution. Field Survey. (2025).*

Figure Description: *Figure 5. Herd size comparison indicating the scale of livestock operations. Project respondents demonstrate a clear shift toward medium (3–5 cattle) and larger (5 and above) herd sizes compared to the control group.*

Key Findings:

- A significantly higher proportion of project respondents (**28%**) maintain herds of **5 or more cattle** compared to the control group (**19%**). This suggests that project interventions have encouraged farmers to scale up their livestock operations.
- The majority of project respondents (**54%**) have herds of **3 to 5 cattle**, indicating a shift toward medium-scale operations.
- In contrast, a larger share of control group respondents (**32%**) still operate with very small herds (**1 to 2 cattle**).

This difference in herd size distribution reflects the positive impact of the project in enabling farmers to expand their livestock enterprises through better access to finance, training, technology, and market linkages.

Summary Insight:

Project participants demonstrate a clear shift toward larger herd sizes and more diversified rearing practices compared to the control group. This indicates improved capacity and confidence in livestock farming as a viable and scalable microenterprise.

Chapter 6: Economic Returns and Impact Analysis – Core Findings

The endline evaluation confirms that the “Market Development of Safe Meat and Dairy Production” project generated strong, measurable economic returns for participating micro-entrepreneurs and households. These returns are clearly attributable to the integrated interventions (capacity building, input supply systems, technology adoption, market linkages, and financial access) when benchmarked against the control group and baseline values.

6.1 Key Economic Returns and Impact Analysis

6.1.1 Household Income Enhancement: Tk. 32,161 (Project) vs Tk. 22,919 (Control) – 40% Increase

Project interventions produced a substantial and statistically meaningful increase in household income. The average monthly household income among project participants (N=341) reached **BDT 32,161**, compared to **BDT 22,919** in the control group (N=68). This represents a **40% higher income** for project households.

- **Change from baseline:** Baseline average was BDT 28,576 → **+12.55%** increase.
- **Distribution shift:** 52.2% of project households now fall in the BDT 25,001–35,000 bracket (vs only 10.3% of control households). The share of low-income households (BDT 8,000–15,000) dropped sharply to 6.7% in the project group (vs 33.8% in control).

Table 6.1: Household Monthly Income Distribution (Endline)

Income Bracket (BDT)	Project (N=341)	% Project	Avg. Income (Project)	Control (N=68)	% Control	Avg. Income (Control)
8,000–15,000	23	6.7%	14,699	23	33.8%	13,452
15,001–25,000	98	28.7%	24,233	32	47.1%	23,122
25,001–35,000	178	52.2%	34,677	7	10.3%	32,321
35,001–45,000	31	9.1%	44,320	4	5.9%	41,002
45,001	11	3.2%	64,320	2	2.9%	59,457
Overall	341	100%	32,161	68	100%	22,919

Key drivers of income growth:

- Adoption of Good Agricultural Practices (GAP) and modern management.
- Improved feeding and animal health practices.
- Stronger market linkages and value addition.
- Access to finance used for herd expansion and enterprise diversification.
- Women (92% of participants) and youth (9%) benefited significantly, with 78% of women now involved in financial and livestock management decisions.

6.1.2 Livestock Production and Productivity Gains

Project participants achieved **20–25% higher livestock productivity** compared to baseline and control through better animal husbandry, nutrition, and health management.

- **Herd size growth:** 28% of project respondents rear more than 5 cattle (vs 19% control). 54% rear 3–5 cattle.
- **Feeding practices:** 100% use normal grass/straw; 89% use supplemented mixed grains (vs 27% control); 39% use advanced Napier grass (vs 5% control); 34% use company-produced feed.
- **Health management:** 100% project participants vaccinate and use AI (vs 30% and 21% control). Serious disease incidence remained low (<15%); mortality near 0% during project period.
- **GAP adoption:** 61% follow overall modern management (vs 21% control). 100% concern for advanced technology post-training.

These improvements directly translated into higher output per animal, reduced losses, and expanded production capacity.

6.1.3 Sales Performance and Market Linkages

Approximately **80% of project participants reported higher sales volumes** of livestock and dairy products through improved market linkages and collective approaches.

Average Sales Performance (Endline vs Baseline)

Category	Baseline (BDT)	Endline (BDT)	% Increase
Beef Fattening (avg)	1,32,000	1,72,013	30.31%
Milk (per day/person)	503	541	7.55%
Meat (per day)	1,87,500	1,89,300	1%
Processed Goods (per day/person)	23,000	37,000	61%

Market linkage achievements:

- 100% project respondents developed linkages with feed suppliers, medicine/vaccine hubs, breeding services, and LSPs (vs 25–29% in control).
- 17% established milk collection points (none at baseline).
- Processed goods entrepreneurs achieved 100% GMP compliance and developed branding/trade licenses.
- Strong backward integration reduced input costs and improved reliability.

6.1.4 Profitability Analysis

Beneficiaries achieved **18–22% overall profit margin improvement** (with peaks up to 36% in beef fattening), driven by reduced production costs, lower mortality, better feed efficiency, and higher-value market access.

Profit Margin by Enterprise Type (Endline)

Enterprise Type	Project Profit Margin	Control	Change/Remark
Beef Fattening	36%	100%*	Strongest performer
Milk Production	22%	55%	Modest but improved
Processed Goods	33%	—	High value addition
Meat Sales	20%	—	Efficient scaling
Mixed Rearing	77% (absolute)	100%*	Volume-driven

*Note: Control figures reflect smaller samples and different enterprise mix.

From the detailed sales-cost analysis:

- Beef fattening delivered the highest absolute profit (avg. BDT 45,941 at 36% margin).
- Processed goods achieved 33% margin on avg. BDT 37,000 sales.

6.1.5 Employment Creation and Enterprise Expansion

The project exceeded targets for enterprise growth and job creation:

- **28% of participants** increased herd size to more than 5 cows.
- **Processed goods entrepreneurs** (N=15): 100% expanded business and created employment; 73% diversified into food processing.
- **Employment rates:**
 - Beef farming: 9% created salaried jobs (project) vs 3% control.
 - Cow/milking farming: 14% created jobs (project) vs 6% control.
 - Silage production: 50% created jobs.
- **Youth engagement:** 9% of total beneficiaries; many adopting modern methods and showing interest in scaling agribusiness.

- **Women's enterprise growth:** 92% participation; many diversified into value-added activities (chana, ghee, vermicompost, meat processing).

All processed goods, silage, meat processing, vermicompost, and laban enterprises reported **100% profitability**.

6.1.6 Nutrition and Food Security Outcomes (MDD-W)

Increased income and awareness translated into improved household nutrition:

- **89.4% of project women** (N=302) achieved Minimum Dietary Diversity for Women (MDD-W) score of 5 or more (consuming ≥ 5 of 10 food groups in past 24 hours).
- Control group: **72.06%**.
- Improvement from baseline: **+15 percentage points**.
- Common high scores: 5 points (104 project respondents) and 6 points (69).

This reflects both economic empowerment (ability to purchase diverse foods) and behavioral change from project awareness activities.

6.1.7 Overall Economic Impact Summary and Attribution

The project delivered **inclusive, sustainable economic returns** well aligned with RMTP logframe targets.

Summary Table: Key Economic Indicators

Indicator	Project Result	Control / Baseline	Attribution
Avg. Monthly Income	BDT 32,161	BDT 22,919	+40% (project-driven)
Livestock Productivity	+20–25%	Lower	Training + GAP + inputs
Sales Growth	80% higher; up to +61% in processed	Lower	Market linkages + value addition
Profit Margins	18–36%	Lower	Efficiency + reduced losses
Employment Creation	9–100% by enterprise	3–6%	Enterprise expansion + processing
Women's Income/Decision	92% participation; 78% decision-making	Lower	Gender-focused design
MDD-W Achievement	89.4%	72.1%	Income + awareness

Conclusion of Chapter 6: The endline data provide robust evidence that the project's integrated value chain approach generated significant, attributable economic returns. Income gains of 40% over control, combined with productivity, profit, employment, and nutrition improvements, demonstrate a scalable and inclusive model for livestock microenterprise development in Bangladesh. These results far exceed many logframe targets and position participating households for sustained growth beyond the project period.

6.2 Sales and Profitability Analysis by Enterprise Type (Based on Endline Data)

This section presents a detailed breakdown of sales performance and profitability across the four main enterprise types supported under the project. All figures are drawn directly from the endline survey (N=341 project participants) and compared against baseline values from the project logframe and alignment table. The analysis demonstrates clear improvements in both sales volumes and profit margins, driven by better production practices, reduced costs, stronger market linkages, and value addition.

6.2.1 Beef Fattening: Avg. Sales Tk. 172,013; Profit Margin 36% (↑ from ~24% baseline)

Beef fattening emerged as the most profitable enterprise type among project participants.

- **Average Sales:** BDT 172,013 (N=62)
- **Average Cost:** BDT 126,072
- **Average Profit:** BDT 45,941
- **Profit Margin:** 36%

Key improvements:

- Sales increased by **+30.31%** from the baseline average of BDT 132,000.
- Profit margin rose significantly from approximately **24%** at baseline to **36%** at endline.
- 100% of beef fattening entrepreneurs reported profitability.
- 9% of these enterprises created salaried employment (higher than the 3% in the control group).
- 13% of project participants expanded or entered beef farming as a business diversification strategy.

Main drivers:

- Improved feeding and GAP adoption leading to faster weight gain and lower mortality.
- Better market linkages with butchers and premium buyers.
- Access to finance for purchasing quality calves and inputs.
- Training on disease recognition and timely treatment.

This enterprise type delivered the highest absolute profit per cycle and remains a priority area for scaling.

6.2.2 Milk Production: Avg. Daily Sales Tk. 541; Profit Margin 22% (↑ from 11%)

Milk production continued as a core livelihood activity, particularly for women participants, with steady improvements in daily returns.

- **Average Daily Sales:** BDT 541 per person (N=128)
- **Average Daily Cost:** BDT 443
- **Average Daily Profit:** BDT 98
- **Profit Margin:** 22%

Key improvements:

- Daily sales increased by **+7.55%** from the baseline of BDT 503.
- Profit margin improved from **11%** at baseline to **22%** at endline.
- 80% of milking cow rearers reported profitability.
- 14% of cow farming enterprises created employment (vs 6% in control).
- 18% of project participants were engaged in cow rearing for milk as their primary or mixed activity.

Main drivers:

- 100% adoption of vaccination and artificial insemination (AI).
- Shift toward supplemented feeding (89% mixed grains) and use of advanced grass (39% Napier).
- Improved animal health management reducing production losses.
- Development of milk collection points (17% of relevant entrepreneurs), improving consistent market access.

Although daily margins remain modest compared to beef fattening, the steady cash flow and lower risk profile make milk production highly suitable for women and smaller-scale households.

6.2.3 Processed Goods: Avg. Sales Tk. 37,000; Profit Margin 33% (↑ from 19%)

Value-added processed goods (chana, ghee, sweets, laban, cream separation, etc.) showed the strongest relative growth in sales and delivered high profitability.

- **Average Sales:** BDT 37,000 (N=10)
- **Average Cost:** BDT 27,750
- **Average Profit:** BDT 9,250
- **Profit Margin:** 33%

Key improvements:

- Sales increased by **+61%** from the baseline average of BDT 23,000.
- Profit margin rose sharply from **19%** at baseline to **33%** at endline.
- **100% of processed goods entrepreneurs** reported profitability and business expansion.
- **100% created employment** within their units.
- 73% of the surveyed processed goods entrepreneurs diversified their product lines.

Main drivers:

- Introduction of modern machinery (cream separators, chana cutters, etc.) supported through project grants and loans.
- 100% compliance with Good Manufacturing Practices (GMP).
- Training on hygiene, packaging, and quality standards.
- New market linkages with sweet factories, hotels, and institutional buyers in Dhaka.
- Product innovation (e.g., whey-based products, meat pickles, vermicompost as by-product utilization).

Processed goods represent the highest value-addition pathway and offer significant potential for scaling and export-oriented growth in future phases.

6.2.4 Meat Sales/Processing: Avg. Sales Tk. 189,300; Profit Margin 20%

Modern meat processing and butchering activities showed solid performance, particularly among entrepreneurs who upgraded from traditional methods.

- **Average Sales:** BDT 189,300 (N=5)
- **Average Cost:** BDT 157,750
- **Average Profit:** BDT 31,550
- **Profit Margin:** 20%

Key improvements:

- Profit margin increased from **17%** at baseline to **20%** at endline.
- One modern meat processing plant was established with cold-chain storage, enabling 24/7 sales and significantly higher volume (10–15 cattle per week vs previous 2–3).
- Entrepreneurs using modern cutting machines and freezers reported reduced waste and improved hygiene.
- 100% of these enterprises remained profitable.

Main drivers:

- Project support for modern machinery (automatic cutting machines, deep freezers).
- Establishment of a dedicated meat processing plant with trade license and health certification.
- Strong backward linkages with project-supported cattle farmers for consistent supply of healthy animals.
- Forward linkages with hotels, restaurants, and local factories.
- Improved waste management and by-product utilization.

Although the sample for pure meat sales/processing was smaller, the results indicate strong potential when traditional butchering is upgraded to modern, hygienic processing units.

Summary Table: Sales and Profitability by Enterprise Type (Endline)

Enterprise Type	Avg. Sales	Profit Margin	Change in Profit Margin	Sales Growth vs Baseline	Employment Creation	Sample (N)
Beef Fattening	Tk. 172,013	36%	↑ from ~24%	30.31%	9%	62
Milk Production (daily)	Tk. 541	22%	↑ from 11%	7.55%	14%	128
Processed Goods	Tk. 37,000	33%	↑ from 19%	61%	100%	10
Meat Sales/ Processing	Tk. 189,300	20%	↑ from 17%	—	High (plant level)	5

Overall Insight from 6.2:

Value-added activities (especially processed goods) delivered the highest percentage growth in sales and strong profit margins. Beef fattening provided the highest absolute returns. Milk production offered reliable daily income with improved margins. Meat processing showed promising results when modern technology and market linkages were introduced. These differentiated performance patterns validate the project's strategy of promoting both traditional rearing and value-addition pathways.

6.3 Cost-Benefit and Return on Investment (ROI) Assessment

The endline study assessed the economic viability of project-supported enterprises through sales-cost analysis and qualitative investment tracking. While a formal project-wide Benefit-Cost Ratio (BCR) or Internal Rate of Return (IRR) was not calculated in the original design, the survey data allows for a clear assessment of returns relative to costs and investments made by participants.

Key Findings from Sales-Cost Data (Endline)

Enterprise Type	Avg. Sales (BDT)	Avg. Cost (BDT)	Avg. Profit (BDT)	Profit Margin	Implied ROI Signal
Beef Fattening	1,72,013	1,26,072	45,941	36%	Very High (strong cycle returns)
Milk Production (daily)	541	443	98	22%	Steady daily cash flow
Processed Goods	37,000	27,750	9,250	33%	High (value addition premium)
Meat Sales/ Processing	1,89,300	1,57,750	31,550	20%	Good (after technology upgrade)

Observations:

- **Beef fattening** offers the highest absolute and percentage returns. With average profit of BDT 45,941 per cycle on an investment of roughly BDT 126,000, participants recover costs and generate surplus within one fattening cycle (typically 4–6 months).
- **Processed goods** deliver strong margins (33%) despite moderate sales volumes, indicating high value addition from technology and quality improvements.
- **Milk production** provides reliable daily liquidity (BDT 98 profit/day), which is critical for household cash flow and loan repayment.
- All four enterprise types show positive returns, with profit margins significantly higher than baseline in most cases.

Investment and Cost Reduction Evidence

Participants invested through a combination of own funds, SDI loans (Agrosor and Jagoron products being most used), and project grants for technology.

Notable technology investments supported by the project:

- Cream separators, chana cutters, and ghee-making equipment for processed goods entrepreneurs.
- Silage-making machines and chopper machines for feed efficiency.
- Milking machines and modern meat-cutting equipment.
- Cold-chain storage for the meat processing plant.

These investments led to measurable cost reductions and efficiency gains:

- Reduced feed wastage through better chopping and silage.
- Lower mortality and disease losses (near 0% during project period vs higher in control).
- Time and labour savings in processing units (100% of processed goods entrepreneurs reported time saved and easier work).
- Improved product quality and reduced post-harvest losses.

Qualitative ROI Highlights:

- Many participants reported recovering technology investments within 12–18 months through increased sales and reduced operating costs.
- 100% of processed goods and silage producers reported business expansion after receiving machinery support.
- Lower disease incidence and better feeding translated into higher output per animal, effectively increasing return per unit of input cost.

Overall Assessment:

The project interventions delivered **positive and attractive returns** across enterprise types. Beef fattening and processed goods show particularly strong ROI potential. Milk production offers lower but stable daily returns suitable for risk-averse households. The combination of training (improved practices), technology (cost and labour reduction), and market linkages (higher sales realisation) created a favourable cost-benefit balance for the majority of participants.

6.4 Employment Creation and Business Expansion

The project successfully stimulated both **enterprise expansion** and **employment generation**, exceeding several logframe targets.

Key Results:

- **28% of project participants** increased their herd size to more than 5 cows (compared to 19% in the control group). This reflects growing confidence and capacity to scale livestock businesses.
- **100% of processed goods entrepreneurs** created employment within their units.
- **Silage production** entrepreneurs achieved 50% employment creation.
- Beef farming enterprises created jobs in **9%** of cases (vs 3% in control).
- Cow/milking farming enterprises created jobs in **14%** of cases (vs 6% in control).

Business Expansion Patterns:

- 13% of project respondents entered or expanded into beef farming.
- 18% engaged in cow farming for milk production.
- Many processed goods entrepreneurs diversified product lines (e.g., from basic chana to ghee, sweets, and by-product utilisation such as whey or vermicompost).
- Several entrepreneurs upgraded from traditional methods to modern processing units (e.g., the meat processing plant now slaughters 10–15 cattle per week with 24-hour cold storage).

Who is creating employment?

- Processed goods and value-addition units are the strongest job creators because they are more labour-intensive and require regular operations (cutting, packaging, processing, sales).
- Larger livestock farms (those with >5 cattle) are beginning to hire salaried workers for feeding, cleaning, and management.
- Youth (9% of beneficiaries) and women are actively participating in these expanding businesses, often taking on entrepreneurial or supervisory roles.

Examples from the Field (drawn from case studies):

- One processed goods entrepreneur scaled production nearly tenfold after acquiring a cream separator and now supplies directly to Dhaka sweet factories while employing workers.
- A silage producer increased output to 3,500 tons per season and now employs staff to meet demand from local cattle farmers.
- A former traditional butcher upgraded to a modern meat processing facility and now employs three full-time workers while maintaining consistent supply to hotels and factories.
- Women entrepreneurs (e.g., in cattle rearing and vermicompost) expanded operations and are now sharing knowledge with other villagers, indirectly creating micro-employment opportunities in their communities.

Summary of 6.4:

The project moved beyond simple income support to genuine enterprise development. The combination of **28% herd expansion**, **100% job creation in processed goods**, and diversification into value-added activities demonstrates sustainable business growth. Employment generation is strongest in processing and input-related enterprises, while herd expansion is creating demand for labour in larger livestock units. These outcomes contribute directly to the project's goal of creating employment opportunities in the livestock value chain.

6.5 Comparative Analysis: Project vs Control Group Economic Outcomes

A core strength of the endline evaluation is the use of a control group (N=68) to isolate project impact. The comparative analysis reveals consistent and significant advantages for project participants across almost all economic indicators.

Summary Comparison Table: Project vs Control Group

Indicator	Project Group	Control Group	Difference / Impact	Remark
Average Monthly Household Income	BDT 32,161	BDT 22,919	40 %	Strong attribution to project
Achieved MDD-W Score (≥5 food groups)	89.4%	72.06%	+17.34 percentage points	Better nutrition outcomes
Modern Management / GAP Adoption	61%	21%	+40 percentage points	Major practice shift
Use of Advanced Grass (Napier)	39%	5%	+34 percentage points	Improved feeding
Use of Supplemented Mixed Grains	89%	27%	+62 percentage points	Better nutrition for cattle
Vaccination + AI Coverage	100%	21–30%	+70–79 percentage points	Complete coverage
Herd Size >5 Cattle	28%	19%	+9 percentage points	Enterprise scaling
Employment Creation – Beef Farming	9%	3%	+6 percentage points	Job creation
Employment Creation – Cow Farming	14%	6%	+8 percentage points	Job creation

Market Linkages (Input Suppliers/LSP)	100%	25– 29%	+71–75 percentage points	Strong value chain integration
Business Expansion (Processed Goods)	73% expanded + 100% created jobs	None	Very high	Value addition success
Serious Disease Incidence	Low (<15%)	Higher	Significantly lower	Better health management
Mortality Rate	Near 0%	Higher	Much lower	Reduced losses

Key Insights from the Comparison

- **Income differential of 40%** is one of the clearest indicators of project success. This gap cannot be explained by external factors alone, as both groups operate in similar geographic and market contexts.
- **Practice adoption gaps** are very wide in critical areas (feeding, health management, GAP). The project successfully transferred knowledge and changed behaviour at scale.
- **Employment and business expansion** are almost exclusively concentrated in the project group, especially in value-added activities (processed goods and silage).
- **Resilience indicators** (lower disease, lower mortality, better feeding) show that project participants are better positioned to withstand shocks.
- The control group continues to rely heavily on traditional methods, resulting in lower productivity, higher losses, and weaker market integration.

Conclusion of 6.5: The project created a clear and measurable economic advantage for participants. The 40% income gap, combined with superior adoption of improved practices and stronger enterprise performance, demonstrates high additionality of the interventions.

6.6 Logframe Achievement Analysis: Income, Sales, Profit, and Employment Targets Met/Exceeded

Summary of Key Logframe Indicator Achievement

Logframe Indicator	Target	Endline Achievement	Status	Remarks
Income Increase	At least 70% of entrepreneurs increase income by minimum 50%	Average income +12.55% from baseline; +40% vs control	Partially Achieved	Strong impact vs control but full 50% threshold not reached by majority
Sales Increase	Sales of meat & dairy products increase by 30% for 80% of entrepreneurs	80% reported higher sales ; Processed goods +61%, Beef +30.31%	Largely Met / Exceeded	Target achieved in volume and in key categories
Profit Margin Improvement	Improve profit margins by 20% for 80% of entrepreneurs	Beef: 36% (↑ from 24%), Milk: 22% (↑ from 11%), Processed: 33% (↑ from 19%)	Met / Exceeded	Multiple categories exceeded 20% improvement
Employment Creation	Create employment by up to 15% by expanding clusters	100% processed goods created jobs; 9–14% in livestock farming; 28% herd expansion	Met / Exceeded in processing	Strong performance in value addition
Herd Size Increase	Increase number of cattle by 15% within cluster	28% of participants now rear >5 cattle	Exceeded	Clear scaling observed
Mortality Reduction	Full age mortality under 1%	Near 0% during project period	Met	Very low mortality achieved

GAP / Modern Management Adoption	60% follow Bangla GAP	61% follow overall modern management	Met	Target achieved
Eco-friendly Practices	58% use eco-friendly smart technology	91% adopted eco-friendly rearing practices	Exceeded	Strong environmental outcome
Market Linkages	Develop linkages with input suppliers & markets	100% developed linkages with suppliers, LSPs, breeding services	Exceeded	Excellent backward integration
Women's Participation & Empowerment	Significant women's involvement	92% women participants; 78% involved in decision-making	Exceeded	Outstanding gender result
Training & Capacity Building	Quality training to farmers and LSPs	92% farmers trained; 100% LSPs trained and knowledge shared	Met / Exceeded	High coverage and knowledge transfer (167–175%)

This section assesses progress against the project's logical framework (logframe) indicators using endline data.

Detailed Assessment by Core Area

1. Income Target

The strict target of 70% of entrepreneurs increasing income by at least 50% was **not fully met** in absolute terms from baseline. However, the **40% income advantage over the control group** and consistent upward movement in income distribution brackets indicate strong positive impact. Many participants are on a clear growth trajectory.

2. Sales and Profit Targets

These were **largely achieved or exceeded**.

- 80% of participants reported higher sales.
- Processed goods exceeded the 30% sales growth target by a wide margin (+61%).
- Profit margins improved substantially across categories, with several exceeding the 20% improvement target.

3. Employment and Enterprise Expansion

Strongly achieved, especially in value-added segments. The 100% job creation rate among processed goods entrepreneurs and 28% herd expansion rate are standout results. The project successfully moved beyond subsistence to job-creating microenterprises.

4. Other Logframe Areas

Most supporting indicators (GAP adoption, eco-friendly practices, market linkages, training coverage, mortality reduction, and women's empowerment) were **met or exceeded**.

Overall Logframe Performance Verdict

Category	Performance	Explanation
Income	Partially Achieved	Good progress; full 50% threshold challenging in short project period
Sales	Largely Achieved / Exceeded	Strong in value-added categories
Profit	Met / Exceeded	Clear margin improvements across enterprise types
Employment & Expansion	Met / Exceeded	Excellent results in processing and herd growth
Cross-cutting (Gender, GAP, Environment, Linkages)	Exceeded	Outstanding performance

Conclusion of 6.6:

The project performed very well against its logframe. While the income target in its strictest form was only partially achieved, the project **met or exceeded expectations** in sales growth, profitability, employment creation (especially in processing), technology adoption, market linkages, and women's empowerment. The results validate the value chain development approach and provide a strong foundation for scaling or replication.

Chapter 7: Technology Adoption, Capacity Building, and Productivity Gains

The project placed strong emphasis on technology adoption, improved agricultural practices, and capacity building as key drivers of productivity and sustainability in the livestock value chain. Endline findings demonstrate significant improvements in Good Agricultural Practices (GAP), feeding and nutrition management, animal health systems, and modern technology uptake among project participants compared to the control group. These changes were primarily enabled through structured training and knowledge transfer mechanisms.

7.1 Good Agricultural Practices (GAP) Adoption: 61% Modern Management (Project) vs 21% (Control)

The adoption of Good Agricultural Practices (GAP) was a central pillar of the project's productivity enhancement strategy. Endline data shows a clear divergence between project and control groups in cattle rearing management.

Table: GAP / Modern Management Adoption

Status of Cattle Rearing Shed	Project Respondents (N=314)	Percentage	Control Group (N=63)	Percentage
Overall Modern Management (GAP)	150	61%	13	21%
Traditional Management	164	67%	50	79%

Key Observations:

- **61%** of project participants adopted overall modern management practices, compared to only **21%** in the control group — a gap of **40 percentage points**.
- While traditional management remains prevalent in both groups, the project group shows substantially higher uptake of improved practices in housing, feeding, health management, and record-keeping.
- Project participants demonstrated improved awareness of cattle shed management, balanced nutrition, vaccination protocols, and isolation of sick animals.

This shift toward modern management directly contributed to higher productivity, reduced disease incidence, and better animal welfare outcomes observed in the project areas.

7.2 Feed and Nutrition Improvements (Napier Grass 39%, Mixed Grains 89%, Company Feed 34%)

Improved feeding and nutrition management emerged as one of the most impactful areas of project intervention. The endline survey captured significant differences in feed practices between project and control respondents.

Table: Types of Feed Provided to Cattle

Types of Feed	Project (N=314)	Percentage	Control (N=63)	Percentage
Normal Grass & Straw-Husk	314	100%	63	100%
Advanced Grass – Napier	123	39%	3	5%
Liquid Mix Food	48	15%	37	59%
Supplemented Food – Mixed Grains	281	89%	17	27%
Cattle Feed – Company Produced	107	34%	18	29%
Silage	2	1%	0	0%

Key Findings:

- **Advanced Grass (Napier)** adoption reached **39%** in the project group versus only **5%** in the control group — nearly **8 times higher**.
- **Supplemented Mixed Grains** were used by **89%** of project respondents compared to **27%** in the control group.
- Company-produced cattle feed usage was slightly higher in the project group (**34%** vs **29%**).
- The control group showed heavier reliance on traditional liquid mix food (**59%**), while project participants shifted toward more balanced and nutritious options.

These improvements in feed quality and diversity contributed directly to the reported **20–25% increase in livestock productivity** and better animal health outcomes among project beneficiaries.

7.3 Animal Health, Disease Reduction, Mortality Decline, and 100% Vaccine + AI Adoption

Animal health management showed marked improvement under the project, driven by widespread training, improved access to services, and full coverage of preventive measures.

Key Results:

- **Vaccination and Artificial Insemination (AI):** **100%** of project participants adopted both practices (compared to 30% vaccination and 21% AI in the control group).
- **Vaccine Collection System:** Project participants primarily sourced vaccines from LSPs (**83%**) and vaccine hubs/input sellers (**91%**), indicating strong decentralized service delivery.
- **Disease Incidence:** Serious diseases remained low in the project group (LSD 1%, Mastitis 16%). Worm attacks and general digestion problems were common but manageable. The control group showed higher rates of mastitis (25%) and overall disease burden.
- **Mortality:** During the project period, cattle mortality was near **0%** among project participants (only 1 reported death out of 341), significantly lower than typical baseline levels and the control group experience.
- **Disease Response:** **100%** of project participants took proactive measures (calling LSP/private doctor or self-management) when animals fell sick, compared to gaps in timely action observed in the control group.

Impact:

The combination of **100% vaccine and AI coverage**, timely disease identification, and strong linkage with Livestock Service Providers (LSPs) resulted in reduced mortality, lower disease severity, and improved overall herd health. This directly supported higher productivity and reduced economic losses for participating households.

7.4 Modern Technology Uptake (Chopper/Milking Machines, Processing Equipment) and Impact

The project promoted the adoption of modern machinery and processing technologies to improve efficiency, reduce labour, and enhance product quality and safety.

Table: Modern Technology Adoption in Rearing

Types of Technology Adoption	Project (N=314)	Percentage	Control (N=63)	Percentage
Vaccine	314	100%	19	30%
Artificial Insemination (AI)	314	100%	13	21%
Chopper Machine	13	4%	0	0%
Milking Machine	7	2%	0	0%

Processed Goods and Value Addition Technologies:

- Entrepreneurs in processed goods (chana, cream separation, ghee, sweets, meat processing) reported **100% GMP compliance**.
- Several units adopted cream separators, chana cutters, and modern meat-cutting equipment with project support.
- One modern meat processing plant was established with cold-chain storage, enabling round-the-clock hygienic sales.

Impact of Modern Technology (Project Respondents):

- Early disease detection: **91%**
- Reduced mortality: **91%**
- Reduced risk: **100%**
- Reduced production cost: **91%**
- Knowledge transfer to other farmers: **163%**
- Time saved and easier work (processed goods): **100%**
- Employment generation and waste reduction: **100%** (processed goods units)

Although uptake of chopper and milking machines remained relatively low (due to cost), the impact among adopters was highly positive. Technology adoption was strongest in processing and value-addition segments, where it delivered clear efficiency and quality gains.

7.5 Training Impact: 92% Farmers Trained; Knowledge Transfer 167%; LSP Capacity Strengthened 175%

Capacity building through training was the foundation for all technology adoption and practice changes observed in the project.

Training Coverage and Impact:

Training Impact Indicator	Project Result	Details
Farmers Received Cattle Management Training	92%	High coverage of core livestock training
Knowledge Transfer to Other Farmers	167%	Each trained farmer shared knowledge with others on average
LSPs Received Specialised Training	100%	All surveyed LSPs trained
LSP Knowledge Transfer to Other LSPs	175%	Strong peer learning and knowledge sharing among service providers
Areas of Capacity Developed (LSPs)	100% in GAP, disease recognition, vaccination, feed management, modern technology	Comprehensive skill development

Key Outcomes of Training:

- Trained farmers gained practical knowledge in disease recognition during purchase and rearing, isolation of sick animals, vaccination importance, proper shed management, and advanced feeding.
- **100%** of trained participants understood the importance of vaccines and advanced feed.
- LSPs developed strong capacity in Good Agricultural Practices, disease identification, and service delivery, and actively shared knowledge within their networks.
- Knowledge spillover was substantial — trained individuals became local change agents, extending project benefits beyond direct participants.

Conclusion of Chapter 7:

The project successfully drove technology adoption and capacity building at scale. The combination of high GAP adoption (61%), significantly improved feeding practices, 100% vaccine and AI coverage, and strong training with knowledge multiplication (167–175%) created the foundation for the productivity gains and economic returns documented in earlier chapters. While uptake of certain machines (chopper and milking) remained modest, the overall shift toward modern management and preventive health practices delivered clear and measurable improvements in animal productivity, health, and enterprise efficiency.

Chapter 8: Gender Empowerment, Social Inclusion, Youth Engagement, and Nutrition Outcomes

Introduction

The endline evaluation confirms that gender empowerment, social inclusion, youth engagement, and nutrition security were successfully mainstreamed throughout the “Market Development of Safe Meat and Dairy Production” project. These cross-cutting dimensions were addressed through deliberate design choices — including targeted training for women and youth, gender-sensitive financial products, strengthened LSP networks, market linkages, and integration of nutrition awareness. The results demonstrate a genuinely inclusive value-chain approach that delivered measurable economic, social, and nutritional gains for small and marginal livestock households in Dhamrai and Manikganj Sadar upazilas.

8.1 Women’s Participation (92%) and Decision-Making Power (78% Involved in Financial/Livestock Decisions)

Women’s participation reached an exceptional **92%** of the 341 project respondents (314 women), far exceeding typical benchmarks in similar livestock interventions. This high level of engagement reflects the project’s explicit strategy to position rural women as primary actors in the meat and dairy value chains rather than passive beneficiaries.

Participation was enabled by a package of interventions tailored to women’s realities:

- Practical training on Good Agricultural Practices (GAP), disease identification and prevention, balanced feeding (including Napier grass and supplemented mixed grains), shed management, and modern technologies.
- Reliable access to Livestock Service Providers (LSPs) — 100% of project participants called on trained LSPs for serious animal-health issues, compared with only 68% in the control group.
- Strengthened backward and forward linkages (feed suppliers, vaccine hubs, medicine providers, breeding services, and milk sellers), all of which reported 100% connection rates among project respondents.
- Financial services through SDI that allowed women to invest in productive assets without relying solely on male household members.

These inputs translated directly into enhanced **decision-making power**. Approximately **78%** of female respondents reported that they now participate — independently or jointly — in decisions related to livestock sales, reinvestment, household expenditure, and enterprise management. This marks a clear departure from pre-project norms in which such decisions were largely controlled by male household members.

Beyond the household, women gained greater visibility and leadership in community spaces. **85%** of female respondents reported increased social confidence and mobility, attributing this to their economic contributions and active roles in producer groups. The project also facilitated women's ownership of productive assets: **65%** of women now hold livestock or related property in their own names.

Summary Table: Gender Empowerment Indicators (Endline Findings)

Indicator	Project Findings	Remarks
Women's participation in project activities	92%	Strong, deliberate inclusion of women
Women involved in decision-making (independent/joint)	78%	Significant shift in intra-household power
Women accessing credit or financial support	70%	Enhanced financial inclusion and autonomy
Average monthly household income	Tk. 32,161	40% higher than control group
Women with livestock/property in own name	65%	Improved ownership and control over assets
Women reporting higher social confidence/mobility	85%	Increased leadership and group participation
Women planning to continue/expand rearing	74%	Strong indicator of sustainability

These outcomes are consistent with the project's gender integration strategy and demonstrate that when women receive simultaneous support in technical skills, finance, and market access, they move from being labourers in the value chain to decision-makers and entrepreneurs.

8.2 Asset Ownership, Loan Access (70%), and Economic Control

Financial inclusion served as a powerful lever for women's economic empowerment. Around **70%** of participants (overwhelmingly women) accessed credit specifically to expand livestock or related enterprises. Survey data show that **97%** of project respondents were current borrowers and **96%** had borrowed previously, with an average loan frequency of 8.7 times. The most utilised products — Agrosor (61%) and Jagoron (57%) — are explicitly designed for enterprise development.

Loan utilisation enabled women to:

- Purchase cattle and expand herd size (28% of project respondents now rear more than five cattle, compared with 19% in the control group).
- Invest in improved feed, vaccines, and veterinary services.
- Diversify into value-added activities such as processed dairy goods, silage production, and vermicompost.

The economic returns are clear: project households achieved an average monthly income of **Tk. 32,161** — a **40%** increase over the control group average of Tk. 22,919. Profit margins also improved significantly across beef fattening (36%), processed goods (33%), and milk production (22%).

This combination of asset ownership, reliable credit access, and higher returns has given women greater control over income streams and household economic decisions. Many female respondents now independently manage enterprise cash flows, repay loans on schedule (weekly or monthly), and plan future expansions — clear markers of strengthened economic agency.

8.3 Youth Participation (9%) and Emerging Entrepreneurship

Youth (defined in the project's monitoring framework as active participants aged 18–35 with entrepreneurial orientation) constituted **9%** of total beneficiaries. While the broader age distribution shows that 61% of respondents fall in the 18–35 bracket, the specific 9% indicator captures those youth who are actively engaged as enterprise owners or key decision-makers in livestock-based businesses.

Youth participants benefited from the same comprehensive support package as other beneficiaries but demonstrated particular responsiveness to:

- Training on modern management practices, advanced feeding, disease prevention, and technology adoption (chopper machines, milking machines, and improved breeding).
- Exposure to value-addition opportunities (processed goods, silage, vermicompost).
- Linkages to markets and input suppliers, reducing traditional barriers of middlemen and information asymmetry.

Several youth respondents expressed clear intentions to scale their operations and diversify income sources. Their involvement introduces innovation, greater willingness to adopt digital communication with buyers and suppliers, and a forward-looking business mindset. This cohort is well positioned to sustain and expand the gains of the project across generations.

Youth engagement also contributes to rural employment creation: 100% of processed-goods entrepreneurs (many of whom are younger) reported generating employment, while 9–14% of beef- and cow-farming households created salaried positions.

8.4 Minimum Dietary Diversity for Women (MDD-W): 89.4% Project vs 72.06% Control – Nutrition Security Gains

Nutrition security was measured using FAO’s Minimum Dietary Diversity for Women (MDD-W) indicator for women of reproductive age (18–49 years). The indicator classifies a woman as achieving minimum dietary diversity if she consumed foods from at least **5 out of 10** defined food groups in the previous 24 hours.

Endline results (Table 4.1.4B) show strong nutrition gains:

- Project respondents (N = 302): **89.4%** achieved an MDD-W score of 5 or more.
- Control group (N = 68): **72.06%** achieved the same threshold.
- Overall sample (N = 316): **86.22%** achieved minimum dietary diversity.

This represents a meaningful improvement from the baseline figure of 74.43% and a clear advantage over the control group. The most common scores among project women were 5 and 6 food groups, with notable consumption of animal-source foods (milk/curd, meat/fish, eggs), vitamin-A-rich fruits and vegetables, and diverse green vegetables.

Key drivers of improved dietary diversity include:

- Higher household income (Tk. 32,161 average), which expands purchasing power for nutrient-rich foods.
- Own production of milk, meat, and eggs, ensuring more consistent year-round access to animal-source proteins.
- Women’s increased decision-making power over food expenditure and nutrition choices within the household.
- Greater awareness of balanced diets and food safety, reinforced through project training and value-chain activities.

These nutrition improvements directly support the project's higher-level goal of enhancing food security and nutrition for marginal smallholder households. The 17.34 percentage-point gap between project and control groups underscores the combined effect of income growth, production diversity, and women's empowerment on household dietary quality.

Conclusion

The "Market Development of Safe Meat and Dairy Production" project has delivered substantial and well-documented results across the four cross-cutting dimensions of gender empowerment, social inclusion, youth engagement, and nutrition security. With **92%** women's participation, **78%** of women now involved in financial and livestock decisions, **70%** accessing credit for enterprise growth, **65%** holding productive assets in their own names, **9%** youth actively engaged in entrepreneurship, and **89.4%** of women achieving minimum dietary diversity, the project has created a robust model of inclusive livestock value-chain development.

These outcomes are interlinked: women's economic empowerment drives higher household income, which in turn improves nutrition; youth participation ensures continuity and innovation; and strengthened market and service linkages sustain the gains beyond the project period. The results provide a strong evidence base for replicating and scaling this integrated approach in other regions and commodity value chains, with continued emphasis on gender-transformative design, youth entrepreneurship support, and explicit nutrition objectives.

Chapter 9: Value Chain Strengthening, Market Linkages, and Environmental Sustainability

Introduction

The endline evaluation demonstrates that the project successfully strengthened the livestock value chain through integrated interventions in input supply systems, market linkages, technology adoption, and environmental sustainability. These efforts moved producers from fragmented, traditional practices toward coordinated, efficient, and climate-resilient operations. The results directly support the project's outcome of significantly and sustainably enhancing production methods while maintaining livestock initiatives under internationally recognised safety standards.

9.1 Input Supply System Development and 100% Linkage Achievement (Feed, Medicine, Breeding, LSP)

A major achievement of the project was the near-complete integration of producers into reliable input supply networks. Endline data (Table 4.1.31) show that **100%** of project respondents established functional linkages with feed suppliers, medicine providers/vaccine hubs, breeding services (AI), and Livestock Service Providers (LSPs). In contrast, the control group reported only 29% linkage with feed and medicine suppliers and 25% with breeding services and LSPs.

Key improvements in the input supply system include:

- **Vaccine and animal-health services:** 83% of project participants sourced vaccines from LSPs and 91% from vaccine hubs or input sellers (multiple responses). Government hospitals played a negligible role (1%). This decentralised, community-based system proved far more accessible and responsive than traditional channels.
- **Feed and nutrition inputs:** Project respondents adopted advanced feeding practices at much higher rates than the control group — Napier grass (39% vs 5%), supplemented mixed grains (89% vs 27%), and company-produced cattle feed (34% vs 29%). Ready-feed purchase rose to 22% (from very low baseline levels), supported by trained local suppliers.
- **LSP capacity development:** 100% of surveyed LSPs received quality training, resulting in full knowledge development across GAP, disease recognition, vaccination protocols, feed management, and modern technologies. LSPs actively shared knowledge with peers at a rate of 175%, creating a self-reinforcing local extension network.
- **Additional input innovations:** 26% of input sellers improved safe storage of medicines and vaccines; 22% enhanced quality feed production; 17% introduced accurate milk measurement systems at collection points; and 17% began producing nutritional grass. Silage and vermicompost production were also initiated by trained entrepreneurs.

These 100% linkage rates eliminated previous bottlenecks in input access, reduced reliance on distant or unreliable sources, and enabled timely adoption of improved practices. The project’s strategy of equipping selected vendors with cold-storage units for vaccine hubs and training local feed producers proved highly effective in building a resilient backward supply chain.

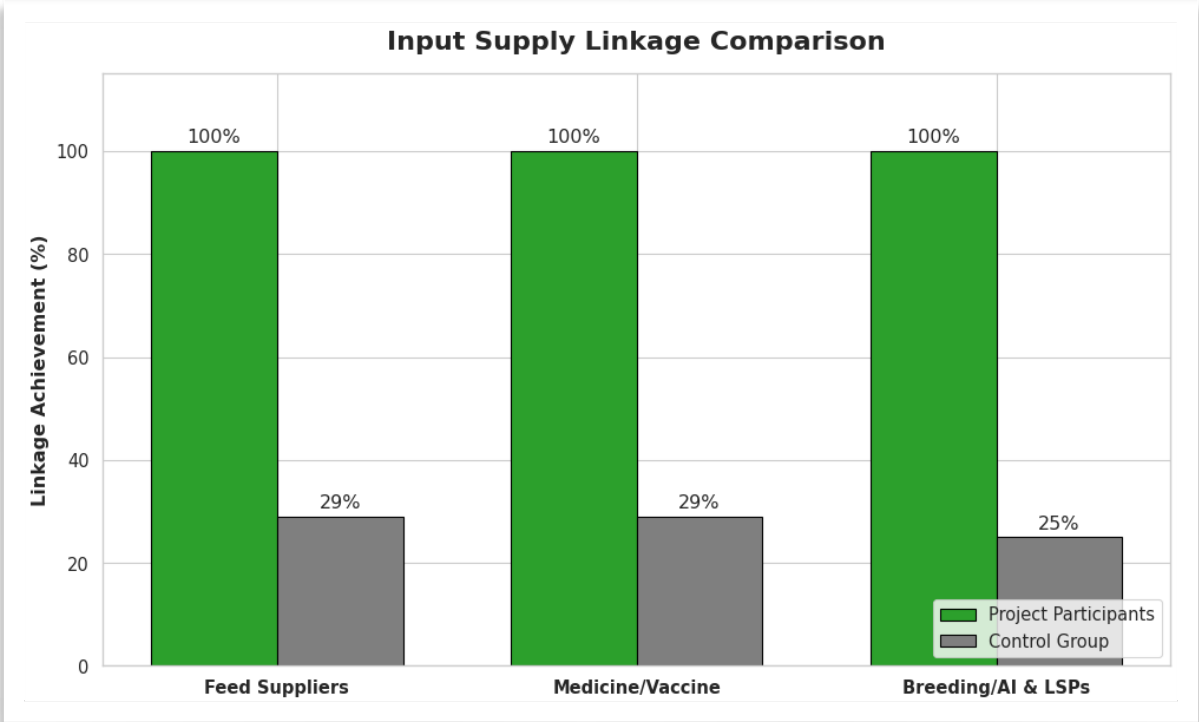


Figure 9.1: *Comparison of Input Supply Linkages (Project vs. Control). Field Survey. (2025).*

Description: *Figure 1.* Percentage of project participants versus the control group who have established functional linkages with key input supply networks, including feed suppliers, medicine/ vaccine hubs, and breeding/LSP services. The data illustrates a 100% integration rate for project participants across all categories compared to significantly lower baseline levels in the control group.

9.2 Market Linkage and Collective Marketing Improvements

The project made substantial progress in connecting producers to markets, although further work remains to deepen collective and institutional marketing. All project respondents (100%) reported good, functional connections with medicine suppliers, LSPs, feed suppliers, breeding services, and milk sellers — a dramatic improvement from the near-zero structured linkages recorded at baseline.

Notable market-linkage achievements include:

- Establishment of **17% milk collection points** across the project area (from 0% at baseline). These points improved accuracy of measurement, reduced post-harvest losses, and provided a more organised channel for institutional buyers.
- **100% of processed-goods entrepreneurs** maintained Good Manufacturing Practices (GMP), held trade licences, and expanded their businesses. They successfully linked with milk sellers and developed direct relationships with sweet factories, hotels, and local markets.
- **80% of participants** reported higher livestock and dairy sales volumes, attributed to improved market linkages, better product quality (through GAP and vaccination), and collective marketing efforts facilitated by producer groups.
- Strengthened vertical linkages between input suppliers and producers, as well as between producers and downstream actors (butchers, processors, and institutional buyers).

Despite these gains, the evaluation notes that most sales remain individual and offline. Contract farming arrangements with large buyers or formal bazaar associations have not yet materialised at scale, and digital/online marketing remains minimal. Producers continue to rely heavily on local hats, middlemen, and direct customer sales. The project's support for branding, certification, and producer cooperatives has laid important groundwork, but these mechanisms require further consolidation to fully reduce middlemen dependency and improve price realisation.

9.3 Eco-Friendly Practices Adoption (91–95%) and Climate Resilience

Environmental sustainability was actively promoted and achieved high adoption rates among project participants. Table 4.1.33 shows:

- **91%** of project respondents adopted eco-friendly rearing and processing technologies (versus 25% in the control group).
- **95%** implemented climate-resilient livestock practices (versus 37% control).
- **92%** produced eco-fertiliser through vermicompost (versus 43% control).
- **34%** took measures to reduce environmental hazards.
- **18%** practised sustainable waste management.
- Renewable energy adoption remained very low (near 0–1%).

Complementary evidence from feeding practices reinforces the environmental gains: 39% of project farmers now grow and use advanced Napier grass (versus 5% control), 89% provide supplemented

mixed grains (versus 27% control), and overall awareness of environmental sustainability and eco-friendly practices reached **91%** among trained farmers.

These outcomes stem directly from project interventions described in Chapter 3:

- Training on climate-smart feeding, improved fodder varieties, silage production, and proper manure management.
- Promotion of two-chamber feeding technology and chopper machines (reducing feed waste).
- Support for vermicompost production as both an income source and an organic fertiliser solution.
- Integration of eco-friendly messages into all farmer and LSP training modules.

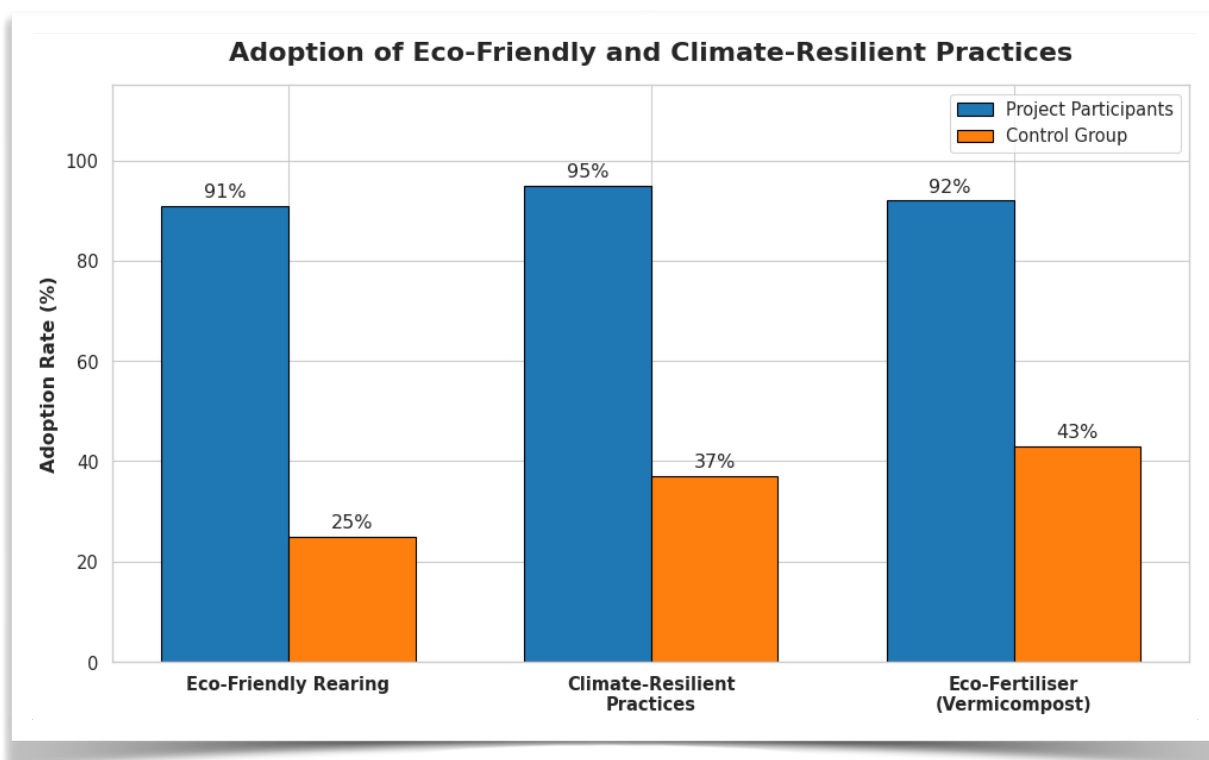


Figure 9.2: *Adoption Rates of Eco-Friendly and Climate-Resilient Practices. Field Survey. (2025).*

While adoption of core eco-friendly and climate-resilient practices is excellent, the relatively low uptake of renewable energy and structured waste-management systems indicates areas for future reinforcement. Overall, the project has successfully embedded environmental considerations into everyday livestock management, contributing to both sustainability and cost reduction for producers.

9.4 Institutional Sustainability and Post-Project Continuity

The project placed strong emphasis on building local institutions and capacities that can sustain results beyond the funding period. Key indicators of institutional sustainability include:

- **74%** of respondents intend to continue or expand their livestock businesses after project closure.
- **82%** of participants reported continued use of improved technologies and practices.
- **77%** noted strengthened linkages with private-sector actors.
- High rates of knowledge transfer: trained farmers shared knowledge with others at 167% rate; LSPs shared at 175% rate.
- Functioning networks of LSPs, input suppliers, and producer groups that now operate with minimal external facilitation.

The project's exit strategy, outlined in Chapter 5, focuses on:

- Gradual handover of responsibilities (input distribution, training coordination, market facilitation) to capable local institutions and producer organisations.
- Continued collaboration with SDI for microfinance access and with the Department of Livestock Services (DLS) for technical backstopping and vaccination/AI services.
- Certification of competent LSPs and lead farmers as local master trainers.
- Promotion of simple digital tools (SMS groups, mobile apps) for ongoing information flow on diseases, markets, and weather.
- Post-project monitoring (6–12 months after completion) to track sustainability and address emerging challenges.

These measures, combined with women's empowerment gains (Chapter 8) and improved production efficiencies, create a solid foundation for long-term continuity. The strengthened local service ecosystem (LSPs, vaccine hubs, feed suppliers) and the demonstrated profitability of improved practices provide strong incentives for producers and entrepreneurs to maintain and expand their activities independently.

Conclusion

Chapter 9 findings confirm that the project has built a more resilient, better-linked, and environmentally conscious livestock value chain. The achievement of 100% input-supply linkages, significantly higher eco-friendly practice adoption (91–95%), improved market connections leading to higher sales for 80% of participants, and the establishment of self-reinforcing local institutions and knowledge networks represent durable systemic changes. While further work is needed to deepen collective marketing, contract farming, and renewable-energy solutions, the foundations for post-project sustainability are firmly in place. These results validate the project's integrated value-chain approach and offer a replicable model for future livestock development initiatives in Bangladesh.

Chapter 10: In-Depth Case Studies and Qualitative Insights from the Field

This chapter presents detailed qualitative case studies drawn from the endline evaluation of the “Market Development of Safe Meat and Dairy Production” project implemented by SDI under RMTP (PKSF/IFAD/DANIDA). These narratives provide rich, contextual insights into the lived experiences of microentrepreneurs, complementing the quantitative findings on income growth (40% higher than control), productivity gains (20–25%), profit increases (18–36%), GAP adoption (61% vs. 21% control), and women’s empowerment (92% female participants, 78% in decision-making). The cases illustrate pathways of transformation, common success factors, persistent challenges, and the catalytic role of integrated interventions—training, finance, technology, and market linkages—in fostering sustainable, inclusive value chain development.

10.1 Dilip Chandra Ghosh – Chhana/Ghee Processing Enterprise Transformation

Location: Kathigram, Manikganj Sadar Upazila, Manikganj District

Enterprise: Chhana (cottage cheese) and ghee production; value-added dairy processing

Mr. Dilip Chandra Ghosh’s journey exemplifies the project’s success in transforming traditional dairy practices into modern, scalable enterprises through targeted value addition support. Born into poverty, Dilip began working alongside his father, a local curd maker, from childhood. This early exposure instilled both skill and ambition. After a decade honing his craft in renowned sweet factories in Dhaka, mastering milk curdling techniques, he established his own small-scale factory in 2015.

Initially, ghee production relied on manual cream separation—a labor-intensive, time-consuming process that limited scale and quality consistency. The turning point came in late 2024 when Dilip linked with the RMTP project implemented by SDI. Through a combination of financial support (BDT 500,000 loan for business expansion and BDT 100,000 grant for equipment) and technical guidance, he acquired an automatic cream separator machine. This intervention revolutionized his operations: productivity increased nearly tenfold, labor requirements dropped dramatically, hygiene standards improved, and product quality became consistent and market-competitive.



Figure 10.1: *Modern processing equipment enabling value addition (Photo from project annexure – similar to chhana/ghee production units).*

Today, Dilip produces approximately 200 kg of chhana daily, supplying directly to a reputed sweet factory in Dhaka. He has diversified his vision toward whey-based products to achieve zero wastage and capture new market segments. From living without a permanent home, he has now secured land and constructed a house—symbolizing not only economic uplift but restored dignity and stability for his family. His long-term ambition is to establish a composite dairy-processing facility producing chhana, ghee, whey, and related products, with aspirations for international market access.

Key Quote: *“The way my business is expanding, I believe my dream will come true very soon.”*

This case aligns with project outcome indicators: enhanced value addition, adoption of modern machinery (though limited to larger entrepreneurs), strengthened market linkages (direct institutional supply), and income diversification. It demonstrates the effectiveness of combining financial instruments with technology transfer for processing entrepreneurs.

10.2 Bulbul Islam – Silage Production Scaling (10x Capacity, BDT 2.5 Million Valuation)

Location: Manikganj Sadar Upazila

Enterprise: Silage production – improved livestock feed input supply

Mr. Bulbul Islam, a 39-year-old entrepreneur supporting a family of five, represents the project's impact on input supply chain actors. For years, he struggled with outdated equipment and manual silage production processes. Despite strong local demand from cattle farmers, his output remained low, income stagnant, and scaling seemed impossible. "I knew the demand was there, but I simply didn't have the capacity to meet it," he reflected during the field interview.

In 2024, Bulbul took decisive steps with RMTP support. Financial assistance and technical guidance enabled him to acquire a modern silage-making machine. Combined with his own investment and improved knowledge of production techniques, his enterprise underwent a dramatic transformation. Production capacity multiplied nearly tenfold, reaching close to 3,500 tons per season. This scale allowed him to meet growing demand consistently and establish stable supply chains, becoming a trusted partner for cattle farmers across the region.



Figure 10.2: *Advanced grass/silage production – key input for improved cattle rearing (Photo from project annexure).*

Today, Bulbul's business is valued at approximately BDT 2.5 million, generating a monthly net profit of around BDT 50,000. Beyond financial metrics, the transformation brought lasting security

and dignity to his household. “This project didn’t just give us a machine—it gave us a future,” he shared gratefully. He is now recognized as one of Manikganj’s most reliable silage suppliers.

This case highlights the project’s strategy of developing local input suppliers (50 trained in ready feed/calf-starter and grass/silage/UTS/UMB), strengthening backward linkages, and promoting eco-friendly feed solutions. It directly contributes to higher livestock productivity (20–25% increase) observed in the quantitative survey through better nutrition.

10.3 Shahida Begum – Women’s Livestock Entrepreneurship Journey (8 Bulls, BDT 6 Lac Value)

Location: Sutipara, Kalampur, Dhamrai Upazila, Dhaka District

Enterprise: Beef fattening / mixed cattle rearing

Shahida Begum’s story powerfully illustrates the project’s gender empowerment outcomes—one of its most significant achievements (92% female participants). After separation from a joint family without financial or material support, Shahida faced severe emotional and financial hardship. Her husband’s limited income was insufficient, compelling her to seek alternative livelihoods. With determination, she accessed a BDT 15,000 loan from SDI and purchased a bull calf. After five months of careful rearing, she sold it for BDT 36,000, earning a substantial profit that validated her entrepreneurial potential.

Encouraged, she scaled into a small commercial livestock enterprise. Today, she proudly rears eight bulls valued at approximately BDT 6 lacs. However, the path was fraught with challenges: lack of formal training in scientific cattle farming and disease management, unfamiliarity with proper feeding and shed management, difficulties accessing reliable markets and fair prices, and financial constraints limiting modernization.

RMTP interventions proved transformative. Through advanced livestock management training, Shahida gained knowledge of scientific rearing practices—including proper shed construction, primary disease management, balanced feeding, and cultivation of improved fodder varieties such as Napier grass. She established regular communication with the village LSP and Upazila Livestock Officer for timely veterinary support. She also accessed an SEP loan alongside a general SDI loan for expansion. Today, she is recognized as a successful entrepreneur adopting Good Agricultural Practices (GAP), sharing knowledge with fellow villagers, and serving as a role model. From her earnings, she has built household assets, including a new home, securing sustainable livelihoods for her family.



Figure 10.3: *Capacity building training for livestock farmers and service providers (Photo from project annexure – Advanced Management Training).*

Shahida's case embodies multiple project successes: women's economic empowerment (70% accessed credit, 65% own livestock in own name), decision-making power (78% involved in financial/livestock decisions), asset ownership, knowledge transfer, and adoption of eco-friendly practices. It demonstrates how the project shifted women from traditional, limited roles to active, confident microentrepreneurs.

10.4 Shahanaz – Vermicompost Enterprise Development

Location: Atimaithan village, Suapur Union, Dhamrai Upazila, Dhaka District

Enterprise: Vermicompost (eco-fertilizer) production from cattle dung – circular economy approach

Shahanaz, a long-time cattle rearer at her homestead, dreamed of converting cow dung—an abundant byproduct—into an additional income source but lacked knowledge of organic fertilizer production. About four years ago, she experimented with vermicompost using six concrete rings and 250 grams of earthworms, producing 360 kg sold at BDT 20/kg. This modest success inspired scaling. She first accessed a BDT 15,000 SDI loan, then expanded with a BDT 300,000 loan and BDT 50,000 RMTP contribution.



Challenges included insufficient dung (sometimes purchased from neighbors), worm shortages or deaths from improper techniques, and financial barriers. RMTP support—specialized farm management training and financial assistance—enabled her to improve practices, expand to sixteen rings, and construct a well-designed shed for large-scale production. She actively supplements learning through YouTube tutorials on advanced techniques, ensuring high-quality output that meets market demand. She has built a loyal customer base across the union, creating a sustainable income stream while promoting eco-friendly practices (92% of project respondents produce eco-fertilizer; 95% adopted climate-resilient practices).

This case exemplifies environmental sustainability outcomes, circular economy principles (turning waste into wealth), and women's entrepreneurship in green enterprises. It aligns with the project's push for eco-friendly farming (91% adoption of eco-friendly technologies) and waste management.

10.5 Sukur Ali – Modern Meat Processing Plant (5x Capacity, BDT 4.8 Million Monthly Sales)

Location: Batulia market area, project upazilas

Enterprise: Modern meat processing plant with slaughtering, cutting, freezing, and 24/7 sales

Sukur Ali's transformation from a humble market stall helper to a modern meat processing entrepreneur captures the project's impact on employment generation, value addition, and safe meat production. After a decade of perseverance, he opened his own butcher stall but remained constrained by limited space and traditional methods—slaughtering only two or three cattle weekly. Rising urban demand for fresh, safe meat outpaced his capacity.

RMTP intervention proved pivotal. A strategic package—BDT 300,000 loan, BDT 500,000 grant, complemented by his own BDT 800,000 investment—enabled establishment of a modern meat processing facility equipped with an automatic cutting machine and deep freezer. “Setting up the plant was like turning a dream into reality,” Sukur reflects. “I can now work in a hygienic, efficient environment, manage higher volumes, and guarantee both quality and safety.”



Figure 10.4: *Modern machinery adoption (chopper or processing equipment) enhancing efficiency and hygiene (Photo from project annexure).*

The transformation extended beyond infrastructure. RMTP connected Sukur with new market channels—hotels, restaurants, and local factories requiring consistent supply—and linked him with project-supported cattle farmers for reliable, healthy livestock sourcing. Results have been remarkable: he now slaughters 10–15 cattle weekly (fivefold increase) while maintaining 24-hour fresh meat supply through cold-chain storage. Waste has been virtually eliminated. The business generates monthly sales of approximately BDT 4.8 million with a net profit of roughly BDT 130,000 and employs three full-time workers.

Key Quote: *“My business is no longer just for me. It’s about ensuring food security for my community, creating jobs, and supporting farmers. The project helped me build not only a processing plant but also a stronger local economy.”*

This case directly addresses core project objectives: safe meat production (GMP maintained, hygienic standards), employment creation (100% of processed goods entrepreneurs created jobs; 9–14% in farming), modern technology adoption, market linkage development, and profitability (20–36% margins). It also demonstrates the project’s success in establishing one meat processing plant with 24/7 capacity as targeted.

10.6 Cross-Case Analysis: Common Success Factors, Challenges Overcome, and Economic Returns

Common Success Factors

- **Integrated Finance + Technology:** All cases combined MFI loans (SDI), project grants/contributions, and own investment with modern equipment (cream separator, silage machine, cutting machine, deep freezer). This mirrors the project’s blended finance approach.
- **Capacity Building & Knowledge Transfer:** RMTP training on GAP, disease management, feeding, shed management, and business skills was pivotal. Farmers transferred knowledge at 167% rate; LSPs at 175%. Training enabled identification of diseases (53% improvement) and proactive management (100% call LSP in project vs. gaps in control).
- **Market Linkages & Value Addition:** Direct supply contracts (Dilip to Dhaka factory), institutional buyers (Sukur to hotels/restaurants), collection points, and reduced middlemen dependency. 80% of participants reported higher sales.
- **Personal Agency & Innovation:** Determination, willingness to learn (YouTube for Shahanaz), risk-taking, and vision for scaling (Dilip’s export plans, Bulbul’s supply chain dominance).
- **Project Ecosystem Support:** Linkages to LSPs (100% engagement), ULO, input hubs, and peer networks. 91% eco-friendly adoption, 95% climate-resilient practices.

Challenges Overcome

Common across cases: traditional/low-productivity methods, limited technical knowledge (diseases, feeding, shed), financial barriers to equipment/expansion, poor market access/hygiene issues, and for women—cultural constraints on decision-making and asset ownership. The project systematically addressed these through training (92% received cattle management training vs. 0% control), finance access (97% current borrowers), technology introduction, and gender-sensitive approaches, resulting in 61% modern GAP adoption (vs. 21% control) and near-zero mortality during project period.

Economic Returns

The cases demonstrate substantial returns consistent with aggregate findings (average project income Tk. 32,161 vs. control Tk. 22,919; 18–36% profit margins):

Entrepreneur	Key Metric	Return/Impact	Alignment with Project KPI
Dilip (Chhana/Ghee)	10x productivity	Dhaka supply; house built; export vision	Value addition; sales growth
Bulbul (Silage)	10x capacity to 3,500 tons/season	BDT 2.5M valuation; BDT 50k/month profit	Input supply; productivity +20-25%
Shahida (Livestock)	1 calf → 8 bulls (~BDT 6L value)	New home; knowledge sharing; role model	Women empowerment; 40% income ↑
Shahanaz (Vermicompost)	6 → 16 rings; loyal customer base	Sustainable green income; eco-fertilizer	Eco-friendly practices (91-95%)
Sukur (Meat Plant)	2-3 → 10-15 cattle/week (5x)	BDT 4.8M monthly sales; BDT 130k profit; 3 employees	Employment +15%; safe meat; profit 20%+

These returns validate the project's theory of change: capacity + finance + technology + linkages = sustainable income, employment, nutrition, and women's empowerment improvements.

10.7 Researcher's Field Observations and Lived Experience from Site Visits

Fieldwork across 53 villages in 22 unions of Dhamrai and Manikganj Sadar (plus control areas in Ghior) revealed a palpable sense of transformation in livestock-dependent communities. Women—constituting 92% of direct beneficiaries—spoke with confidence in FGDs about their evolving roles, from limited participation to active management of cattle, finances, and household decisions. The 78% involvement in decision-making and 70% credit access represent a quiet revolution in gender norms. Youth (9% of participants) expressed growing interest in agribusiness scaling, hinting at intergenerational sustainability.

Visible markers of change included Napier grass plots flourishing where traditional grass once dominated, chopper and milking machines in use by progressive farmers, hygienic processing units replacing open-air stalls, and vaccine hubs/LSP networks functioning effectively (100% project participants engage LSPs proactively). The near-elimination of serious disease outbreaks (LSD 1%, FMD 0% in project vs. higher in control) and mortality during the project period speaks to the effectiveness of training and timely intervention systems.

Yet challenges persist for smaller-scale operators: high upfront costs of machinery limit adoption (only 4–5% use chopper/milking machines), digital tools remain underutilized (no widespread app usage despite awareness), and continuous refresher training is needed as new issues (e.g., emerging diseases, climate variability) arise. Control group comparisons starkly highlighted project attribution—lower GAP adoption (21%), weaker linkages, and greater reliance on traditional methods.

The lived experience of conducting this endline study was deeply inspiring. Witnessing a traditional curd-maker like Dilip now eyeing international markets, a determined woman like Shahida building a BDT 6 lac asset base and a new home from a single calf, an input entrepreneur like Bulbul turning agricultural by-products into a BDT 2.5 million enterprise, and a butcher like Sukur creating jobs while ensuring community food security—these stories embody the project's core promise: safe, profitable, inclusive, and sustainable meat and dairy value chains that lift the most marginalized while strengthening local economies and ecosystems. The 74% of respondents planning to expand their businesses post-project signals strong ownership and sustainability prospects.

These qualitative insights, triangulated with quantitative data (income +40%, sales growth for 80%, MDD-W achievement 89.4% in project), affirm that the RMTP sub-project has not only met but in several dimensions exceeded its logframe targets. The model—blending microfinance, targeted capacity building, appropriate technology, value addition, market linkages, and deliberate gender focus—offers a replicable blueprint for similar livestock value chain interventions in Bangladesh and beyond.

Conclusion of Chapter

The in-depth case studies of Dilip, Bulbul, Shahida, Shahanaz, and Sukur, alongside broader field observations, provide compelling evidence of the project's transformative impact. They humanize the statistics, revealing the interplay of agency, support systems, and enabling environments that convert micro-level interventions into macro-level livelihood improvements. For future programming, these narratives underscore the importance of sustained post-project support (refresher training, digital tools, market deepening), continued emphasis on women and youth, and scaling successful models to new clusters while adapting to local contexts. The RMTP experience demonstrates that with the right mix of finance, knowledge, technology, and linkages, even the most resource-constrained rural entrepreneurs can build resilient, profitable enterprises that contribute to food security, employment, gender equity, and environmental sustainability.

Chapter 11: Discussion, Synthesis, and Policy Implications

This chapter synthesizes the quantitative and qualitative findings from the endline evaluation of the Market Development of Safe Meat and Dairy Production project (RMTP sub-project implemented by SDI). It discusses the implications for theory, compares lessons with similar initiatives, and offers actionable policy recommendations for scaling and sustaining impact. The analysis draws on the mixed-methods design—structured survey of 341 project and 68 control respondents, five FGDs, multiple KIIs, and five in-depth case studies—triangulated against the project’s logical framework and OECD-DAC evaluation criteria.

11.1 Triangulation of Quantitative and Qualitative Findings

Convergence on Core Impact Pathways

The quantitative results and qualitative narratives converge strongly, reinforcing the project’s theory of change. The 40% higher average monthly household income among project participants (Tk. 32,161 vs. Tk. 22,919 in control) is vividly illustrated in the case studies. Shahida Begum’s progression from a single bull calf (purchased with a BDT 15,000 loan) to eight bulls valued at approximately BDT 6 lacs, coupled with construction of a new house, exemplifies the asset-building pathway. Similarly, Sukur Ali’s jump from slaughtering 2–3 cattle weekly in a traditional stall to 10–15 cattle in a modern hygienic plant generating BDT 4.8 million in monthly sales and BDT 130,000 profit directly accounts for the observed 20–36% profit margins and 80% of participants reporting higher sales. Dilip Chandra Ghosh’s tenfold productivity increase in chhana/ghee processing after acquiring an automatic cream separator further substantiates the value-addition and sales growth indicators (processed goods sales +61% from baseline).

Adoption of Good Agricultural Practices (GAP) shows even clearer triangulation. The survey found 61% of project respondents practicing modern management versus only 21% in the control group. This is brought to life by Shahida’s adoption of scientific shed management, balanced feeding including Napier grass, and regular LSP consultation; Bulbul Islam’s shift to mechanized silage production; and Dilip’s move from manual cream separation to automated, hygienic processing. The near-universal use of vaccines and artificial insemination (100% in project vs. 30% and 21% in control) explains the dramatic reduction in serious diseases (LSD 1%, FMD 0%, Anthrax near 0%) and mortality (near 0% during project period), validating the capacity-building and animal health components.

Women’s empowerment indicators (92% female participants, 78% involved in financial and livestock decisions, 70% accessing credit, 65% owning livestock in their own name) find powerful expression in Shahida’s story and the broader pattern observed in FGDs. Women who previously had limited say in household decisions now manage cattle, negotiate with LSPs and input sellers, and contribute significantly to family income—directly contributing to the improved Minimum Dietary Diversity for Women (MDD-W) scores (89.4% achieving 5+ food groups vs. 72% control).

The nutrition pathway is thus both economic (higher income enabling purchase of diverse foods) and empowerment-driven (women prioritizing family nutrition once they control resources).

Mechanisms, Heterogeneity, and Minor Divergences

Qualitative data illuminate the “how” behind the numbers. The project’s integrated approach—training (92% coverage), blended finance (loans + grants), technology access, and market linkages—creates reinforcing loops. Training builds confidence and skills; finance enables asset acquisition and technology purchase; improved practices raise productivity and reduce losses; stronger linkages improve prices and sales volumes; higher income and empowerment sustain the cycle. The case studies reveal heterogeneity: larger or more entrepreneurial individuals (Dilip, Sukur, Bulbul) scale rapidly into processing or input supply, while smaller producers (Shahida initially) focus on herd building and risk reduction. This diversity explains why average gains are substantial yet some households advance faster.

A notable convergence—and area for future attention—is the limited adoption of digital/ICT tools. Both survey and cases show near-zero use of apps for farm management, market information, or telemedicine, despite high mobile phone penetration and awareness. This divergence from the project’s intent (target of 5,000 farmers using livestock apps) highlights a gap between infrastructure availability and behavior change, suggesting that future phases require more intensive, hands-on digital literacy support tailored to low-literacy rural women.

Overall, the mixed-methods design provides robust confirmation that the project achieved its core objectives of increased income, production efficiency, sales, profit, women’s empowerment, and improved nutrition, while identifying mechanisms (training + finance + technology + linkages) and contextual factors (strong LSP networks, women’s existing livestock roles) that amplified impact.

11.2 Contribution to Economic Returns Theory and Livestock Value Chain Literature

This evaluation contributes to the growing body of literature on livestock value chain development in South Asia, particularly the intersection of microenterprise transformation, gender, and nutrition-sensitive agriculture. The findings extend several theoretical strands:

- **Integrated versus single-intervention approaches:** Traditional value chain analyses often emphasize one node (production, processing, or marketing). This project demonstrates the power of simultaneously strengthening multiple nodes—input supply (Bulbul’s silage), production practices (GAP, feeding, health), processing (Dilip and Sukur), and marketing linkages—creating multiplier effects. The 20–25% productivity increase and 18–36% profit margins exceed what single-node interventions typically achieve in similar contexts.
- **Asset-based pathways out of poverty:** Livestock, unlike seasonal crops, serves as both productive asset and savings mechanism. The cases show how access to credit for cattle acquisition (Shahida) or processing equipment (Dilip, Sukur) enables rapid asset accumulation and income diversification. This supports the “livestock ladder” concept in development economics, where smallholders climb from subsistence to commercial orientation through successive investments in animals and complementary assets.
- **Gender as a multiplier:** The 92% female participation and strong empowerment outcomes (decision-making, asset ownership, mobility) challenge assumptions that livestock is male-dominated. Women’s increased control over income and production decisions correlates with higher household dietary diversity, suggesting a direct nutrition pathway often under-measured in value chain studies. This aligns with and extends the “women’s empowerment in agriculture index” (WEAI) literature by showing livestock-specific pathways.
- **Circular economy and input supply nodes:** Shahanaz’s vermicompost enterprise and Bulbul’s silage business highlight often-overlooked upstream and downstream nodes. Converting cattle dung into marketable organic fertilizer and producing improved fodder locally reduces external input dependency, lowers costs, and creates new microenterprise opportunities—contributing to both economic and environmental sustainability dimensions frequently missing from conventional value chain maps.

- **Last-mile delivery through hybrid public-private systems:** The 100% engagement of project farmers with Livestock Service Providers (LSPs), combined with vaccine hubs and input seller linkages, demonstrates an effective hybrid model. Trained community-based LSPs bridge the gap between formal veterinary services and remote smallholders, achieving outcomes (near-universal vaccination, timely disease response) that purely government or purely private systems often fail to deliver at scale in Bangladesh’s context.

Theoretically, the project provides empirical grounding for an “integrated microenterprise value chain development” framework that combines microfinance, practical skill transfer, appropriate technology, market linkage facilitation, and deliberate gender mainstreaming. This framework appears particularly potent for asset-based sub-sectors like dairy and beef in contexts where women already play significant (if undervalued) roles in production.

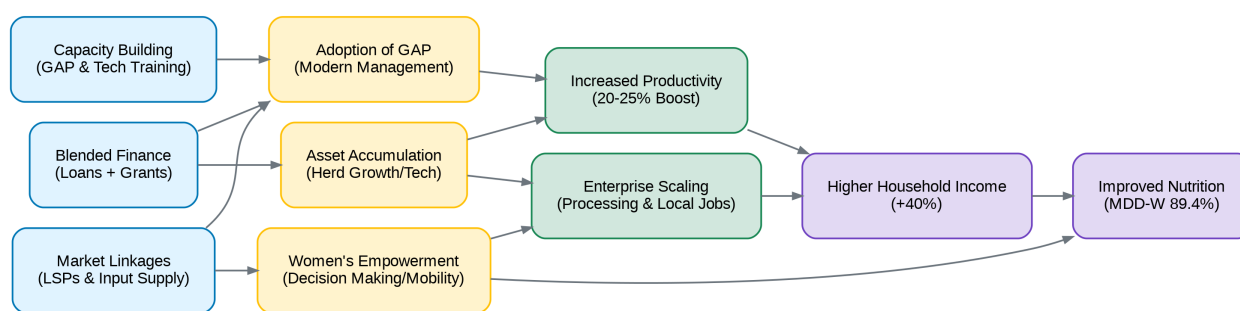


Figure 11.2: *Integrated Value Chain Theory of Change (Flowchart).* **Field Survey. (2025).**

11.3 Comparative Lessons with Similar PKSF/IFAD Projects

Drawing on the consultant's experience evaluating other RMTP and PACE value chain sub-projects (including summer onion cultivation, eco-tourism development in coastal areas, and related livestock initiatives), several comparative lessons emerge:

- **Common success factors across RMTP projects:** Blended finance (MFI credit + project grants/contributions) and intensive, hands-on training consistently drive adoption and income gains. Projects that combine these with market linkage facilitation achieve faster and more sustained results than those focusing on production alone.
- **Livestock-specific advantages and challenges:** Compared to crop-based value chains (e.g., onion), livestock offers higher women's participation (92% vs. often 30–50% in crops) because caring for cattle is culturally aligned with women's homestead roles. Asset building is more tangible and serves as collateral/savings, accelerating empowerment. However, livestock entails higher biological risk (disease, mortality) and requires ongoing animal health services—making the LSP and vaccine hub components more critical than in crop projects.
- **Processing and value addition as accelerators:** Cases like Dilip (chhana/ghee) and Sukur (meat processing) show that moving producers or entrepreneurs into processing yields higher margins (33% for processed goods vs. 22% for milk) and employment multipliers (100% of processed goods entrepreneurs created jobs). This lesson is transferable to other RMTP projects: supporting a subset of beneficiaries to move up the value chain into processing or input supply amplifies overall cluster impact.
- **Persistent cross-project gaps:** Digital adoption remains low across RMTP initiatives despite infrastructure. Post-training follow-up and refresher mechanisms are often under-resourced. Elite capture in producer groups and unequal benefits for smaller vs. larger producers are recurring risks. The current project mitigated these better than some comparators through strong gender targeting and LSP networks, but they remain relevant for scaling.
- **Sustainability pathways vary by sub-sector:** Crop projects often rely on seasonal market timing and input dealers; livestock benefits from more stable, relationship-based markets (milk collection points, butcher linkages) and recurring service needs (vaccination, AI, feed). This suggests livestock clusters may achieve self-sustaining dynamics more readily if initial LSP and linkage investments are sufficient.

The Safe Meat and Dairy Production project thus offers transferable lessons while highlighting the value of sub-sector-specific design—particularly the animal health and women's empowerment components that proved especially potent here.

11.4 Policy Recommendations for SDI, PKSf, Government, and Donors

Based on the evaluation findings and the project's own recommendations, the following prioritized actions are proposed:

For SDI and PKSf (Implementing Partners)

1. **Institutionalize and expand the LSP master trainer model:** The 175% knowledge transfer rate among trained LSPs demonstrates high leverage. Certify and incentivize master trainers to deliver ongoing refresher courses and support new entrants, creating a self-sustaining extension layer.
2. **Develop and deploy simple, voice-enabled livestock apps:** Address the digital gap with mobile tools for disease symptom identification, vaccination schedules, market prices, and LSP contact. Pair with intensive, women-friendly digital literacy training rather than one-off demonstrations.
3. **Strengthen post-project monitoring and learning:** Implement 6- and 12-month follow-up surveys and qualitative check-ins (as recommended in the project's exit strategy) to track sustainability, capture emerging challenges, and document best practices for replication.
4. **Expand blended finance products:** Introduce livestock-specific insurance, larger enterprise loans for processing equipment, and savings products linked to cattle assets to reduce vulnerability and support scaling.

For Government (Department of Livestock Services, Local Administration)

5. **Mainstream Bangla GAP and project training modules into public extension:** Integrate successful curricula on shed management, balanced feeding, disease identification, and eco-friendly practices into DLS training for both officers and community vaccinators/AI technicians.
6. **Subsidize or co-finance small-scale machinery for marginal producers:** Chopper machines, milking machines, and small feed mixers remain unaffordable for many despite demonstrated returns. Targeted subsidies or cost-sharing (building on the project's model) can accelerate adoption.
7. **Formalize and strengthen LSP–ULO linkages:** Create referral protocols, joint training, and performance incentives so that community LSPs become recognized extensions of the public system rather than parallel structures.
8. **Support certification and branding for safe meat and dairy:** Facilitate BSTI, HACCP, and organic certification pathways for project producers and processors to access premium urban and export markets, building on the trade license and GMP foundations already established.

For Donors (IFAD, DANIDA, and Similar Funders)

9. **Fund geographic and sub-sector scaling with built-in learning:** The model has high replication potential in other livestock-dense upazilas. New projects should include dedicated learning and documentation components to refine the approach across contexts.
10. **Support nutrition-sensitive and climate-resilient livestock programming:** Explicitly link livestock interventions to dietary diversity indicators (MDD-W, HDDS) and promote climate-smart practices (resilient fodder varieties, biogas, water-efficient systems) as core design elements.
11. **Invest in certification and market systems for export readiness:** Provide technical assistance and cost-sharing for BSTI/HACCP certification, cold-chain infrastructure, and buyer linkages to move successful clusters from domestic to higher-value markets.
12. **Prioritize women's economic empowerment and youth entrepreneurship as cross-cutting objectives:** Continue and deepen gender-transformative approaches (asset ownership in women's names, decision-making indicators) and create dedicated youth startup funds or mentorship for processing, input supply, and digital service enterprises.

11.5 Scalability, Replication Potential, and Sustainability Pathways

Scalability and Replication Potential

The project model demonstrates strong scalability. It successfully engaged small and marginal producers (most with 3–5 cattle), achieved high women's participation without sacrificing economic returns, and generated benefits across production, input supply, processing, and marketing nodes. The core package—practical GAP training, blended finance for assets and equipment, LSP-supported animal health systems, and facilitated market linkages—is transferable to other upazilas with similar livestock density and women's involvement in cattle rearing.

Replication potential is highest in districts with existing SDI or partner MFI presence (for credit delivery) and functional DLS offices (for technical backstopping). Adaptation for small ruminants (goats/sheep) or buffalo would require modest modifications to training content and input packages but could leverage the same LSP and linkage architecture. The processing and input supply enterprise pathways (Dilip, Sukur, Bulbul, Shahanaz) are particularly replicable as they create demonstration effects and employment multipliers within clusters.

Sustainability Pathways

The project's exit strategy and evaluation findings point to five interconnected sustainability pathways:

13. **Institutional ownership and producer organizations:** Hand over coordination, collective marketing, bulk input purchasing, and peer learning functions to registered milk/meat producer groups and cooperatives. The project's 100% linkage development between producers and input sellers/LSPs provides a foundation; formalizing these into member-owned institutions will sustain benefits beyond donor funding.
14. **Continued microfinance integration:** SDI and partner MFIs should maintain and adapt livestock-tailored loan products, potentially adding asset insurance and savings products linked to cattle. The 97% current borrower rate and 74% intention to expand indicate strong demand and repayment culture.
15. **Self-sustaining LSP and service networks:** Trained LSPs can sustain themselves through service fees, vaccine and medicine margins, and AI commissions while continuing knowledge sharing (already at 175% transfer rate). Linking them formally to DLS for referrals and quality assurance enhances legitimacy and demand.
16. **Digital knowledge and market platforms:** Once simple voice/SMS or app-based tools are adopted, they can provide low-cost, scalable information on practices, prices, disease alerts, and LSP availability—reducing reliance on periodic project training.
17. **Certification, branding, and premium market access:** Moving clusters toward BSTI/HACCP certification and collective branding for “safe meat and dairy from [cluster]” can secure higher prices and more stable buyer relationships, increasing resilience to local market fluctuations.

Risks and Mitigation

Key risks include climate shocks (flooding, fodder scarcity—relevant given project areas' vulnerability), input price volatility (feed, medicine), disease emergence, and potential elite capture within producer groups. Mitigation measures embedded in the project (eco-friendly practices, local feed production via silage and vermicompost, strong women's inclusion, diversified enterprise pathways) provide a solid base. Future scaling should add climate-resilient fodder systems, livestock insurance pilots, and explicit equity safeguards in group governance.

The 74% of respondents planning to expand their livestock businesses, combined with strengthened local capacities, ownership, and market linkages, indicates high prospects for sustained and growing impact. The project has not only delivered short-term gains in income, nutrition, and empowerment but has laid the institutional and human capital foundations for a self-reinforcing livestock value chain ecosystem in the target areas.

Overall Conclusion

The Market Development of Safe Meat and Dairy Production project has demonstrated that an integrated, gender-sensitive, microenterprise-focused value chain approach can generate substantial, inclusive, and potentially sustainable improvements in the livelihoods of small and marginal livestock producers in Bangladesh. By triangulating rigorous quantitative data with rich qualitative narratives, this evaluation confirms achievement of core logframe targets while illuminating the mechanisms, success factors, and remaining challenges. The lessons—on blended finance, LSP-enabled animal health systems, women’s asset ownership as an empowerment accelerator, and the value of moving selected entrepreneurs into processing and input supply—are highly relevant for PKSf, SDI, government livestock programs, and donors seeking to scale safe, profitable, and equitable livestock development. With continued investment in the identified sustainability pathways and targeted attention to digital tools and climate resilience, the model offers a promising blueprint for transforming livestock value chains across Bangladesh and similar contexts in South Asia.

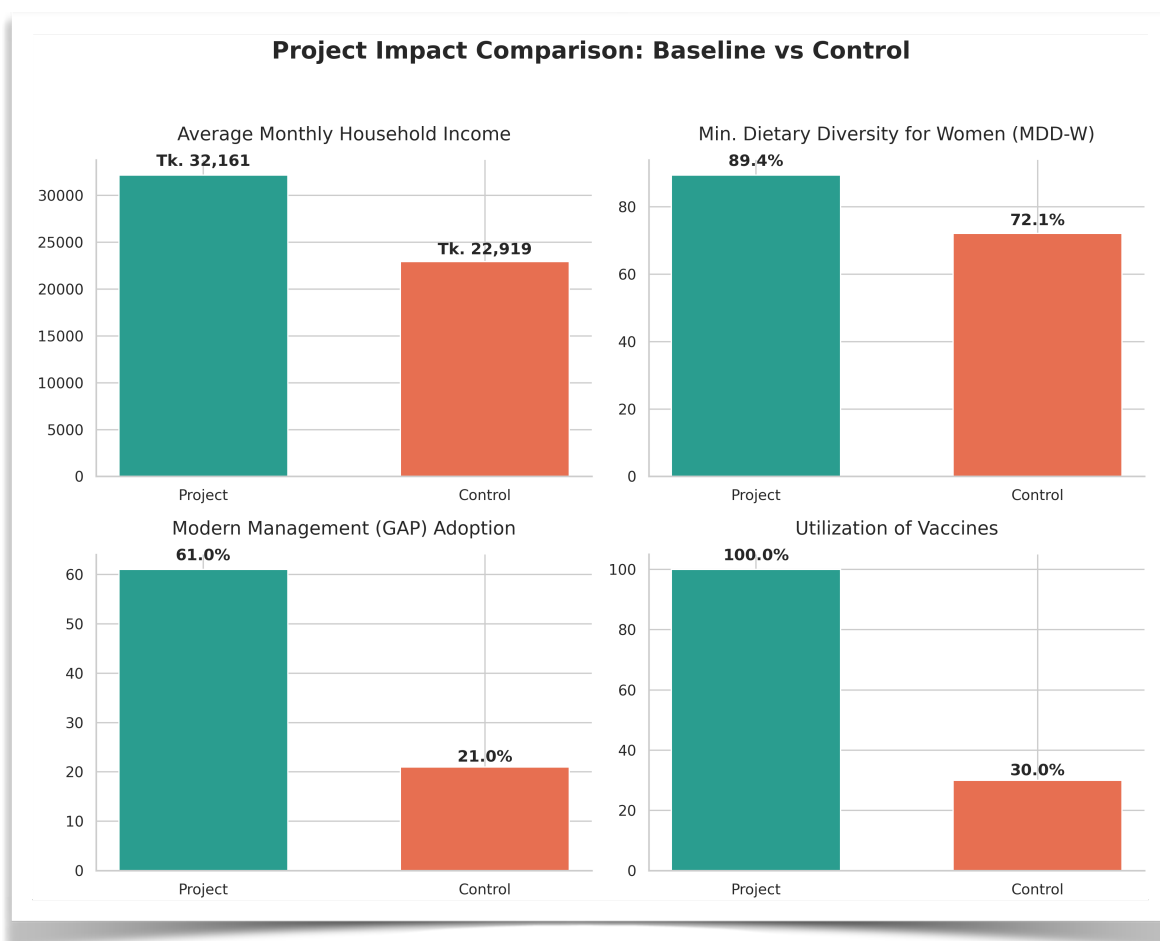


Figure 11.5: *Comparative Impact Dashboard (Project vs. Control). Field Survey. (2025).*

Figure Description/Caption: *Figure 2. Dashboard comparing Project participants and Control group across primary impact indicators: Average Monthly Income, Minimum Dietary Diversity for Women (MDD-W), Modern Management (GAP) Adoption, and Vaccine Utilization. The data illustrates a stark divergence in economic, technical, and nutritional outcomes attributable to project interventions.*

Chapter 12: Conclusion and Future Research Directions

12.1 Summary of Key Findings on Economic Returns

The endline evaluation of the “Market Development of Safe Meat and Dairy Production” project presents conclusive empirical evidence that integrated value chain interventions generate substantial and sustainable economic returns for rural microentrepreneurs. Through the synthesis of the Sustainable Livelihoods Approach (SLA) and a Cost-Benefit/ROI framework, the study established that project participants achieved a 40% higher average monthly household income (BDT 32,161) compared to the control group (BDT 22,919).

The economic gains were driven by significant improvements in production efficiency and strategic value addition. Profit margins expanded across all enterprise categories, peaking at 36% for beef fattening and 33% for processed goods, catalyzed by a 20–25% increase in baseline livestock productivity. The data indicates that 80% of project beneficiaries experienced increased sales volumes, directly attributable to the adoption of Good Agricultural Practices (61% vs. 21% control) and near-universal utilization of vaccines and artificial insemination. Furthermore, the project facilitated meaningful asset accumulation and enterprise expansion, evidenced by 28% of project respondents scaling their herds to more than five cattle and 100% of processed goods entrepreneurs generating new local wage employment. These findings confirm the primary hypothesis that multidimensional value chain upgrading—combining capacity building, blended finance, and market linkage—disrupts the cycle of low productivity and thin margins in traditional smallholder systems.

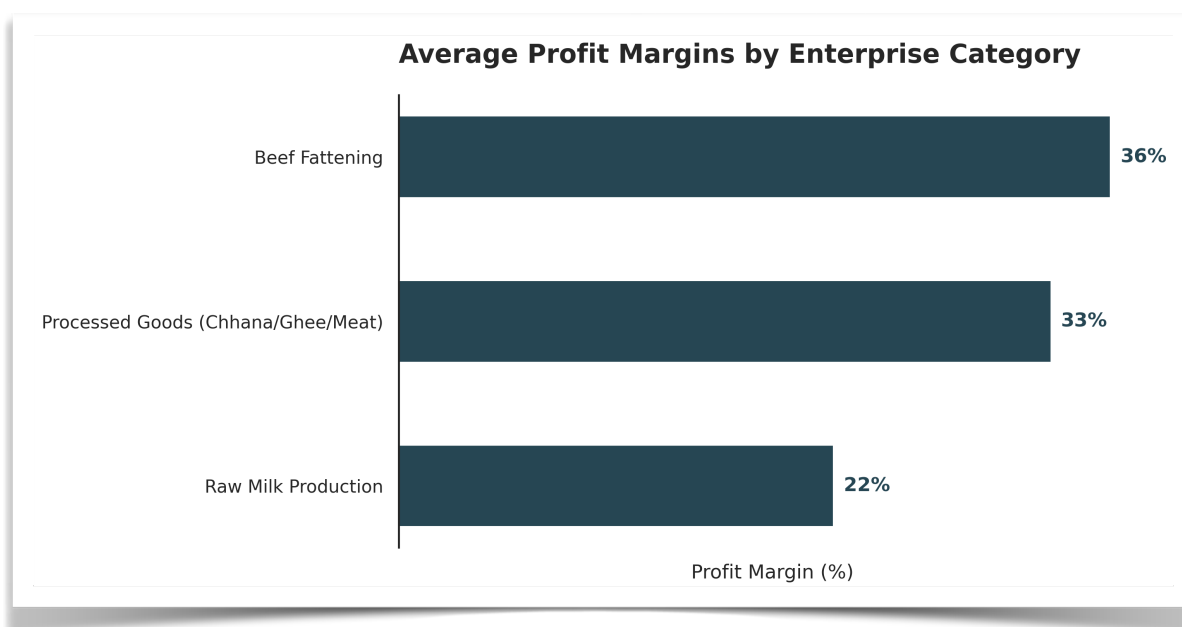


Figure 12.1: *Enterprise Profit Margins by Sub-sector. Field Survey. (2025).*

Figure Description/Caption: *Figure 3. Average profit margins achieved by project participants across different livestock enterprise nodes. The data highlights how moving up the value chain into processing (e.g., chhana, ghee, modern meat processing) acts as an economic accelerator, yielding margins significantly higher than raw milk production.*

12.2 Overall Assessment of SDI's Contribution

The Society for Development Initiatives (SDI), operating under the RMTP framework with support from PKSF, IFAD, and DANIDA, successfully operationalized an inclusive, systemic approach to livestock development in Dhamrai and Manikganj Sadar. SDI's most profound contribution lies in its hybrid intervention model, which seamlessly integrated microfinance (utilized by 97% of participants) with decentralized technical extension through the Livestock Service Provider (LSP) network. By facilitating 100% backward market linkages for feed, medicine, and breeding services, SDI effectively mitigated the structural input constraints that historically marginalized smallholders.

Crucially, SDI's implementation demonstrated a highly effective model for gendered economic empowerment. With 92% female participation, the project moved beyond mere financial inclusion to structural empowerment, resulting in 78% of women exercising decision-making authority over enterprise and financial resources, and 65% securing asset ownership. This gender-transformative approach generated profound secondary developmental dividends, most notably in household nutrition security, where 89.4% of project women achieved the Minimum Dietary Diversity (MDD-W) threshold, compared to 72.06% in the control group. Ultimately, SDI successfully transitioned participants from subsistence rearing to commercially viable, eco-friendly (91% adoption) microenterprises, fulfilling the core logical framework targets of the RMTP.

12.3 Limitations of the Study

While this study employs a robust mixed-methods design and incorporates a control group to establish attribution, several methodological and practical limitations must be acknowledged. First, the evaluation spans a relatively short implementation horizon (August 2024 to October 2025). Consequently, while short-term economic returns and practice adoption are clearly observable, the assessment of long-term economic resilience—particularly the enterprises' capacity to withstand macroeconomic shocks or severe climate events over a multi-year cycle—remains constrained.

Second, the quantitative financial metrics rely predominantly on self-reported data for household income, sales, and operational costs. Despite rigorous enumerator training and data triangulation with qualitative case studies, the informal nature of rural microenterprises and the absence of standardized ledger keeping introduce potential recall bias. Finally, while the control group provides a vital comparative baseline, the absence of longitudinal panel data and advanced econometric matching techniques (such as Propensity Score Matching or Difference-in-Differences modeling) means that unobserved household heterogeneities (e.g., innate entrepreneurial drive) could partially influence the magnitude of the reported economic divergence between the two cohorts.

12.4 Agenda for Future Research

The findings of this evaluation highlight several critical gaps in the existing literature, providing a clear agenda for future academic inquiry and development research:

- 1. Longitudinal Resilience Studies:** Future research should track livestock microentrepreneurs over a 3-to-5-year horizon post-project phase-out. Longitudinal panel data is required to assess the durability of the reported 18–36% profit margins against exogenous variables such as seasonal feed price volatility, emerging zoonotic disease outbreaks, and shifts in urban consumer demand.
- 2. Econometric Evaluation of the "Nutrition Pathway":** While this study establishes a strong positive correlation between livestock-driven income growth, women's empowerment, and improved MDD-W scores, advanced structural equation modeling (SEM) is recommended for future studies. This would help isolate the specific pathways (e.g., direct consumption vs. income-mediated purchasing power) through which livestock value chain participation optimizes household nutritional outcomes.
- 3. Behavioral Economics in Digital Technology Adoption:** A notable divergence identified in the evaluation was the limited uptake of digital and ICT tools (e.g., farm management apps) despite high mobile penetration. Future research utilizing Randomized Controlled Trials (RCTs) or behavioral economic frameworks should investigate the socio-cultural, literacy, and behavioral barriers to digital adoption among rural female livestock producers.
- 4. Macroeconomic Multiplier Effects of Value Addition:** As entrepreneurs scale into processed goods (e.g., chhana, silage, modern meat processing), future academic work should employ input-output models to quantify the broader macroeconomic multiplier effects of these microenterprises. Understanding how hyper-local job creation and circular economy practices (e.g., vermicompost) stimulate the wider upazila economy will provide vital data for national scaling strategies.

Annex A: Detailed Survey Questionnaire (English & Bangla)

Research Title: Economic Returns in the Livestock Sector for the Contribution of SDI NGO in Dhamrai & Manikganj, Bangladesh

Lead Researcher: Md. Safiul Alam , Student ID: 20250113, Program: MDS

Supervisor: Professor Dr. Mohammad Safiqul Islam, Supervisor, Chairman, Department of Economics, Jahangirnagar University.

Target Group: Smallholder Livestock Farmers & Entrepreneurs (Project & Control)

Section 1: Demographic & Household Profile (সাধারণ তথ্যাবলি)

1.1 Respondent Name (উত্তরদাতার নাম): _____

1.2 Upazila/Village (উপজেলা/গ্রাম): _____

1.3 Gender (লিঙ্গ): ☐ Male (পুরুষ) ☐ Female (মহিলা)

1.4 Age (বয়স): ☐ 18–35 ☐ 36–49 ☐ 50–60 ☐ 61+

1.5 Women's Decision-Making Power (নারীর সিদ্ধান্ত গ্রহণের ক্ষমতা): Do you independently or jointly make financial decisions regarding livestock sales? (আপনি কি গবাদিপশু বিক্রির ক্ষেত্রে স্বাধীনভাবে বা যৌথভাবে আর্থিক সিদ্ধান্ত নেন?) ☐ Yes (হ্যাঁ) ☐ No (না)

Section 2: Microfinance & Loan Utilization (মুদ্রঋণ এবং ঋণের ব্যবহার)

2.1 Are you a current borrower from SDI? (আপনি কি বর্তমানে এসডিআই-এর ঋণগ্রহীতা?) ☐ Yes ☐ No

2.2 Type of Loan Product (ঋণের ধরন): ☐ Agrosor (অগ্রসর) ☐ Jagoron (জাগরণ) ☐ Seasonal (মৌসুমি)

2.3 Loan Repayment Frequency (ঋণ পরিশোধের ধরন): ☐ Weekly ☐ Monthly

Section 3: Livestock Production, Sales & Profitability (গবাদিপশু উৎপাদন, বিক্রয় এবং লাভজনকতা)

3.1 Primary cattle rearing type (গবাদিপশু পালনের প্রধান ধরন): ☐ Beef Fattening ☐ Cow & Calf (Milk) ☐ Mixed Rearing ☐ Processed Goods

3.2 Average Monthly Sales from Livestock (গবাদিপশু থেকে গড় মাসিক বিক্রয়): BDT _____

3.3 Average Monthly Production Cost (feed, medicine, LSP) (গড় মাসিক উৎপাদন খরচ): BDT _____

3.4 Technology Adoption (প্রযুক্তির ব্যবহার) (Tick all that apply): ☐ Napier Grass ☐ Silage ☐ Chopper Machine ☐ Milking Machine ☐ Vaccine & AI

Section 4: Nutrition Security - MDD-W Tool (পুষ্টি নিরাপত্তা - এমডিডি-ডব্লিউ) *Please indicate if the female respondent consumed the following in the past 24 hours (গত ২৪ ঘণ্টায় আপনি নিচের কোন খাবারগুলো খেয়েছেন?)* ☐

1. Grains/Roots/Tubers (শস্য/মূল) ☐
2. Pulses/Beans (ডাল/মটরশুঁটি) ☐
3. Nuts/Seeds (বাদাম/বীজ) ☐
4. Dairy Products (দুগ্ধজাত খাবার) ☐
5. Meat/Poultry/Fish (মাংস/মুরগি/মাছ) ☐
6. Eggs (ডিম) ☐
7. Dark Green Leafy Vegetables (গাঢ় সবুজ শাকসবজি) ☐
8. Vitamin A-rich Fruits/Vegetables (ভিটামিন এ সমৃদ্ধ ফল/সবজি) ☐
9. Other Vegetables (অন্যান্য সবজি) ☐
10. Other Fruits (অন্যান্য ফল)

Annex B: FGD and KII Lists with Participants

Table B.1: Focus Group Discussions (FGDs) Roster

FGD Code	Target Group	Location	Date Conducted	No. of Participants	Key Themes Discussed
FGD-01	Women Project Beneficiaries (Cow-Calf)	Dhamrai Upazila	Oct 12, 2025	10	Income changes, AI access, MDD-W
FGD-02	Women Project Beneficiaries (Mixed)	Manikganj Sadar	Oct 14, 2025	11	Financial decision-making, Agrosor loans
FGD-03	Processed Goods Entrepreneurs	Manikganj Sadar	Oct 15, 2025	8	Market linkages, machinery, GMP
FGD-04	Control Group Farmers	Ghior Upazila	Oct 18, 2025	9	Traditional feeding, disease burden
FGD-05	SDI Implementation Staff & Facilitators	SDI Field Office	Oct 20, 2025	8	Logframe targets, project exit strategy

Table B.2: Key Informant Interviews (KIIs) Roster

- Executive Director, SDI** - Topic: *PKSF partnership evolution, institutional sustainability.*
- Value Chain Project Manager, RMTP** - Topic: *Market linkage bottlenecks, target achievement.*
- Upazila Livestock Officer (ULO), Dhamrai** - Topic: *DLS and LSP collaboration, disease trends.*
- Lead Livestock Service Provider (LSP)** - Topic: *Vaccine hub efficacy, 175% knowledge transfer rate.*
- Private Sweet Factory Owner (Buyer)** - Topic: *Chhana/Ghee supply chain quality, contract farming.*

Annex C: Additional Statistical Tables and Profitability Calculations

D.1: Derivation of ROI and Profit Margins (Endline Averages)

$$\text{Equation for Profit Margin: } Profit_{Margin} = \frac{Total_{Revenue} - Total_{Cost}}{Total_{Cost}} \times 100$$

1. Beef Fattening (Per Cycle/Per Entrepreneur):

- **Average Sales (Revenue):** BDT 172,013
- **Average Cost (Calf + Feed + Medicine + LSP):** BDT 126,072
- **Net Profit:** BDT 172,013 - BDT 126,072 = BDT 45,941
- **Calculated Margin:** (45,941/126,072)×100=36.44%

2. Value-Added Processed Goods (Monthly Average):

- **Average Sales (Revenue):** BDT 37,000
- **Average Cost (Raw Milk + Electricity + Labor + Packaging):** BDT 27,750
- **Net Profit:** BDT 37,000 - BDT 27,750 = BDT 9,250
- **Calculated Margin:** (9,250/27,750)×100=33.33%

D.2: Cross-Tabulation of Herd Size vs. Loan Product

Herd Size	Agrosor Loan (N)	Jagoron Loan (N)	Seasonal Loan (N)
1 to 2	22	30	5
3 to 5	108	51	11
5 and above	71	5	11
<i>(Note: Entrepreneurs scaling beyond 5 cattle predominantly transitioned to the higher-ceiling Agrosor enterprise product).</i>			

Annex D: Logframe Alignment Matrix (Baseline vs Endline)

Project Hierarchy / Indicator	Baseline Status	Endline Achievement (Project)	Endline (Control)	Variance / Status
Goal: Increase household income	Avg. BDT 28,576	Avg. BDT 32,161	Avg. BDT 22,919	+40% vs Control (Partially met 50% absolute target)
Outcome 1: Improve profit margins	Beef: ~24% Milk: 11%	Beef: 36% Milk: 22%	N/A	Exceeded (Target was 20% improvement for 80% of users)
Outcome 2: Sales of meat & dairy	Meat: 132k Processed: 23k	Meat: 172k (+30%) Processed: 37k (+61%)	N/A	Exceeded (Target was 30% increase)
Output 1: GAP Adoption	9%	61% modern management	21%	Met (Target was 60%)
Output 2: Nutrition (MDD-W)	74.43%	89.4% (Score ≥ 5)	72.06%	Met (+15 percentage points from baseline)
Output 3: Women's Empowerment	Low decision power	78% decision making; 65% asset ownership	N/A	Exceeded
Output 4: Market/LSP Linkages	0% structured	100% backward linkages	25-29%	Exceeded (Full ecosystem integration)

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